

The Economic Cost of Pollinator Loss: Evidence from an Invasive Species*

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Abstract

Insect pollinators are in global decline due to anthropogenic pressures, despite continued growth in the production of pollination-dependent crops. This divergence poses risks to agricultural productivity and food security, yet the economic value of pollination services remains poorly identified. As a result, policymakers face challenges in incorporating pollinators into cost-benefit analyses for conservation efforts and in addressing the negative externalities that contribute to their decline. This paper provides causal estimates of the effects of pollinator loss by exploiting quasi-exogenous variation in pollination supply from the staggered biological invasion of the Asian hornet (*Vespa velutina*) – a predator of insect pollinators – across France. Using a difference-in-differences design, I find that hornet invasion reduces managed beehive counts by 11.4%, oilseed rape yields by 5.2%, and average farmland prices by 4.3%. These results underscore the high economic value of insect pollinators for agriculture and illustrate how ecosystem health can shape economic outcomes.

JEL Codes: Q57, Q51, Q10, Q15

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1 Introduction

Mounting scientific evidence indicates that we are in the midst of a sixth mass extinction event (Ceballos et al., 2015). Across all measurable dimensions, from genetic diversity to species abundance and ecosystem integrity, biodiversity is declining at unprecedented rates. According to the 2019 IPBES report, approximately 25% of animal and plant species are threatened with extinction, 23% of terrestrial ecosystems exhibit significant compositional degradation, and, most strikingly, global wildlife abundance has declined by an estimated 82% since prehistoric times (IPBES, 2019). This collapse is likely driven by anthropogenic pressures, including land-use change, pollution, invasive species, and climate change, which undermine the ecological foundations of human welfare (Díaz et al., 2019).

Biodiversity supports a wide array of ecosystem services that act as inputs to the production of goods and services. Of the 18 *nature’s contributions to people* identified by the IPBES, 14 have declined globally over the past 50 years (IPBES, 2019). These include essential services like climate regulation, pollination, and nutrient cycling. While ecologists have long emphasized biodiversity’s foundational role in sustaining human societies, the economic implications of its collapse remain poorly understood.

This gap is particularly problematic in policy settings where regulatory decisions increasingly rely on quantitative cost-benefit analysis and marginal valuation (see, for instance, Circular A-4 in the United States, OMB, 2023). Because most ecosystem services are not priced in markets, they are often undervalued or ignored in economic evaluations. While economics has developed a variety of tools, from stated preference methods to natural capital pricing models, to value numerically environmental goods (Carson, 2012; Costanza, 2020; Fenichel & Abbott, 2014; Weitzman, 1998), these methods often depend on structural assumptions and calibration, producing widely varying estimates.

Recent contributions argue for combining structural models with causal microeconomic evidence to improve the reliability of policy guidance, as illustrated by the U.S. Environmental Protection Agency’s approach to estimating the social cost of carbon (EPA, 2023). Yet the growing causal inference literature has largely focused on non-biological environmental shocks, such as pollution (e.g., Graff Zivin et al., 2023) or temperature (e.g., Carleton et al., 2022), while biologically mediated services, such as pollination, seed dispersal, and pest control, remain empirically understudied.

This gap is driven primarily by the scarcity of observational data and the ethical and practical infeasibility of experimentally manipulating ecological processes. Consequently, the welfare impacts of biodiversity loss are often inferred indirectly or remain unquantified. This paper contributes to filling that gap by providing causal evidence on the value of one of the

most economically salient animal-mediated ecosystem services: insect pollination (hereafter pollination).

Pollination plays a critical role in natural and agricultural systems. Nearly 90% of wild flowering plants and over three-quarters of food crop species depend, at least in part, on animal pollinators. These crops account for 35% of global production volume, valued at \$235 to \$577 billion annually (2015 USD). Furthermore, pollination-dependent crops include many fruits, vegetables, seeds, nuts, and oil crops that are key sources of micronutrients, vitamins, and minerals essential to human nutrition. Yet both wild and managed pollinators are experiencing marked declines in abundance and diversity, particularly in Europe and North America. These trends are primarily driven by negative externalities associated with land-use change, intensive agricultural practices, pesticide application, environmental pollution, invasive species, pathogens, and climate change. At the same time, global production of pollinator-dependent crops has risen by over 300% in the past five decades, intensifying pollination demand. This divergence raises concerns for food security, particularly under climate change (IPBES, 2016).

Efforts to quantify the value of pollination services are extensive, with one of the most frequently cited global estimates amounting to \$210 billion (2020 USD) (Gallai et al., 2009). However, the majority of studies rely on correlational designs that are susceptible to omitted variable bias, or on field experiments that lack external validity (see Porto et al., 2020 for a review).¹ No existing studies identify causal effects of marginal reductions in pollination supply under real-world conditions².

This paper addresses this key limitation in the literature by leveraging a natural experiment arising from the inadvertent and staggered invasion of the Asian hornet (*Vespa velutina*) across France. A voracious predator of insect pollinators (hereafter, pollinators), the Asian hornet provides a source of plausibly exogenous variation in pollination supply. I exploit this variation to estimate the causal effects of pollinator decline on managed beehives, crop production, and farmland prices. Introduced accidentally in 2004, likely via international trade, the Asian hornet spread rapidly throughout France and into neighboring European countries. Its emergence sparked early policy concern due to its documented impacts on honey bees and other pollinating species, threatening both apiculture and pollination-dependent agriculture. Despite targeted control efforts, the invasion advanced largely unchecked. Importantly, efforts to monitor and document the species' geographic spread have generated high-quality

¹The most common method, the dependency ratio approach, estimates pollination value by multiplying crop market values by crop-specific pollination dependence indices. These indices are typically derived from in-situ experiments that prevent insect pollinators from accessing specific plots—often by placing mesh cages over them—and measure the resulting changes in yield. As such, they capture the effect of a complete (100%) removal of insect pollination services, which may not reflect marginal changes under real-world conditions.

²To the best of my knowledge.

settlement data since its initial introduction. I use this spatial and temporal variation in successful establishment to implement a difference-in-differences strategy, comparing trends in economic outcomes in invaded versus non-invaded regions before and after hornet arrival.

The results reveal substantial economic effects. First, I estimate an average reduction of 11.4% in the number of beehives used in honey production, implying a loss of approximately 103,500 colonies through 2016. Second, I find a 5.2% decline in yields of rapeseed³, a pollination-dependent crop, with no detectable changes in comparable non-pollination-dependent crops. This yield reduction corresponds to 1.55 quintals per hectare⁴, amounting to a national productivity loss of €42 million in 2021 euros. I also find some evidence of land reallocation away from rapeseed, leading to a larger decline in total output. Finally, I estimate a 4.3% average decline in farmland prices, corresponding to a €224 per hectare reduction in land value. Relative to the average agricultural land area in France between 2000 and 2003, this decline implies a cumulative farmland valuation loss of about €7.15 billion in 2021 euros. This total corresponds to roughly 0.36% of France’s average annual GDP over the same period. These findings indicate that pollination services may hold substantial economic value and that their disruption can have wide-ranging effects across the agricultural sector and land markets, implying that the social cost of activities contributing to pollinator decline may be considerable.

Beyond its contribution to the pollination literature, this study offers some of the first causal evidence on the economic impacts of a biological invasion in agriculture (Shackleton et al., 2019). These findings are particularly relevant given the growing frequency of invasive species linked to global trade and climate change. The Asian hornet has also invaded South Korea (J.-K. Kim et al., 2006), Japan (Ueno, 2014), and, more recently, the United States (Georgia Department of Agriculture, 2023), and its suitable habitat range is expected to increase further under projected climate scenarios (Barbet-Massin et al., 2013).

Despite substantial ecological research and policy attention, the economic consequences of the Asian hornet invasion remain understudied. The only prior economic assessment focuses on public control costs, estimating €23 million in nest destruction expenditures between 2006 and 2015, with projected annual costs reaching €11.9 million under full range expansion (Barbet-Massin et al., 2020). That study notes that the damages to apiculture and pollination services likely exceed current control expenditures. This paper complements that work by providing causal evidence on broader economic impacts.

This paper contributes to several strands of the literature. First, it adds to a growing body

³Rapeseed—also known as canola or oilseed rape—is referred to as ‘rapeseed’ throughout the paper for consistency.

⁴A quintal equals 100 kilograms, roughly 220 pounds or 3.7 bushels of wheat. A hectare equals 10,000 square meters, or approximately 2.47 acres. Both units are standard in France and much of Europe.

of work that causally identifies the role of animal-mediated ecosystem services in supporting human welfare. Recent studies have linked variation in key species to adverse impacts on public health, agricultural productivity, and economic development more broadly (Alsan, 2015; E. Frank & Sudarshan, 2024; E. G. Frank et al., 2022; Raynor et al., 2021; Springborn et al., 2022). Closest to the present study is E. G. Frank (2024), who exploit the spread of white-nose syndrome among bats in the United States to show that the collapse of bat populations, which serve as key natural pest controllers, led to increased pesticide use and higher infant mortality.

Second, this paper contributes to the literature on how agricultural productivity responds to environmental shocks, such as weather variability and pollution exposure (Boone et al., 2019; Schlenker & Roberts, 2009; C. Zhao et al., 2017). By focusing on pollination as an input to crop production, I provide new evidence on a biological mechanism that is distinct from commonly studied abiotic stressors.

Third, by decomposing production changes into intensive (yield) and extensive (area) margins, and by comparing pollination-dependent and non-dependent crops, this study contributes to the literature on farmer adaptation and behavioral responses to shocks in production conditions (Annan & Schlenker, 2015; Burke & Emerick, 2016; Gollin & Udry, 2021; Puy & Shrader, 2024).

Fourth, this paper also contributes to the literature examining the effects of invasive species on human welfare. While ecological research has extensively documented the risks posed by biological invasions, their economic consequences remain relatively understudied (Jones & McDermott, 2018; Shackleton et al., 2019). Finally, this paper also adds to the growing body of research on the Asian hornet invasion in France by providing new economic evidence, building on the ecological and entomological findings discussed in the next section.

The remainder of the paper is structured as follows. Section 2 provides background on the Asian hornet invasion, assesses its suitability as a natural experiment, and outlines the economic mechanisms under study. Section 3 describes the data sources and processing steps. Section 4 defines the parameters of interest, discusses the identifying assumptions, and presents the estimation strategy. Section 5 reports the results, addresses potential limitations, and interprets the findings. Section 6 concludes and outlines directions for future research.

2 Ecological Context and Economic Motivation

2.1 The Asian Hornet Invasion As An Ecological Shock

Geographic Spread and Introduction History The Asian Hornet (*Vespa velutina*), also known as the yellow-legged hornet, is an insect species native to parts of East and Southeast Asia, including northeastern India, China, Taiwan, and Indonesia (Perrard et al., 2014). The species was inadvertently introduced into France in 2004, with the first nests reported in the Lot-et-Garonne département⁵. Genetic and ecological evidence confirmed the successful establishment of a self-sustaining population by 2006 (Villemant et al., 2006). Since its introduction, the Asian Hornet has spread rapidly across France, advancing at an average rate of approximately 78 kilometers per year. By 2006, the hornet had established in 13 départements, expanding to 32 by 2009, 56 by 2012, and 67 by 2014. As of 2023, it has been officially recorded in all contiguous French départements, with the last being Haut-Rhin, as illustrated in Figure 1. This rapid expansion has not been confined to France: the hornet has successfully colonized most parts of Western Europe⁶.

Population Density and Environmental Suitability As with many animal species, quantifying the abundance of the Asian hornet in France is challenging due to the absence of high-resolution, systematic nest records across time and space.⁷ Entomological studies have attempted to estimate the species’ carrying capacity,⁸ though these efforts are hindered by underreporting, limited detection, and a lack of coordinated monitoring. The most comprehensive estimate to date is provided by Franklin et al. (2017), who conducted an intensive local survey in Andernos-les-Bains, a municipality in southwestern France nine years after introduction. They report that the number of detected nests increased annually and that growth was expected to continue even a decade after colonization, consistent with logistic population dynamics that most invasive species follow (Robinet et al., 2017). They also estimate a carrying capacity of 10.64 nests per square kilometer in a semi-urban landscape. However, carrying capacity varies across regions with environmental conditions—particularly

⁵Although the precise origin of the invasion remains uncertain, entomological analysis from the Muséum National d’Histoire Naturelle indicates that the most probable introduction pathway was the importation of pottery by a bonsai producer. Genetic evidence further suggests that the entire population may have originated from a single multi-mated female founder (Franklin et al., 2017).

⁶This includes Spain (2010), Portugal (2011), Italy (2012), Germany (2014), Belgium and the United Kingdom (2016), and the Netherlands (2017). See <https://frelonasiatique.mnhn.fr/> for a detailed account.

⁷Several British media outlets (e.g., The Guardian, The Times, The Telegraph, Daily Mail) have reported estimates of 500,000 Asian hornet nests in France two decades after introduction (see, e.g., [The Guardian, 2024](#)). However, these figures are not traceable to verifiable sources and should be interpreted with caution.

⁸Carrying capacity is the maximum population size of a species that an environment can sustain over time.

temperature, precipitation, and urbanization.⁹ Figure 8 shows predicted habitat suitability across Europe under current and future climate conditions. Favorability is highest in southwestern France, where environmental conditions already support higher population densities, and is expected to increase further under climate change.

Ecological Drivers of Invasion Success Overall, while global trade likely facilitated the initial introduction of the Asian hornet into France, climate change has expanded the range of suitable habitats, enabling the species to establish and spread (Robinet et al., 2018). In addition, the absence of natural competitors and the lack of effective defensive behaviors in the European honeybee – which constitutes the majority of the hornet’s diet—have contributed to both the rapid expansion and the sustained population growth of the species in its introduced range (Arca et al., 2014).

Policy Response and Management Efforts Since its introduction, the Asian hornet has triggered a series of legislative and policy responses in France and at the European level. In 2012, the French government designated the species as a *second category health hazard for honeybees* across the national territory, mandating a coordinated strategy for surveillance, prevention, and control¹⁰. In 2014, the European Union listed the hornet as an *Invasive Alien Species of Union concern*. A 2019 EU assessment emphasized that “*this species can have significant knock-on effects on the ecosystem, representing an important obstacle to crop pollination*” (Fedele et al., 2019). In March 2025, the French National Assembly unanimously passed new legislation aimed at intensifying efforts to curb the spread of the Asian hornet. Nevertheless, control measures have remained largely ineffective, and both entomologists and government reports acknowledge that the species’ geographic expansion has proceeded virtually unimpeded (MAA & MTES, 2018; Robinet et al., 2017).

2.2 The Asian Hornet Invasion As A Natural Experiment

This section draws on the ecological literature to justify the relevance of the Asian hornet as a source of quasi-exogenous variation in pollination supply. Specifically, I argue that the Asian hornet invasion can serve as a valid instrumental variable for pollination supply. In this study, I estimate the reduced-form effect of the invasion on agricultural outcomes.

⁹S. H. Kim et al. (2021), for example, estimate an abundance model using 2018–2021 South Korean data, where the Asian hornet established in 2003, and find persistent spatial heterogeneity, with higher densities near the initial point of introduction.

¹⁰In practice, these efforts included funding entomological research, deploying poisoned bait traps, and organizing the detection and destruction of nests—typically carried out by local authorities or certified pest control operators.

Relevance assumption Asian hornets are generalist predators that prey on various taxonomic groups of pollinators especially Apidae (e.g. bumblebees) and Syrphidae (e.g. hoverflies) (O’Shea-Wheller et al., 2023; Villemant et al., 2014)¹¹. Recent estimates suggest that a single colony consumes over 11.3 kilograms of insect biomass per year, which corresponds to more than 90,000 honeybee-sized prey items (Rome et al., 2021)¹². This magnitude of insect predation implies a potentially substantial reduction in pollinator populations where hornets are established. In addition to direct predation, Asian hornets disrupt pollinator activity by targeting areas with continuous insect presence, such as hive entrances. Their presence provokes a defensive response in honeybee colonies, often leading to foraging paralysis and increased homing failure¹³. These behavioral disruptions raise the risk of colony collapse and contribute to a significant reduction in pollination services (Requier et al., 2019; Rojas-Nossa & Calviño-Cancela, 2020). The ecological literature widely recognizes *V. velutina* as a significant and emerging threat to pollination services (S. H. Kim et al., 2021; Monceau et al., 2014). Its combined impact through biomass depletion and behavioral suppression has raised concerns about cascading effects on crop productivity and biodiversity. In both scientific assessments and policy documents, the hornet is frequently cited as an invasive species of concern for ecosystem function in temperate Europe (IPBES, 2016).

Exclusion Restriction To interpret the Asian hornet invasion as a quasi-experimental shock to pollination supply, it must be assumed that its impact on agricultural outcomes operates exclusively through effects on insect pollinators. There is no evidence that Asian hornets provide pollination services to agricultural crops. The only documented case involves limited pollination of *Hedera hibernica*, a wild, non-agricultural species, with the hornet’s presence associated with an overall decline in pollination quality (Rojas-Nossa et al., 2023). One potential concern is survey-based and anecdotal evidence suggesting that hornets may feed directly on fruit crops (Nave et al., 2024). While these reports are limited and context-specific, they raise the possibility of a direct consumption channel in some horticultural settings. This mechanism is unlikely to be relevant in the present study, which focuses on a pollination-dependent field crop not subject to direct hornet damage. A second concern is that the hornet may indirectly affect pest dynamics by reducing populations of beneficial

¹¹Multiple ecological studies have documented that their diet is heavily skewed toward the European honeybee, which can account for up to 70% of consumed prey biomass, depending on local environmental conditions.

¹²The life cycle of the Asian Hornet follows an annual pattern, initiated by a single overwintering queen in early spring. Colonies expand rapidly, reaching up to 15,000 individuals through the season. By late autumn a nest may include up to 2,000 workers and producing over 1,000 future queens and drones (Villemant et al., 2014).

¹³Homing failure refers to the inability of foraging bees to return to the hive, often due to stress, disorientation, or predator avoidance. Foraging paralysis describes the suppression of foraging activity, where bees remain inside the hive or reduce trips due to the persistent presence of predators such as the Asian hornet at the hive entrance.

insects such as hoverflies, which contribute to natural pest control (Villemant et al., 2014). While theoretically plausible, these indirect effects are likely to be minor relative to the hornet’s primary ecological role as a pollinator predator.

Exogeneity For the Asian hornet to serve as a valid instrument for pollination supply, its spread must be exogenous to other determinants of agricultural outcomes. The species was introduced inadvertently into France and was only formally detected once established populations had already formed Villemant et al. (2006). This unplanned introduction reduces concerns about endogenous placement of the invasion with respect to agricultural conditions. The primary determinant of the hornet’s spatial spread is geographical proximity, reflecting a natural diffusion process from the initial point of entry. Secondary factors influencing the speed and extent of spread include habitat suitability – such as local climate, elevation, and urbanization rates (Robinet et al., 2017). However, at the scale of French départements, contiguous areas typically exhibit limited variation in these characteristics. Nonetheless, potential confounding may arise if variables correlated with the hornet’s spread also affect agricultural outcomes. While this poses a concern for identification, many such variables are observable and are directly controlled for in the reduced-form difference-in-differences design employed in this paper. As a result, the analysis relies on weaker identifying assumptions, which are discussed in greater detail in Section 4.

Taken together, these biological facts provide strong evidence that the Asian hornet invasion constitutes a relevant and persistent shock to insect-mediated pollination supply. This motivates its use in the empirical analysis as a quasi-natural experiment for evaluating the economic consequences of pollinator loss.

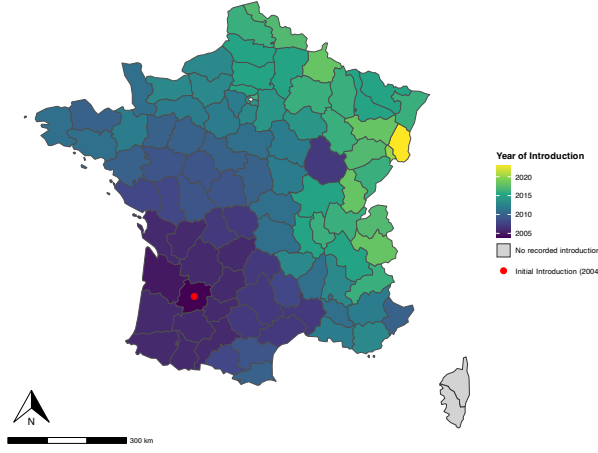
2.3 Economic Mechanisms

In light of this ecological shock, I outline the theoretical mechanisms through which the Asian hornet invasion may affect economic outcomes. Figure 2 presents the conceptual framework that guides the identification strategy and illustrates the hypothesized causal pathways from pollinator decline to agricultural and land market impacts. I adopt a multi-outcome strategy, tracing effects from honey-producing beehives to crop production and farmland prices, to capture both short-term consequences and forward-looking market responses.

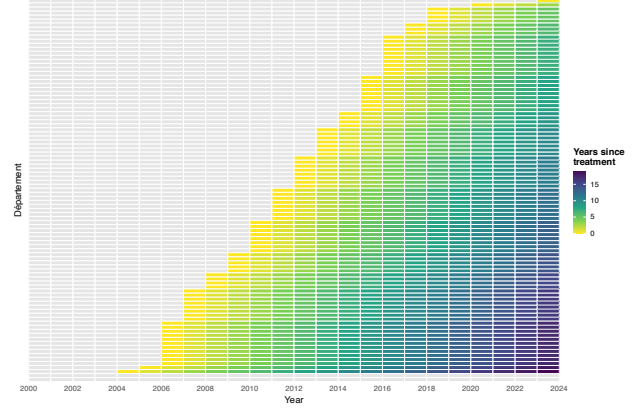
Beehives in Honey Production To validate the hypothesized shock to pollination supply and assess its potential magnitude, I first examine regional counts of honey-producing beehives. These serve as the most direct nationally available proxy for insect pollinator abundance, reflecting ecological pressure from the hornet on managed pollinators. While this is an imperfect indicator for overall pollination supply, it also provides insight into the impact

Figure 1: Spatial and Temporal Spread of the Asian Hornet Invasion in France

(a) Year of first invasion by département

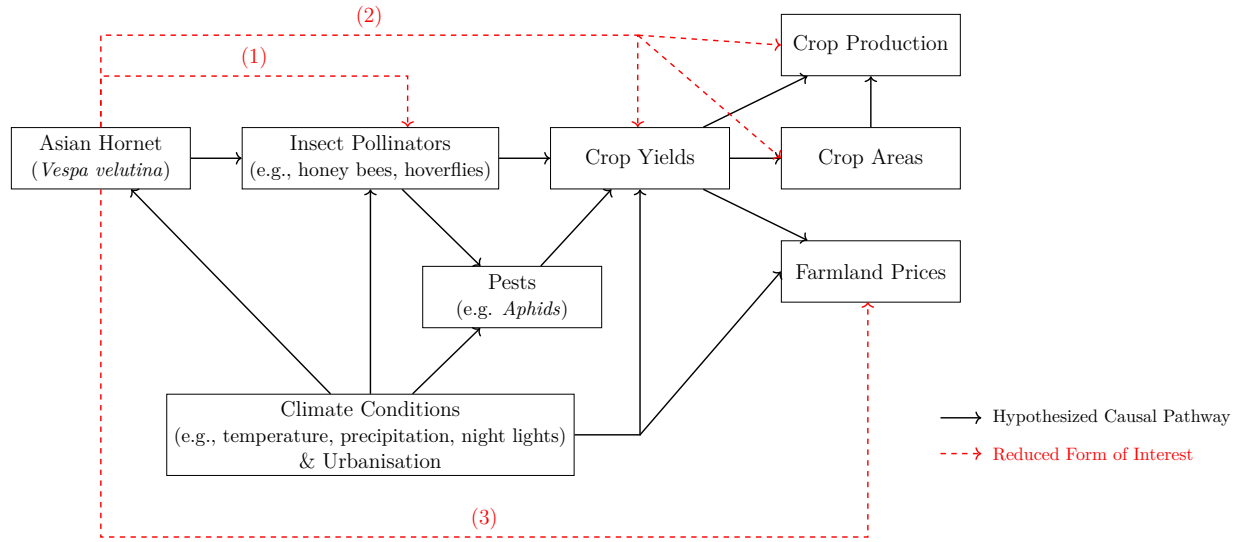


(b) Exposure duration by département and year



Note: Panel (a) maps the year of first recorded presence of the Asian hornet across French départements. Grey areas indicate no recorded invasion through 2023. The red dot marks the site of initial introduction in 2004. Panel (b) visualizes staggered exposure over time: each row is a département, and color indicates years since invasion.

Figure 2: Schematic Pathway Linking Asian Hornet Invasion to Agricultural Outcomes



Note: This figure presents the conceptual framework linking the Asian hornet invasion to agricultural outcomes via its effect on insect pollinators. Solid arrows represent hypothesized causal pathways informed by economic, ecological and agronomic mechanisms. Dashed arrows denote reduced-form relationships estimated in the empirical analysis. Climate conditions and urbanization are included as confounding factors that influence hornet dynamics, pollinator abundance, and crop outcomes. Urbanization & Climate Conditions may also directly affect farmland prices through land market channels. The inclusion of pests highlights a theoretical violation of the exclusion restriction were the Asian hornet to be used as an instrument.

of the invasion on the honey industry, which is the most directly affected sector. Since the hornet’s arrival, beekeepers have reported substantial losses of colonies¹⁴. Yet no academic study has causally linked these losses to the Asian hornet, despite growing pressures on pollinators from potentially confounding factors such as neonicotinoids, *Varroa destructor*, climate change induced phenological calendar disruptions, and habitat loss¹⁵.

Crop Production, Cultivated Area, and Yield Second, I examine the impact of reduced pollination supply on crop production. Pollination services are a key input in the production of many crops, and in some cases, essential for their development (Klein et al., 2007). A negative shock to pollinator abundance should therefore lower the output of pollination-dependent crops¹⁶. To decompose the production response, I separately analyze changes in yield (intensive margin) and cultivated area (extensive margin). These adjustments may reveal adaptation strategies, such as crop switching to non-pollination dependent crops or land reallocation¹⁷. I also examine comparable non-pollination-dependent crops, which serve both as a placebo and to detect potential crop-switching responses.

Farmland Prices Finally, I analyze farmland prices at the level of *Nouvelles Régions Agricoles* using a hedonic framework. According to standard valuation models, farmland prices reflect the net present value of expected agricultural returns (Drescher et al., 2001). If the hornet invasion reduces yields of pollination-dependent crops and adaptation is imperfect, we should observe a decline in land values in affected regions. As a forward-looking outcome, farmland prices incorporate market expectations and are less susceptible to transitory shocks or measurement error than crop-level data. They also aggregate heterogeneous impacts across crops and production systems, offering a concise measure of the broader economic consequences of pollinator decline.

This study estimates the effect of confirmed Asian hornet invasion on each outcome variable, that is, focusing on the extensive margin of exposure (i.e., whether a département has been invaded). Due to the lack of reliable data on nest density or predation rates, I cannot observe the intensity of exposure directly. However, I expect the effects to increase over time following invasion, consistent with the species’ population growth dynamics and the

¹⁴Yaël Serfaty, director of the French beekeepers’ association UNAF, stated: “Chaque année nous perdons 30% des colonies” (“Each year we lose 30% of our colonies”), referring to losses attributed to the Asian hornet (Fleury, 2024).

¹⁵Neonicotinoids are neurotoxic insecticides harmful to bees, while *Varroa destructor* is a parasitic mite that weakens colonies and transmits viruses. Both are commonly cited drivers of honey bee colony collapse (IPBES, 2016).

¹⁶A theoretical exception arises if farmers face no land constraints and demand is inelastic, prompting them to increase cultivated area to offset yield losses.

¹⁷This channel is likely weaker for perennial crops (e.g., fruit trees) due to high sunk costs and delayed returns on investment.

resulting cumulative pressure on pollinators. Spatial heterogeneity is also likely: the invasion originated in the southwest, which combines early exposure with high climatic suitability, suggesting stronger impacts in that region.

3 Data and Descriptive Statistics

In this section, I briefly summarize the data sources involved in this analysis¹⁸. Table 10 provides metadata for each data set.

3.1 Asian Hornet Spread

Annual département-level data on hornet presence are obtained from the *Inventaire National du Patrimoine Naturel* (INPN), the official biodiversity portal operated under the supervision of the French Ministry of Ecological Transition, the National Museum of Natural History (MNHN), and the French Biodiversity Office¹⁹. Records consist of dated GPS observations of viable colonies, reported by a combination of institutions and the public, and validated by entomologists from the MNHN. The data are publicly available through the INPN’s dedicated Asian hornet portal (INPN, 2018).

The dataset covers the full period from the first confirmed successful settlement of the Asian hornet in 2004 through 2023 and includes all metropolitan French départements. Although observation data may reflect variation in reporting effort, the INPN dataset is the most comprehensive and validated source of hornet presence in France and is widely used in ecological and entomological research (e.g. Barbet-Massin et al. (2013), Fournier et al. (2017), and Villemant et al. (2011)). This is the most reliable available proxy of hornet colonization despite potential collection biases.

Rather than aggregating raw geolocated records, I rely on the pre-processed progression map provided by the INPN, which directly reports the year of first colonization for each département. From this, I construct a binary treatment indicator that equals one in the year a département is deemed successfully colonized and remains one thereafter. Figure 1 illustrates the resulting staggered adoption structure. The invasion began in the southwest and proceeded primarily via spatial diffusion, with occasional non-contiguous jumps (e.g., Côte-d’Or). As of 2023, all contiguous metropolitan départements are colonized except for Corsica²⁰. On average, 5.3 départements were newly colonized per year with slower rate at the

¹⁸Note that all data used in this analysis is publicly available

¹⁹The INPN was established in accordance with Article L411-1 A of the French Environmental Code and is mandated to maintain a comprehensive inventory of France’s ecological, faunistic, floristic, and geological heritage.

²⁰To ensure consistency in the treatment definition, I address four cases where départements were initially

start and end of the sample window. The largest cohorts entered in 2006 (13 départements) and 2016 (10 départements), corresponding to the most rapid expansion phases. The resulting variable is a binary, staggered treatment indicator at the département-year level from 2004 to 2023.

3.2 Crop Production & Beehive Counts in Honey Production

I use annual département-level data on crop-specific agricultural output from the *Statistiques Annuelles Agricoles* (SAA), published by AGRESTE, the statistical agency of the French Ministry of Agriculture (AGRESTE, 2024c). The SAA is compiled jointly by the Ministry’s Statistics and Forecasting Department (*Service de la Statistique et de la Prospective*) and its regional statistical services. This dataset serves as the basis for constructing official national and regional agricultural accounts and is produced in accordance with European statistical obligations.²¹

The SAA reports, for each crop and département, annual production (in quintals), cultivated area (in hectares), and yield (quintals per hectare). Overall, it covers the years 2000–2023 and includes a wide range of crops, from disaggregated varieties (e.g., Golden Apples) to aggregate categories (e.g., total cereals)²². Most crop data are available at the département level; however, recorded beehive counts in honey production used to proxy pollination supply are only available at the regional level for 2000–2020²³.

The SAA is a synthetic statistical operation that harmonizes information from multiple sources, including annual agricultural surveys (e.g., Structure survey, Orchard Inventory), administrative records (e.g., CAP subsidy declarations), and reports from local producer organizations (AGRESTE, 2024b). These estimates are validated by regional statistical services and benchmarked periodically against exhaustive sources such as agricultural censuses.

marked as colonized but reclassified as uncolonized in the following year before eventually being re-colonized and stay so until the end of the sample window. For these départements (Ille-et-Vilaine, Nord, Val-d’Oise, and Seine-Saint-Denis), I define treatment onset as the first year of uninterrupted colonization through the end of the sample period. This ensures a staggered treatment path in line with the recent difference-in-differences literature.

²¹As required by Regulation (EU) No. 543/2009 of 18 June 2009 and Regulation (EU) No. 549/2013 of 21 May 2013.

²²In practice, the data are distributed across two distinct yet complementary datasets: one spanning 2000–2020 and aligned with the 2000 and 2010 censuses, and another covering 2010–2023 and incorporating revisions based on the 2020 census. Since the Asian hornet invasion began in the mid-2000s, I require pre-treatment data from 2000 onward to assess baseline trends, as well as post-2010 data to evaluate potential long-term effects. I construct a consistent panel by merging the two series for crops with stable definitions across both datasets, verifying internal consistency by cross-checking overlapping values for the common year of census (2010). This approach ensures continuity over time while preserving alignment with the most recent census benchmarks.

²³In metropolitan France, a region is a higher-level administrative unit comprising an average of 7.4 départements. There are 13 regions and 96 départements in total.

A limitation of the SAA is that crops with low production in some départements are not always directly surveyed; values in these cases are imputed based on past data and national averages. These imputations are most common for fruits and vegetables and may understate local production variability (e.g. Strawberries and Pears). Despite this limitation, the SAA remains the most comprehensive and authoritative source of publicly available crop production data in France.

In total, the 2000–2020 dataset includes 210 distinct crop categories. Of these, I successfully harmonize 140 series across the full sample period using stable crop definitions. However, several features of the raw data necessitate a selective approach for empirical analysis.

First, crop production is reported in physical units (e.g., quintals or hectares), which vary by crop and do not account for differences in economic value. As a result, aggregating or pooling across crops without standardization would lead to misleading comparisons and distort the economic interpretation of treatment effects. Second, agricultural production is highly heterogeneous both across space – some crops are grown in only a subset of départements – and across time, with coverage gaps more common for fruits and vegetables. Third, for reliable identification of both pre-treatment trends and long-run effects, the dataset must offer adequate time-series depth and geographic coverage.

To address these concerns, I retain only crops for which a balanced département-year panel yields at least 1,600 observations over the 2000–2023 years span. This filtering criterion ensures that each retained crop has sufficient spatial and temporal variation to support credible estimation. The resulting selection includes 25 crops, predominantly those of high economic relevance, as more spatially widespread crops tend to correspond to higher aggregate production at the national level. Among these, 16 are field crops, 5 are legumes, and 4 are fruits. According to the pollination dependence index of Klein et al. (2007), 13 of these crop categories depend to some extent on insect pollination.

3.3 Pollination Dependence Index

To assess potential heterogeneity in the impact of the Asian hornet across crops, I rely on the pollination dependence classification developed by Klein et al. (2007). This index, widely used in both ecological and economic research—including the IPBES global assessment of pollinators (IPBES, 2016)—categorizes crops based on experimental evidence quantifying the reduction in yield when pollinators are excluded. Dependence is grouped into five classes: *essential* ($\geq 90\%$ yield reduction), *high* ($40\% \leq \text{reduction} < 90\%$), *modest* ($10\% \leq \text{reduction} < 40\%$), *little* ($0\% < \text{reduction} < 10\%$), and *none* (0% reduction). This classification is based on crop-level meta-analyses of fruit set, weight, and quality under pollinator exclusion. I use this index to define pollination-dependent crops and explore differential treatment effects

accordingly.

3.4 Farmland Prices

I obtain annual data on average farmland prices from the *Valeur Vénale des Terres* (VVT) dataset, published by AGRESTE (AGRESTE, Service de la statistique et de la prospective (SSP), 2024a)²⁴. This dataset fulfills a legal requirement to monitor and publicly report agricultural land values²⁵.

The VVT reports nominal euro prices per hectare of freehold farmland from 1999 to 2021 across Metropolitan France at the level of *Nouvelles Régions Agricoles* (NRAs), which are groupings of adjacent *Petites Régions Agricoles* (PRAs). PRAs are small agricultural zones defined by the intersection of départements and homogeneous agro-climatic areas; each département contains multiple PRAs. Due to the relatively low number of annual land transactions within individual PRAs, the SSP and FNSAFER aggregate these PRAs into approximately 400 NRAs to produce statistically robust price estimates. I use this NRA-level data in the analysis.

Land prices are derived from a comprehensive observatory of rural land transactions maintained by SAFER, based on notifications of proposed sales that notaries are legally required to submit (AGRESTE, Service de la statistique et de la prospective (SSP), 2024b). To ensure relevance and consistency, the sample is restricted to transactions involving land intended to remain in agricultural use, with plots exceeding 70 ares in size. Transactions are further filtered to exclude outliers using an interquartile range method applied to logged prices at the regional level.

The dataset reports average prices over a three-year moving window, stated in nominal euros of the central year. As such, the VVT offers the most granular and systematically validated public data on agricultural land values available in France. Summary statistics for the freehold price variable over the estimation sample are shown in Table 5.

3.5 Agro-Climatic Conditions

To account for weather-related variation in agricultural outcomes, I construct a set of agro-climatic control variables using data from the SAFRAN model developed by Météo-France, the official meteorological agency of France. SAFRAN²⁶ is a high-resolution atmospheric

²⁴The VVT is produced jointly by the Ministry’s Statistics and Forecasting Department, and the National Federation of Land Development and Rural Settlement Companies (FNSAFER).

²⁵Article L.312-4 of the Rural and Maritime Fishing Code.

²⁶*Système d’analyse fournissant des renseignements atmosphériques à la neige*. Initially developed at CNRM/CEN for avalanche forecasting, SAFRAN produces spatially and temporally consistent reanalysis data for mainland France and is widely used for hydrological and surface modeling applications.

reanalysis system that combines in-situ weather station observations with numerical modeling to generate hourly weather estimates across a regular 8,602-cell grid covering all of metropolitan France. I use daily aggregated SAFRAN data on minimum and maximum temperature and total liquid precipitation.

To quantify agro-climatic exposure at the département-year level, I construct two sets of temperature-based indicators, defined over one-degree intervals of daily temperature. The first set is unweighted and records the number of days in each degree temperature bin over the full calendar year. The second set follows the crop-specific methodology of Schlenker and Roberts (2009), applying spatial and temporal weights to reflect the distribution of cropped area and the timing of key phenological stages, thereby capturing exposure during relevant parts of the growing season.

Spatial weights are derived from crop-specific land use shares within each département using satellite-based classifications from the Copernicus CORINE Land Cover inventory (Copernicus Land Monitoring Service, 2012). Temporal weights are based on each crop’s growing season, estimated from phenological data compiled by the TEMPO observatory network.²⁷ This two-dimensional weighting scheme yields agronomically relevant measures of temperature and precipitation exposure during periods most sensitive to crop development. Full construction details are provided in Appendix 6.

Figure 12 illustrates the distribution of growing-season temperature exposure across selected crops and temperature intervals. To validate the construction of these exposure measures, I replicate the non-linear yield–temperature relationship of Schlenker and Roberts (2009) for maize – the only crop from their original study with sufficient coverage in the French dataset. Results confirm a similar non-linear yield response to high temperatures, supporting the validity of the constructed exposure metrics (see Figures 15 and 16; see Appendix 6 for details).

Finally, to reduce dimensionality and avoid overparameterization given the limited number of treated cohorts and the modest sample size, I construct two summary indicators from the derived temperature exposure measures: Growing Degree Days (GDD) and Hot Degree Days (HDD). These standard metrics aggregate daily temperature exposure into a single agronomically meaningful measures of heat accumulation and extreme heat stress, respectively.

²⁷TEMPO is a national platform coordinated by CNRS, INRAE, and the French Ministry of Higher Education and Research, monitoring phenological patterns across cultivated and wild species.

3.6 Urbanization Proxy

Urbanization has been identified as a potential second-order determinant of Asian hornet settlement, possibly influencing nesting sites, food availability, or detection rates. To proxy urbanization intensity at the département level, I use the recent Simulated VIIRS Nighttime Light (SVNL) dataset developed by Chen et al. (2024). This dataset produce consistent global nightlight measures from 1992 to 2023 at a spatial resolution of 15 arc-seconds (approximately 500 meters)²⁸. The SVNL series closely matches observed VIIRS data which has been increasingly used as a proxy for urban activity and economic development in recent literature (e.g., Chen et al., 2020; M. Zhao et al., 2020).

I access the SVNL through Google Earth Engine and compute annual averages of nighttime light intensity at the département level. Figure 11 illustrates both the raw SVNL raster data and the département-level annual averages for selected years. The derived département-level averages serve as a consistent and spatially granular proxy for urbanization, which I include as a control variable to account for potential confounding effects related to habitat suitability or detection likelihood.

4 Empirical Strategy

To estimate the causal effect of the Asian hornet invasion on agricultural outcomes, I exploit quasi-exogenous variation in the timing of its spread across French départements following its accidental introduction in 2004. The invasion unfolded in a staggered manner, generating variation in treatment timing that I use to implement a dynamic difference-in-differences strategy. This section introduces the parameters of interest, discusses the identifying assumptions, and outlines the estimation methodology.

4.1 Parameters of Interest

For each of the outcome categories—beehives in honey production, crop production, and farmland prices—this study aims to recover estimates of three key parameters, each capturing a distinct dimension of the causal effect of the Asian hornet invasion (see Appendix 6 for a formal framework).

I consider the following model:

$$y_{it} = \alpha_i + \delta_t + \sum_{\ell \neq -1} \beta_{\ell} \cdot D_{it}^{\ell} + \mathbf{X}_{it}' \gamma + \varepsilon_{it},$$

²⁸Specifically, it harmonizes and extends the DMSP-OLS (1992–2013) and NPP-VIIRS (2012–2023) products.

where i indexes the geographical unit,²⁹ t denotes the year, and $D_{it}^\ell = 1\{t - g_i = \ell\}$ is an indicator for relative time ℓ , with $\ell = -1$ omitted as the reference period. The model includes unit and year fixed effects, as well as a vector of time-varying controls $\mathbf{X}_{it}$ which I discuss in the next section.

The primary parameter of interest is the event-study coefficient, β_ℓ , which reflects the average treatment effect at each relative time period ℓ since invasion. This allows for an examination of the dynamic response and a test of consistency with the hypothesized causal mechanisms. Second, I consider the average treatment effect on the treated, which summarizes the overall magnitude of the impact across all treated units and post-treatment periods. Third, I examine cohort-specific treatment effects to assess heterogeneity by timing of exposure. These effects help evaluate whether the economic impact varies across invasion cohorts in ways consistent with hornet population dynamics or regional differences in exposure conditions. Together, these parameters provide a comprehensive characterization of the invasion effect, capturing its dynamic evolution, regional and temporal heterogeneity, and average magnitude.

4.2 Identifying Assumptions

To identify the causal parameters described above, I rely on the difference-in-differences framework, comparing outcome trends between treated and untreated départements. Specifically, identification rests on two key assumptions.

No Anticipation *I assume that units do not adjust their outcomes in response to the treatment prior to the actual occurrence of invasion.*

This assumption is plausible in this context for several reasons. First, the Asian hornet was introduced inadvertently in 2004, and the first recorded detection of successful settlement did not occur until 2006. This provides a clear buffer against anticipation effects in the early years. Moreover, most local efforts to control the spread of the hornet were reactive—implemented after invasion had already occurred in a given area and often proved ineffective. However, this assumption is more tenuous in the case of beehive counts. Beekeepers relatively early on recognized the Asian hornet as a major threat to colony health, leading to the development of preventive measures such as homemade hornet-specific traps. Nevertheless, widespread adoption of such measures appears to have occurred only after 2018, with the first commercial hornet traps entering the market around that time³⁰. While

²⁹Beehive counts are measured at the regional level, crop production at the département level, and farmland prices at the level of *Nouvelles Régions Agricoles*.

³⁰The first widely cited manufacturer of commercial hornet traps in France is *Jabeprod*, founded in 2018. See, for example, [Ouest-France \(2023\)](#).

this suggests some scope for preemptive behavior, the risk of significant bias is limited, as traps are primarily deployed in response to invasion and are unlikely to directly influence the outcomes analyzed in this study. A more significant concern arises due to data limitations. Beehive data are available only at the regional level, while the treatment variable is defined at the département level. I assign treatment at the regional level based on the median year of first invasion across départements. As a result, approximately half of the départements in a given region are treated before the region is coded as treated, mechanically introducing potential pre-treatment effects. That said, the hornet’s impact is unlikely to be economically meaningful in the very early stages of colonization, given the expected logistic growth of the population. For crop outcomes, the no-anticipation assumption is more credible. Although farmers may be forward-looking and respond to anticipated shocks in pollination supply, most of the agronomic literature suggests that even salient input disruptions rarely lead to ex-ante adjustments in production decisions. Farmland prices, as outcomes further down the causal chain, are even less likely to respond prior to observed invasion. Finally, this assumption can be partially assessed by examining pre-treatment event-time coefficients. If they are statistically indistinguishable from zero and show no systematic trend, it would lend support to the validity of the no-anticipation assumption.

Parallel Trends Assumption *I assume that, in the absence of the Asian hornet invasion, treated départements would have followed the same outcome trends as the comparison group.*³¹

Identification in event-study designs hinges on the validity of the parallel trends assumption in post-treatment periods, where counterfactual outcomes are unobserved. This assumption is particularly important in the case of the Asian hornet invasion, which involved gradual geographic spread and delayed population growth. Under such dynamics, treatment effects may evolve gradually, increasing the importance of selecting a credible comparison group for which the parallel trends assumption plausibly holds over an extended post-treatment period.

While recent difference-in-differences estimators allow for flexible comparisons (e.g., using either never-treated or not-yet-treated units) this natural experiment presents particular challenges. By 2023, nearly all départements in metropolitan France had been colonized, rendering the never-treated group both small and likely incomparable to ever-treated units. At the same time, relying on not-yet-treated départements as the control group introduces the risk of compositional bias, especially given the combined spatial correlation in invasion patterns and agricultural production. To mitigate these concerns, I implement two main

³¹For both crop outcomes and farmland prices, the parallel trends assumption is expected to hold conditional on agro-climatic conditions and urbanization rates. For farmland prices specifically, I include lagged values of these variables and regional specific linear trends, as discussed in the following section.

restrictions. First, I truncate the sample at 2015, focusing on the period when meaningful untreated variation remains. Second, I restrict the analysis to crops with comparable production profiles across treated and comparison groups.

Sample Window To ensure credible identification in the presence of near-universal treatment and spatial heterogeneity, I truncate the sample to 2015 to construct a more credible comparison group.

As of 2023, the structure of treatment adoption presents several challenges for identification. First, nearly all départements have been treated, with Corsica as the only untreated unit. Using Corsica as a counterfactual is problematic, as its persistent untreated status likely reflects structural differences that shape outcome trends.³² Second, treatment cohorts are highly unbalanced at the tails of the sample period which undermines the reliability of cohort-specific estimates³³. Third, for imputation-based estimators, the shrinking pool of untreated observations over time leads to changes in control group composition. If treatment timing is correlated with unobserved time-invariant characteristics—such as agro-climatic conditions or regional production systems—estimates of untreated potential outcomes may be biased³⁴.

To address these identification challenges, I truncate the sample at 2015 and use only pre-treatment periods of later-treated départements as the comparison group. This improves internal validity by ensuring that all “never-treated” units in the estimation window are in fact eventually treated, making them more comparable to treated units. The year 2015 is selected because 2016 marks the largest one-year expansion in hornet presence, truncating the sample at that point allows for dynamic effect estimation up to ten years post-treatment for early cohorts, and it excludes the problematic single-unit cohorts of 2020 and 2023. Two single-unit cohorts (2004 and 2005) remain; to mitigate noise, I pool them with the 2006 cohort under the assumption of similar exposure dynamics.

Figure 3 summarizes the resulting treatment structure in the estimation sample. The never-treated group comprises 22% of French départements. Among treated cohorts, the smallest (2008 and 2014) include 4 départements each, while the largest (2006) includes 13 départements.

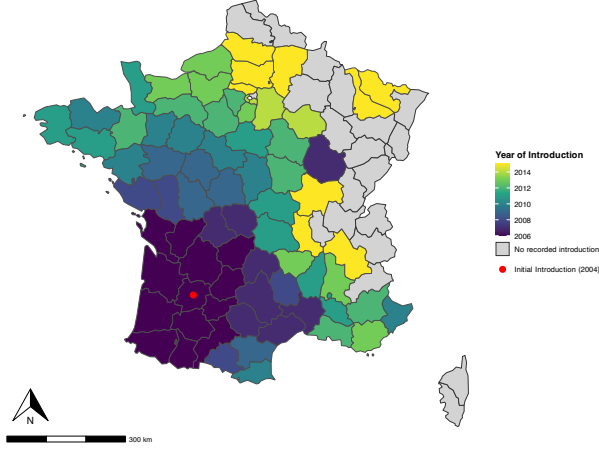
³²From a Roy model perspective, Corsica’s persistent untreated status may reflect selection on structural factors—such as its insular geography—that not only shield it from invasion but also correlate with distinct, time-varying characteristics.

³³The 2004, 2005, 2020, and 2023 cohorts each include only one département, offering no within-cohort variation and thus undermining identification in interaction-weighted estimators, which rely on intra-cohort comparisons for credible inference.

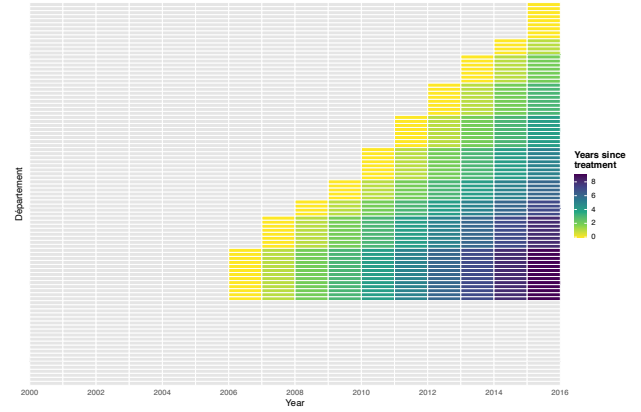
³⁴For example, départements in southern France were both treated earlier and tend to specialize in fruits and oilseeds under warmer climates, whereas northern départements focus more on cereals. If these structural differences are correlated with trends in agricultural outcomes, they may confound treatment estimates.

Figure 3: Spatial and Temporal Treatment Dynamics in the Truncated Analysis Sample

(a) Year of first treatment by département



(b) Treatment exposure duration by département and year



Note: This figure displays treatment dynamics prior to the truncation of the sample in 2015. Panel (a) maps the year of first recorded presence of the Asian hornet across French départements. Grey areas correspond to départements with no recorded invasion by 2015. The red dot indicates the site of initial introduction in 2004. Panel (b) illustrates the staggered exposure structure over time; each row represents a département, and shading reflects years since first invasion. Two single-unit cohorts (2004 and 2005) are pooled with the 2006 cohort to reduce noise, based on the assumption of similar exposure dynamics.

Crop Choices While truncating the sample improves the stability and comparability of potential control groups, the credibility of the parallel trends assumption in the context of crop production further depends on careful crop selection, given the high spatial heterogeneity in agricultural patterns. I restrict the analysis to crops that satisfy two criteria: (i) a sufficiently large average cultivated area – at least 70 hectares for fruits and 560 hectares for field crops³⁵ – in both ever-treated and never-treated départements, to ensure stable estimation; and (ii) limited baseline imbalance in outcome levels, defined as having standardized mean differences below 0.4³⁶. Tables 7, 8, and 9 report pre-invasion comparisons between ever-treated and never-treated units across key variables.

Small-scale crops pose identification risks for several reasons. Département-level aggregates based on a small number of farms are more sensitive to farm-level idiosyncratic shocks, increasing variance and potentially biasing dynamic treatment effects if those shocks correlate with treatment timing. Moreover, the SAA dataset imputes values for underreported crops

³⁵In 2002, the average orchard size in France was approximately 7 hectares, while the average field crop farm covered about 56 hectares in 2010 (AGRESTE, 2003, 2024a). The threshold is thus roughly equivalent to requiring at least 10 farms per département cultivating a given crop, ensuring sufficient representativeness and statistical stability.

³⁶While the identification strategy does not require balance in baseline levels, large differences may suggest differential exposure to unobserved, time-varying confounders, potentially undermining the parallel trends assumption. Event-study estimates for excluded crops are presented in Figures 24, 25 and 26.

using national or historical averages; when coverage is systematically lower in one treatment group, this may induce differential measurement error. Finally, French agriculture is highly spatially clustered, especially for fruits and legumes, increasing the risk that confounding structural differences are mistaken for treatment effects.

Applying these filters yields three candidate crops: rapeseed, table pears, and green beans. Visual inspection of pre-treatment dynamics reveals significant pre-trend violations for pears and beans, leaving rapeseed as the focal pollination-dependent crop.

Rapeseed offers several advantages. First, it is significantly sensitive to insect pollination. While the seminal review by Klein et al. (2007) characterizes rapeseed as only *modestly* dependent – reporting yield increases of 10 to 40% in the presence of insect pollinators – more recent evidence suggests that this substantially underestimates its reliance. For example, experimental studies in Canada find yield increases of 46% when three honey bee hives per hectare are introduced relative to no hive presence (Sabbahi et al., 2005). In France, estimated pollination-induced yield gains reach 36% for conventional varieties and as much as 90% for hybrids, which now dominate cultivation, based on a review of 30 years of studies on rapeseed pollination dependence (Pierre & Renard, 2010)³⁷. Flowering peaks in May, aligning with the Asian hornet’s seasonal foraging activity, and its primary pollinators—honey bees, bumble bees, and solitary bees—are all targeted by the hornet. Second, rapeseed is economically significant. France was the fifth-largest global producer in 2023, with over 1.3 million hectares under cultivation (Ritchie et al., 2023). Its demand continues to grow due to its role in food, animal feeding, biofuels, and even bioplastics. Third, rapeseed is relatively homogeneously cultivated across the country and well-covered in agricultural surveys. Less than 17% of départements fall below the area threshold, and data quality is high due to its status as a major field crop. Finally, rapeseed can be compared to two agronomically similar but non-pollinated crops, barley and soft wheat, which serve as effective placebo outcomes. All three are winter-sown, widely grown, and require comparable inputs, climate, and growing conditions. Figure 9 illustrates this similarity by showing strong correlations between the average area cultivated in the pre-invasion periods for rapeseed and those for barley and soft wheat. This comparability strengthens the empirical strategy by enabling placebo tests: in a partial equilibrium setting, the Asian hornet should not affect non-pollinated crops such as barley and soft wheat. Even under general equilibrium adjustments, such as land reallocation or crop switching, one would expect treatment effects to differ in magnitude and timing across these crops compared to rapeseed.

³⁷Hybrid varieties comprised 80% of certified rapeseed seeds by 2013 and 95% by 2020, according to Terres Inovia, France’s national agronomic institute for oilseed crops.

Remaining Threats to Identification In this context, violations of the parallel trends assumption would arise if time-varying confounders that are correlated with both the Asian hornet invasion and the outcomes evolve differently across treated and untreated units.

A primary concern is agro-climatic variation, especially under climate change. If, for instance, the frequency or severity of heatwaves increased more rapidly in southwestern France, the region where the invasion originated, relative to other areas, observed outcome changes might be spuriously attributed to the hornet. In fact, regional climate projections and historical trends indicate that temperature and precipitation patterns have evolved heterogeneously across France (Ouzeau et al., 2014). To mitigate this concern, I control for agro-climatic conditions in the crop production analysis using Growing Degree Days (GDD) and Hot Degree Days (HDD), helping to isolate the invasion’s effect from confounding climatic variation.

A second threat comes from urbanization, which has been identified as a second-order determinant of Asian hornet spread. Urbanization may also influence agricultural outcomes through land conversion, increased land prices, labor reallocation, environmental pressures, and landscape fragmentation. However, bias from this channel is likely limited in this setting for two reasons. First, while urbanization may influence hornet colonization at a micro-geographic scale, département-level aggregation likely dampens such variation. Second, urbanization trends in France have been relatively stable over the study period, reducing the risk of differential trends driven from this factor that would violate the identifying assumptions. To account for any residual influence, I control for urbanization using night-time light intensity as a proxy.

A third possible concern involves local demand shocks. However, the crop analysis focuses on field crops, which are typically sold in well-integrated national or international markets due to their versatility, storability, and ease of transport. This limits the scope for local demand variation to bias the results.

Finally, differential policy responses to the invasion could confound estimates if, for instance, early-treated départements also implemented more effective control efforts. While this could bias estimated effects upward, two considerations reduce concern. First, policy responses were largely ineffective, and their limited success is well-documented. Second, even if policies had some mitigating effect, estimating the net impact of the invasion remains informative for assessing the overall economic consequences.

4.3 Estimators

Homogeneity of Treatment Effects & Issues with OLS Two-Way Fixed Effects In earlier approaches, event-study parameters have been estimated using a two-way fixed effects

OLS regression. Recent econometric literature has shown that these OLS estimators may yield biased estimates in the presence of treatment effect heterogeneity (de Chaisemartin & D’Haultfoeulle, 2020; Goodman-Bacon, 2021). In static models, the TWFE coefficient may be a non-convex average of underlying effects, potentially with negative weights that can lead to sign reversals. In dynamic models, the coefficients on leads may be contaminated by post-treatment effects, even when the parallel trends assumption holds, thus invalidating standard placebo tests.

In the context of the Asian hornet invasion, assuming homogeneous treatment effects is especially implausible. The binary treatment indicator used—based on confirmed hornet presence—masks variation in invasion intensity across départements, which likely depends on local ecological conditions such as prey density, climate, and landscape features. Further heterogeneity arises from crop varietal differences in pollination dependence, unobserved in the data but likely spatially correlated. Land value responses may also differ with local agricultural potential and expected profitability, which mediate how biological shocks are capitalized. In addition, control measures were implemented with varying intensity across jurisdictions, affecting realized exposure. Crop composition and scale effects introduce further variation, especially if the relationship between pollinator loss and output is nonlinear.

Given these multiple sources of heterogeneity, TWFE is not a credible estimator for the causal parameters of interest. While TWFE results are reported in the Appendix for comparison, the main analysis relies on estimators that are robust to treatment effect heterogeneity.

Heterogeneity-Robust Estimators Recent advances in Difference-in-Differences (DiD) methods address biases that arise in staggered adoption settings with treatment effect heterogeneity (see Roth et al., 2023 for a comprehensive review). These estimators differ in their choice of comparison groups (never-treated vs. not-yet-treated), estimation strategies (imputation vs. aggregation), and treatment of covariates, each implying distinct parallel trends assumptions and exhibiting specific statistical properties. This study implements three leading approaches: the imputation-based estimator of Borusyak et al. (2024), the interaction-weighted of Sun and Abraham (2021) and cohort-time average treatment effect estimator of Callaway and Sant’Anna (2021).

The estimator of Borusyak et al. (2024) imputes counterfactual outcomes for treated units by estimating a two-way fixed effects model by OLS using not-yet-treated observations. It flexibly accommodates time-varying covariates, allowing for conditional parallel trends, but relies on the consistency of the outcome model. In contrast, Callaway and Sant’Anna (2021) and Sun and Abraham (2021) construct estimators by aggregating cohort-specific two-by-two DiD comparisons. The former allows for comparisons using not-yet-treated units, while the latter is restricted to using never-treated units but typically exhibits greater efficiency

due to its interaction-weighted design.

Both Callaway and Sant’Anna (2021) and Sun and Abraham (2021) rely on doubly robust estimation that adjusts for observed covariates under a conditional parallel trends assumption. However, these estimators require sufficient within-cohort variation and may suffer from multicollinearity or finite-sample instability when treatment timing is sparse or unbalanced.

Accordingly, while I report estimates from all three approaches, the baseline specification is selected to balance the empirical features of each outcome with the statistical properties of the estimators. For each outcome class, I present the preferred specification and motivate its choice. Further details regarding the estimators are reported in Appendix 6, which also discusses the relationship between the estimators.

Beehives in Honey Production Beehive data are only available at the regional level, yielding a limited sample of 208 observations within the truncated analysis window. Treatment timing is aggregated at the regional level, defined as the median year of first invasion across départements within each region³⁸. Given this structure, the imputation-based estimator of Borusyak et al. (2024) cannot be applied, as it requires sufficient not-yet-treated observations across time. Instead, I implement the interaction-weighted estimator of Sun and Abraham (2021). While this approach is efficient and robust to treatment effect heterogeneity, it does not allow for covariate adjustment, limiting conditional identification. Nonetheless, given the small sample and clear cohort structure, it offers the most credible option in this setting. To improve precision, I bin relative periods into two-year intervals. I weight observations by the baseline number of hives, computed as the average between 2000 and 2003, to recover population-level effects per hive.

Crop Production Crop production data are available at the départemental level, yielding over 1,400 observations per crop. This larger sample, combined with potential endogeneity concerns, motivates the use of the imputation-based estimator of Borusyak et al. (2024). The baseline specification includes agro-climatic controls: crop-weighted GDD, HDD, a quadratic in precipitation, and night-time lights as a proxy for urbanization. I weight observations by the baseline area cultivated, computed as the average between 2000 and 2003, to recover population-level effects per hectare.

Farmland Prices Farmland price data are available at the level of *Nouvelles Régions Agricoles*, which are subdivisions of départements. This yields a dataset of over 6,500 observations. The relatively large sample size supports the use of the imputation-based estimator

³⁸This results in seven treatment cohorts, each composed of a single region, and two never-treated regions (Grand Est and Corsica).

proposed by Borusyak et al. (2024). In the baseline specification, I include region-specific linear trends and control for contemporaneous and lagged weather conditions, measured by unweighted HDD, GDD, and a quadratic in precipitation, all up to three years prior. Lagged weather is included for two reasons. First, farmland prices are reported as three-year moving averages and thus mechanically reflect weather conditions from previous years. Second, past weather realizations affect both actual agricultural performance and farmers’ expectations about future profitability, thereby influencing land values (Puy & Shrader, 2024). I also control for the current level of urbanization, proxied by night-time light intensity.

Across all specifications, standard errors are clustered at the level of treatment assignment to account for within-group serial correlation. For specifications with weights and covariates, I also report unweighted and unadjusted results to assess robustness.

5 Results

This section presents the main empirical results across the three categories of outcomes examined in the study. First, I examine the impact of the Asian hornet invasion on regional beehive counts, which serves as direct evidence of the hornet’s effect on pollination capacity. I estimate a significant average treatment effect on the treated (ATT) of approximately 10.8% decline in beehive counts. Second, I analyze effects on agricultural production, distinguishing between pollinator-dependent (rapeseed) and independent (barley and wheat) crops. Results show significant negative effects on both the extensive and intensive production margins for rapeseed, with no comparable impacts for non-dependent crops. Finally, I estimate the effect on farmland prices, finding a negative post-invasion ATT of around 4.5%, consistent with reduced land productivity due to diminished pollination services.

5.1 Beehive Counts in Honey Production

Baseline Results Figure 4a presents the event study estimates. Pre-treatment coefficients fluctuate around zero with no significant trends, supporting the parallel trends assumption. Following the invasion, the estimated treatment effect becomes increasingly negative, consistent with the progressive biological dynamics of the species. The effect reaches a statistically significant decline of 3.6% two years post-invasion and appears to stabilize in later periods. After nine years of exposure, the cumulative ATT reaches -23.86%³⁹. The average effect across all treated units is a significant 11.4% decline (see Table 1).

A potential concern is that earlier-treated cohorts also experienced longer exposure to the

³⁹Percentage effects are calculated as $(\exp(\hat{\beta}) - 1) \times 100$, where $\hat{\beta}$ is the estimated coefficient from a semi-logarithmic model.

Asian hornet, which could drive the estimates due to their larger weight in the sample and the shifting composition of the event-study estimates. If the effects are truly due to the hornet, one would expect consistently negative impacts across all cohorts. To assess this, I estimate average treatment effects by cohort (see Figure 5b). Results are negative across all groups, supporting the interpretation that the decline is attributable to the Asian hornet. The larger effects observed in earlier-treated regions are consistent with greater exposure intensity, likely due to two factors: first, longer exposure allows for population buildup under logistic growth; second, the climate of southwestern France, with warmer temperatures and milder winters, is more conducive to hornet establishment and proliferation.

Robustness Checks I conduct five robustness checks, summarized in Figure 18. First, I estimate an unweighted specification in which all regions receive equal weight to test whether the results are driven by regions with higher baseline hive counts. While pre-treatment coefficients show a mild downward trend, particularly up to six years prior to invasion, they remain statistically indistinguishable from zero. In contrast, the post-treatment estimates show a stronger and significant decline, in line with the weighted specification. Second, to address concerns regarding the small size of the never-treated group (limited to Corsica and Grand Est), I implement the estimator of Callaway and Sant’Anna (2021), which uses not-yet-treated units as controls. Although this approach is sensitive to the changing composition of the comparison group over time, it enables the use of a larger control set in the short run. The results are consistent with the baseline estimates. Third, I redefine treatment timing based on the latest year of invasion across composing départements. This alternative definition attenuates the estimated effects and increases uncertainty. The likely reason is that the wide dispersion in invasion dates leads to contamination of the control group, which now includes many units that are already treated. This pre-treatment contamination biases estimates toward zero. Fourth, I assess whether binning post-treatment periods masks underlying heterogeneity by applying the unbinned Sun and Abraham (2021) estimator. The resulting estimated dynamic effects remains consistent with the baseline specification. Finally, I estimate the Borusyak et al. (2024) specification. However, the small number of untreated observations makes the consistency of the imputed counterfactual less credible, and results should be interpreted with caution.

Discussions and Limitations Some limitations remain. The inability to incorporate covariates in the Sun and Abraham (2021) specification limits the ability to condition on relevant confounders⁴⁰. In particular, weather shocks may both influence the invasion dy-

⁴⁰While the `fixest` package allows the inclusion of covariates using the `sunab()` function, discussions with the package author, Laurent Bergé, indicate that covariates may not be handled as intended in the original design of Sun and Abraham (2021). I report specifications with controls in Figure 18a using freezing (FDD), growing (GDD), and hot (HDD) degree days.

namics and directly affect bee colony health. Colonies are especially sensitive to extreme cold, and climatic conditions influence the phenology of pollinated crops, potentially creating mismatches with pollinator cycles, especially under climate change.

Another limitation is the use of regional-level data, which creates imperfect alignment between treatment assignment and observed outcomes. This may lead to partial treatment exposure even before assignment and limits the size and representativeness of the control group.

Despite these limitations, results consistently indicate a post-invasion decline in beehive counts that cannot be explained by time-invariant factors or common year effects. The magnitude of the estimated impact aligns with anecdotal evidence on colony losses attributed to the Asian hornet. Relative to the baseline average number of hives in honey production in France, an average treatment effect of -11.4% implies a loss of approximately 103,500 colonies due to the Asian hornet through 2016.

Table 1: Average Treatment Effects of the Asian Hornet Invasion on Log Honey-Producing Beehive Count

	Specification					
	<i>Baseline</i>	<i>Unweighted</i>	<i>No Year Binning</i>	<i>Callaway & Sant'Anna (2021)</i>	<i>Max. Invasion Year</i>	<i>Borusyak et al. (2021)</i>
<i>ATT Estimate (SE)</i>	-0.1081*** (0.0308)	-0.1080** (0.0287)	-0.1275*** (0.0478)	-0.0993* (0.0590)	-0.0346 (0.0201)	-0.0220 (0.0416)
<i>Estimator</i>	SA	SA	SA	SC	SA	BJS
<i>Comparison Group</i>	Never Treated	Never Treated	Never Treated	Not-Yet-Treated	Never Treated	Not-Yet-Treated

Note: This table reports average treatment effects on the treated for the logarithm of the number of honey-production beehives. Event-study estimates corresponding to the baseline specification are shown in Figure 4a; results from alternative specifications are presented in Figure 18. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Crop Production

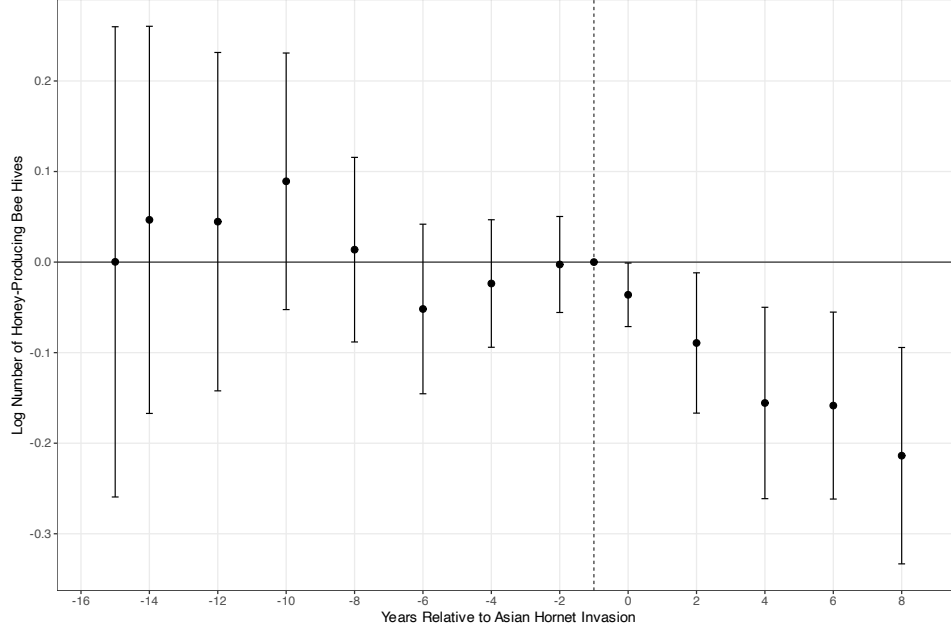
Baseline Results Event-study estimates are reported in Figure 6, with corresponding average treatment effects in Table 2.

I estimate a statistically significant decline in rapeseed production of approximately 13.3%, significant at the 10% level, while no significant effects are found for barley or wheat. Event-study dynamics show no evidence of pre-treatment trends for non-pollinated crops. For rapeseed, pre-treatment coefficients are statistically indistinguishable from zero but display a mild downward trend in point estimates, which continues through the third post-treatment year. Afterward, production losses accumulate to nearly 30% by year four and then stabilize.

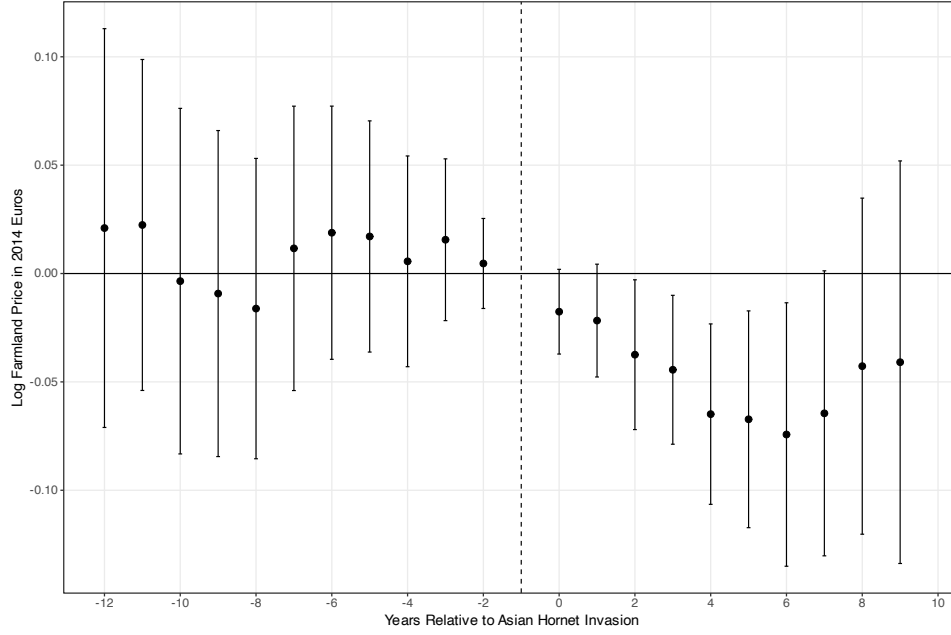
On the intensive margin, I estimate a statistically significant decline in rapeseed yield of

Figure 4: Event-Study Estimates of the Effect of Asian Hornet Invasion

(a) Number of Honey-Production Beehives

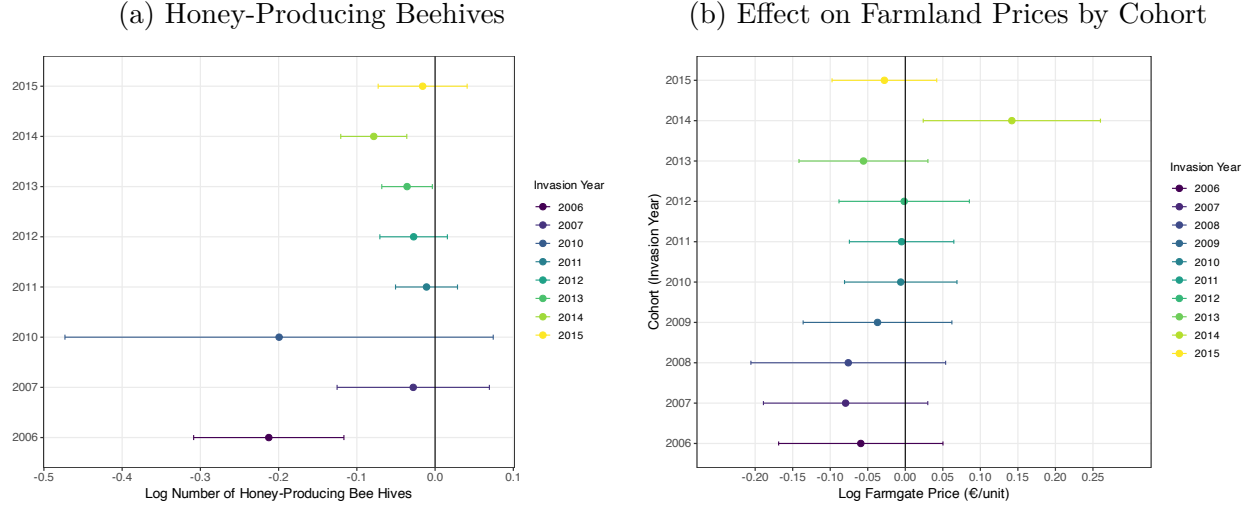


(b) Farmland Prices



Note: Panel (a) presents event-study estimates from a Sun and Abraham (2021) specification using never-treated units as the comparison group. The dependent variable is the log number of honey-producing beehives at the regional level. Treatment is assigned using the median year of first recorded invasion across départements within each region. Event time is grouped into two-year bins, with the year prior to invasion as the reference. Observations are weighted by the region's baseline average number of hives; standard errors are clustered at the regional level. Panel (b) reports event-study estimates of the effect of the Asian hornet invasion on log farmland prices, based on the baseline specification using the Borusyak et al. (2024) estimator. Controls include lagged and contemporaneous agro-climatic conditions (GDD, HDD, and a quadratic in precipitation up to three years), a proxy for urbanization based on night-time light intensity, and region-specific linear trends. Standard errors are clustered at the département level. The vertical dashed line marks the year before the first recorded invasion.

Figure 5: Cohort-Specific Average Treatment Effects of Asian Hornet Invasion



Note: Estimated average treatment effects on the treated by invasion cohort, defined as the year of first recorded Asian hornet presence. Panel (a) reports estimates for honey-producing beehive counts using the baseline specification and the estimator of Sun and Abraham (2021), with standard errors clustered at the region level. Panel (b) presents estimates for farmland prices based on the baseline specification and the imputation-based estimator of Borusyak et al. (2024). Standard errors are constructed using a nonparametric cluster bootstrap with 1,000 resamples, drawing départements with replacement. Ninety-five percent confidence intervals are based on the normal approximation of the bootstrap distribution. See Appendix 6 for further details on inference procedures.

5.2%, while barley and wheat yields show no significant change. Pre-treatment estimates are near zero for all crops. The rapeseed yield trajectory declines steadily through year four, reaching approximately -10%, before flattening.

For the extensive margin, no significant pre-treatment trends are detected in area cultivated across crops, supporting the parallel trends assumption. However, rapeseed displays a gradual decline in area beginning several years before treatment, with point estimates continuing to fall up to year three post-invasion. This trend is followed by a more marked decline, reaching a cumulative reduction of around 20%. In contrast, barley remains stable throughout the entire event window, with estimates centered around zero. Wheat shows no significant changes until year six, after which point estimates indicate a modest significant increase.

To explore heterogeneity in treatment timing, I estimate cohort-specific average treatment effects for production, yield, and area cultivated using the Borusyak et al. (2024) estimator with nonparametric cluster bootstrap standard errors (Figure 7).⁴¹

For rapeseed, earlier-treated cohorts that were exposed to longer durations of hornet invasion experience larger and more negative effects on production. Barley and wheat, in contrast, show small and statistically insignificant effects across cohorts. This divergence suggests

⁴¹See Appendix 6 for further details.

that the observed losses in rapeseed are unlikely to be driven by common shocks. That said, estimates remain imprecise, and confidence intervals are wide.

Patterns in area cultivated mirror those in production: larger declines for rapeseed in earlier-treated cohorts and limited variation for barley and wheat. One exception is the 2014 cohort, which exhibits high variance and instability in estimates. These cohort-level results highlight the potential role of unobserved confounders affecting land-use decisions, possibly shared across crops.

Yield estimates reinforce the pattern of heterogeneity. Rapeseed exhibits larger and more precisely estimated losses in earlier-treated cohorts, while yield trajectories for non-pollinated crops remain flat. An exception is soft wheat in the 2006 cohort, which shows a modest decline. Later cohorts exposed for only one or two years display weaker effects, consistent with a cumulative impact mechanism.

Overall, the results from the baseline specification are internally coherent. A 13.3% decline in production, combined with a 7.76% decline in area cultivated, implies a yield decline of approximately 6% based on baseline national estimate, closely matching the directly estimated 5.2% drop in yield.

Taken together, treatment effects are most robust and consistent for yields, while estimates for area cultivated are noisier and may partially reflect common cohort-level trends not attributable to pollination.

Robustness Checks To assess robustness, I implement several alternative model specifications. Figure 19 reports event-study estimates from three variants: (i) excluding covariates, (ii) excluding weights, and (iii) excluding both. I also include a specification with regional linear time trends. The corresponding average treatment effects are presented in Table 2.

Estimates for rapeseed yield are consistently negative and statistically significant across all variants, ranging from a 4.5% to 6.6% decline. Yield effects for barley and wheat remain near zero and statistically insignificant, with the exception of soft wheat in the no-covariates specification, which shows a minor 1.5% decline. In contrast, production and area estimates are more sensitive. All crops display upward pre-trends in the no-covariates and no-weights specifications, potentially reflecting omitted variable bias or a breakdown in the parallel trends assumption. Including regional linear trends yields stable results for production and area but substantially alters yield estimates. Notably, due to spatial clustering in treatment timing, regional trends may absorb a substantial portion of the identifying variation. Even so, we observe a clear break in the rapeseed yield trend after treatment, with point estimates rising for all crops post-invasion.

I also examine sensitivity to estimator choice. Figure 20 presents event-study results using three alternative methods: a two-way fixed effects (TWFE) estimator, the Sun and Abraham (2021) estimator using never-treated units as controls, and the Callaway and Sant’Anna (2021) estimator using not-yet-treated units. The TWFE estimates for rapeseed yield closely align with the imputation-based baseline. The other two estimators yield smaller and statistically insignificant effects for rapeseed yield but maintain the same rank-order: effects for rapeseed are consistently more negative than for barley and wheat.

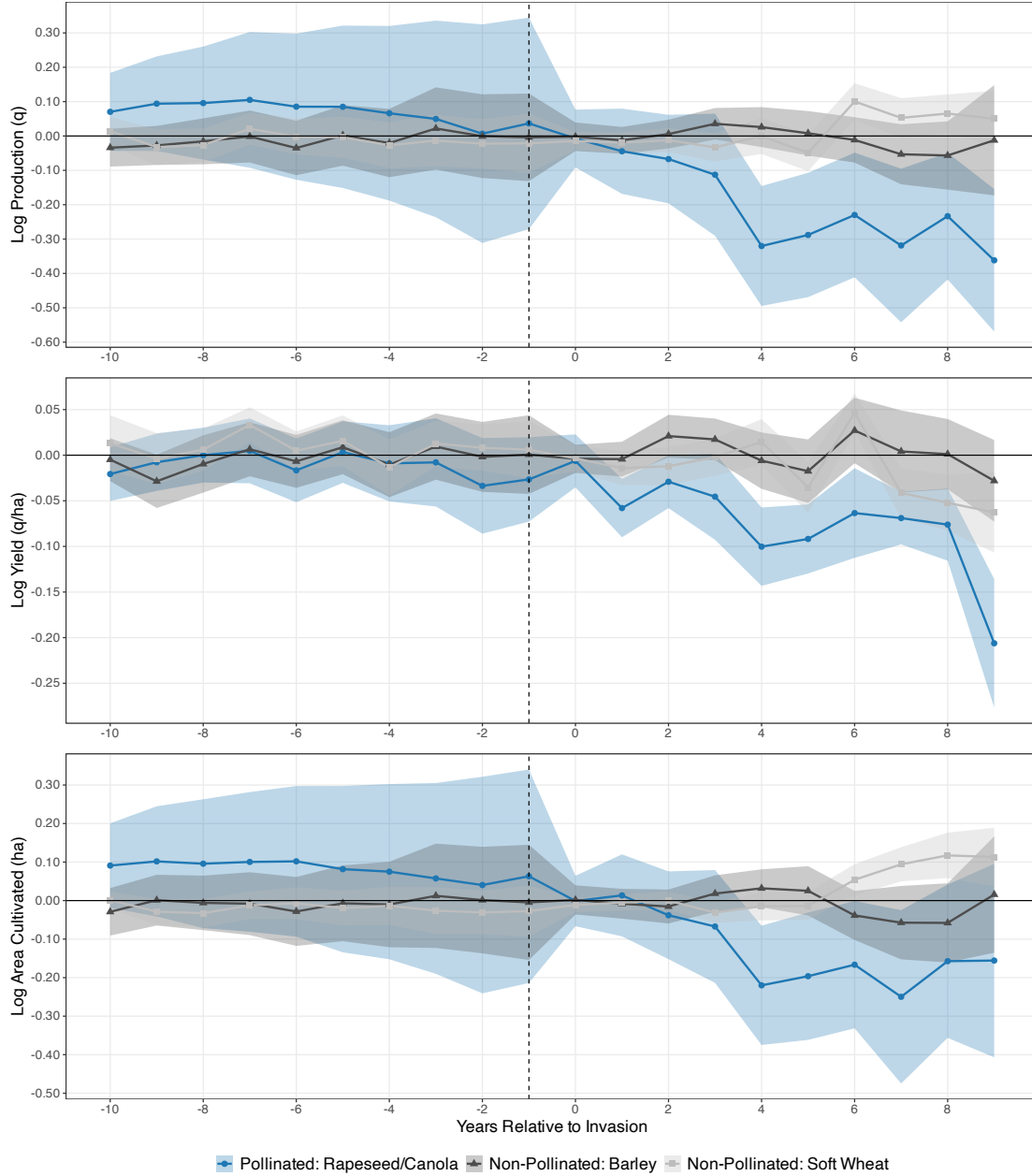
Discussion and Limitations The results suggest a robust negative effect of the Asian hornet invasion on rapeseed yields, observed in both weighted and unweighted specifications. These findings support the interpretation that the hornet has adversely affected rapeseed productivity across France. Moreover, point estimates tend to be correlated across crops in both pre- and post-treatment periods, comforting the argued exposure to shared underlying factors. However, neither the Sun and Abraham (2021) nor the Callaway and Sant’Anna (2021) estimators detect statistically significant declines in yield. Covariates cannot be included in these specifications due to limited within-cohort variation, which may partially explain the discrepancy.

Estimates for cultivated area are noisier, but the trajectory for rapeseed suggests a gradual decline pre-invasion followed by a sharper drop after four years of treatment, possibly reflecting land reallocation in response to lower productivity. The estimated increase in area dedicated to wheat after six years, coupled with a non-significant rise in wheat production, may indicate crop switching in response to reduced rapeseed profitability.

Estimated effects on total production are quantitatively consistent with findings along both margins but remain noisy. Moreover, differences between weighted and unweighted estimates, along with the presence of mild downward trends in both area and output, raise the possibility of pre-existing declines in large producers more likely to be invaded.

To contextualize the economic magnitude, an estimated 5.2% decline in rapeseed yield corresponds to a reduction of 1.55 quintals per hectare relative to pre-invasion levels. Applied to the national average cultivated area of 1,096,044 hectares, this implies a total loss of 169,447 tonnes attributable to reduced productivity following the Asian hornet invasion. At 2004 prices of €192 per tonne of rapeseed, this amounts to an estimated economic loss of approximately €42 million in 2021 euros. By contrast, the 13.3% decline in production also reflects potential behavioral responses, such as land reallocation away from rapeseed. As such, it does not solely indicate a loss in value but rather a broader adjustment in cropping decisions. Based on this estimate, total rapeseed output would have declined by 433,392 tonnes.

Figure 6: Effect of Asian Hornet Invasion on Crop Production



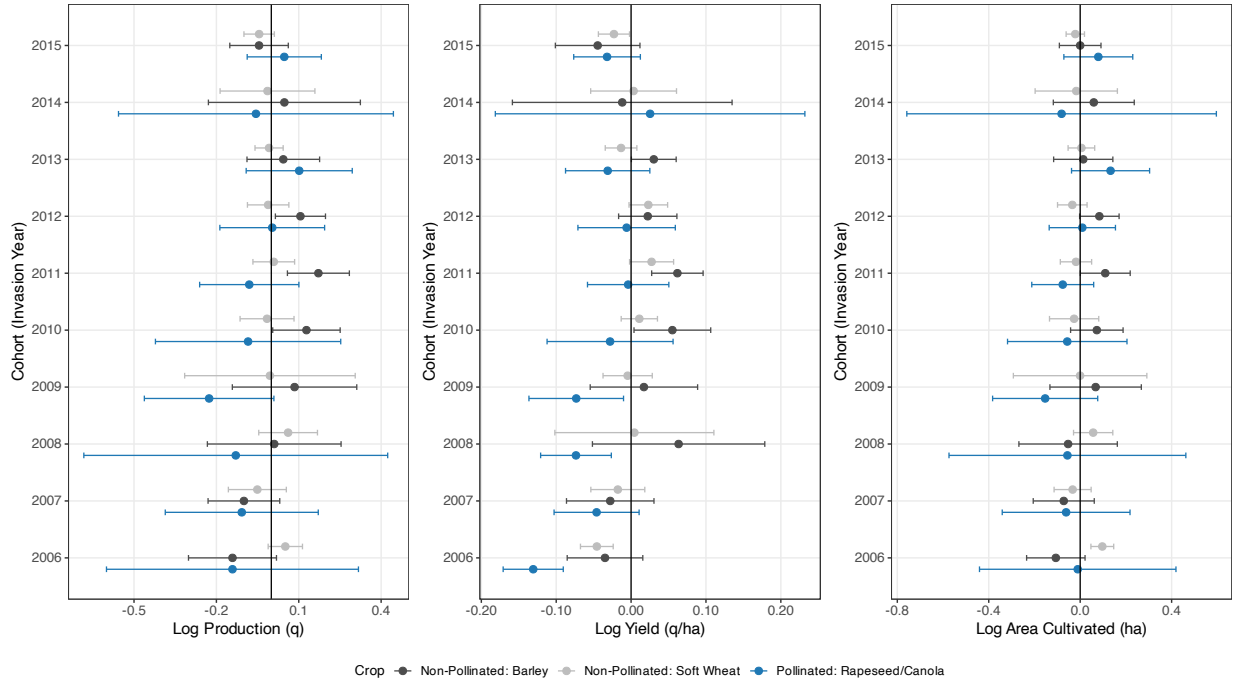
Note: This figure displays event-study estimates of the impact of the Asian hornet invasion on three major crops: rapeseed (blue), barley (dark gray), and soft wheat (light gray). Each panel corresponds to a different logged outcome at the département-year level: total production (top), yield per hectare (middle), and cultivated area (bottom). Estimates are obtained using the imputation-based estimator of Borusyak et al. (2024), with not-yet-treated units serving as the control group. The specification includes controls for crop-weighted Growing Degree Days (GDD), Hot Degree Days (HDD), a quadratic in precipitation, and average night-time light intensity. Observations are weighted by baseline average cultivated area, and standard errors are clustered at the département level. The sample is a balanced panel truncated at 2015. Shaded bands represent 95% confidence intervals. The vertical dashed line marks the year before the first recorded invasion.

Table 2: Average Treatment Effects of the Asian Hornet Invasion on Crop Production

Outcome	Baseline			Without Covariates			Without Weights			Without Covariates and Weights		
	Pollinated		Non-Pollinated	Pollinated		Non-Pollinated	Pollinated		Non-Pollinated	Pollinated		Non-Pollinated
	Rapeseed	Wheat	Barley	Rapeseed	Wheat	Barley	Rapeseed	Wheat	Barley	Rapeseed	Wheat	Barley
Log Production (q)	-0.1250* (0.0671)	-0.0059 (0.0156)	0.0005 (0.0223)	-0.1489** (0.0691)	-0.0114 (0.0126)	0.0114 (0.0241)	0.1914* (0.1021)	0.1249*** (0.0467)	0.1248*** (0.0398)	0.1839* (0.1027)	0.1428*** (0.0458)	0.1344*** (0.0395)
Log Yield (q/ha)	-0.0503*** (0.0146)	-0.0083 (0.0073)	0.0032 (0.0086)	-0.0451*** (0.0134)	-0.0151*** (0.0055)	-0.0052 (0.0089)	-0.0665*** (0.0152)	-0.0037 (0.0103)	0.0099 (0.0134)	-0.0598*** (0.0141)	0.0032 (0.0089)	0.0105 (0.0125)
Log Area Cultivated (ha)	-0.0747 (0.0587)	0.0024 (0.0112)	-0.0027 (0.0232)	-0.1037* (0.0585)	0.0036 (0.0101)	0.0166 (0.0253)	0.2580*** (0.0999)	0.1287*** (0.0440)	0.1149*** (0.0386)	0.2437*** (0.0999)	0.1397*** (0.0430)	0.1240*** (0.0386)
<i>Weights</i>												
Baseline Area Cultivated	X	X	X	X	X	X						
<i>Covariates</i>												
Agro-climatic Conditions & Urbanisation Proxy	X	X	X				X	X	X			

Note: This table reports average treatment effects on the treated by crop and outcome, estimated using the imputation-based estimator of Borusyak et al. (2024). The treatment is defined as the first recorded invasion of the Asian hornet in each département. All outcome variables are expressed in natural logarithms. Standard errors (in parentheses) are clustered at the département level. Baseline specifications include crop-weighted Growing Degree Days (GDD), Hot Degree Days (HDD), a quadratic in precipitation, and average night-time light intensity as covariates. Observations are weighted by average cultivated area in the pre-invasion period. Production is measured in quintals (q), where 1 quintal = 100 kg; area is measured in hectares (ha), where 1 ha = 10,000 m². Event-study estimates for the baseline specification are shown in Figure 6; robustness checks without covariates and without weights are shown in Figure 19. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 7: Average Treatment Effects on Crop Production by Invasion Cohort



Note: Estimated average treatment effects on the treated (ATT) by invasion cohort (year of first recorded Asian hornet presence) for three agricultural outcomes: log production (left), log yield per hectare (center), and log area cultivated (right). Estimates are from the baseline specification using the imputation-based estimator of Borusyak et al. (2024), weighted by average pre-invasion cultivated area. Each point reflects the ATT for a given cohort, with color indicating crop type: rapeseed (blue, pollination-dependent), barley (black), and soft wheat (gray), both non-pollinated crops. Standard errors are constructed via a non-parametric cluster bootstrap with 1,000 resamples, where départements (the unit of treatment variation and clustering) are drawn with replacement. Ninety-five percent confidence intervals are constructed using the normal approximation applied to the distribution of estimates obtained from a nonparametric, cluster-level bootstrap. Appendix 6 provides further details on the standard error estimation procedure.

5.3 Farmland Prices

Baseline Figure 4b presents event study estimates from the baseline specification. The parallel trends assumption in pre-treatment periods appears to hold, as estimated coefficients are statistically insignificant and fluctuate around zero. Post-treatment estimates decline up to six years after the invasion, with significant effects from the second to the sixth year. The trend then reverses and stabilizes around a 4% reduction nine years post-invasion. This accumulation of effects over time is consistent with the logistic population growth dynamics of the Asian hornet. The average treatment effect is estimated at -4.3% . Cohort-specific treatment effects are negative for all groups except the 2014 cohort. Point estimates are larger for earlier-treated cohorts and tend to decrease for more recently treated cohorts. This pattern is broadly consistent with the hypothesized ecological mechanism, whereby earlier exposure allows more time for hornet populations to expand and potentially reach higher carrying capacities.

Robustness Figure 21 reports event study estimates from robustness checks, and Table 3 presents average treatment effects. The first two checks sequentially remove covariates and then both covariates and region-specific linear trends. While post-treatment dynamics remain similar, pre-treatment coefficients become negative and statistically significant from 12 to 9 years before invasion. A similar pattern is observed when both covariates and regional trends are omitted. This may indicate that regional climatic trends or other time-varying factors correlated with Asian hornet invasion are confounding the untreated potential outcome, underscoring the importance of controlling for region-specific dynamics.

Next, I estimate the interaction-weighted estimator of Sun and Abraham (2021), which uses never-treated units and accommodates region-specific trends. The resulting event study estimates display a similar post-treatment dynamic, with an average treatment effect of -4.5% , consistent with the baseline. I then apply the estimator of Callaway and Sant’Anna (2021), which uses not-yet-treated units as controls. Since this estimator does not allow for region-specific trends, it may be more vulnerable to omitted variable bias. Nonetheless, the estimated trajectory remains similar in shape, albeit less precisely estimated and statistically insignificant, with post-treatment effects of similar magnitude to the baseline without covariates or trends.

Discussion and Limitations Overall, the estimates suggest that the Asian hornet invasion negatively affects farmland prices. While the cumulative decline over time aligns with ecological expectations, the relatively rapid onset of the effect is somewhat surprising. A notable feature is the attenuation of the negative impact after six years. This could reflect adaptive responses, such as compensation for losses in pollination-dependent perennial

crops or shifts toward less affected land uses. Alternatively, it may indicate some degree of adaptation to reduced pollination. However, large standard errors limit the precision of these interpretations. The instability of pre-treatment coefficients in some specifications also suggests sensitivity to model assumptions and potential omitted trends. These findings underscore the importance of accounting for regional heterogeneity and time-varying confounders in estimating causal effects in this setting. Relative to the national average farmland price over the 2000–2003 period, an estimated decline of 4.3% corresponds to a decrease of approximately 224 euros per hectare in 2021 terms. Relative to the average agricultural land area in France between 2000 and 2003, approximately 31.9 million hectares, this decline implies a cumulative farmland valuation loss of roughly €7.15 billion in 2021 euros. This represents about 0.36% of France’s average annual GDP over the same period.

Table 3: Average Treatment Effects of the Asian Hornet Invasion on Farmland Prices

	Specification				
	<i>Baseline</i>	<i>Without Covariates</i>	<i>Without Covariates and Region Trends</i>	<i>Sun & Abraham (2021)</i>	<i>Callaway & Sant’Anna (2021)</i>
<i>ATT Estimate (SE)</i>	−0.0429*** (0.0165)	−0.0458** (0.0232)	−0.0133 (0.0486)	−0.0449** (0.0229)	−0.0283 (0.0360)
<i>Region Linear Trends</i>	X	X		X	
<i>Covariates</i>					
Lagged and Contemporaneous Weather & Urbanization	X				
<i>Estimator</i>	BJS	BJS	BJS	SA	SC
<i>Comparison Group</i>	Not-Yet-Treated	Not-Yet-Treated	Not-Yet-Treated	Never-Treated	Not-Yet-Treated

Note: This table reports average treatment effects on the treated for log farmland prices, measured in 2014 constant euros. The baseline specification uses the imputation-based estimator of Borusyak et al. (2024) with not-yet-treated units as comparison group, and includes controls for lagged weather conditions (up to three years) measured by Growing Degree Days (GDD), Hot Degree Days (HDD), and a quadratic in precipitation, as well as a proxy for urbanization based on night-time light intensity. Region-specific linear trends are also included. Standard errors are clustered at the region level. Baseline event study estimates are presented in Figure 4b; robustness checks are shown in Figure 21a. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion and Further Research

This paper provides causal evidence on the economic consequences of pollinator loss by leveraging quasi-exogenous variation from the staggered invasion of the Asian hornet across France.

I estimate that hornet presence led to an 11.4% reduction in the number of beehives used for honey production, consistent with substantial losses reported by the beekeeping industry amid broader pollinator pressures. I also find a 5.2% decline in rapeseed yields, with no comparable decline in non-pollination-dependent crops, implying a national productivity loss of approximately €42 million in 2021 euros. The estimated productivity loss in oil rapeseed

alone exceeds the €23 million in control expenditures reported over the same period (Barbet-Massin et al., 2020). These findings underscore the potentially substantial contribution of insect pollinators as inputs in agricultural production functions. Additionally, I present suggestive evidence of land reallocation and possible crop switching from pollination-dependent crops such as rapeseed to less dependent crops like wheat, highlighting potential implications for crop composition and food policy. Finally, using a hedonic framework, I estimate a 4.3% decline in farmland prices, corresponding to a €224 per hectare reduction in land value. Relative to the land dedicated to agricultural production before the invasion of the asina hornet, this would imply a valuation loss of about 7.15 billion euros in 2021 terms – equivalent to around 0.36% of France’s average annual GDP over the 2000–2003 period. This suggests that the economic effects of pollinator loss extend beyond immediate productivity impacts to broader land use and valuation outcomes and potentially has a tremendous value.

Nonetheless, the analysis is subject to several limitations that future research could address. These estimates reflect effects measured up to 2016, twelve years after the initial introduction of the hornet, while the species has continued to expand in both range and population density. The longer-run economic impacts may therefore be substantially larger. In addition, the absence of fine-grained, farm-level data limits the crop coverage of this analysis, particularly for perennial or fruit crops. Future work could incorporate micro-level production data, such as from the EU Farm Accountancy Data Network (FADN), to extend the analysis across more crop types and regions.

Furthermore, identifying high-quality proxies for pollination supply, such as colony-level data from France’s mandatory *téléruchers* register, could enable instrumental variable strategies to estimate the marginal value of pollinators. Such estimates would offer more reliable inputs for cost-benefit analyses and the design of effective conservation policy. Overall, this study highlights the importance of accounting for biodiversity and ecosystem service losses in economic policy evaluation.

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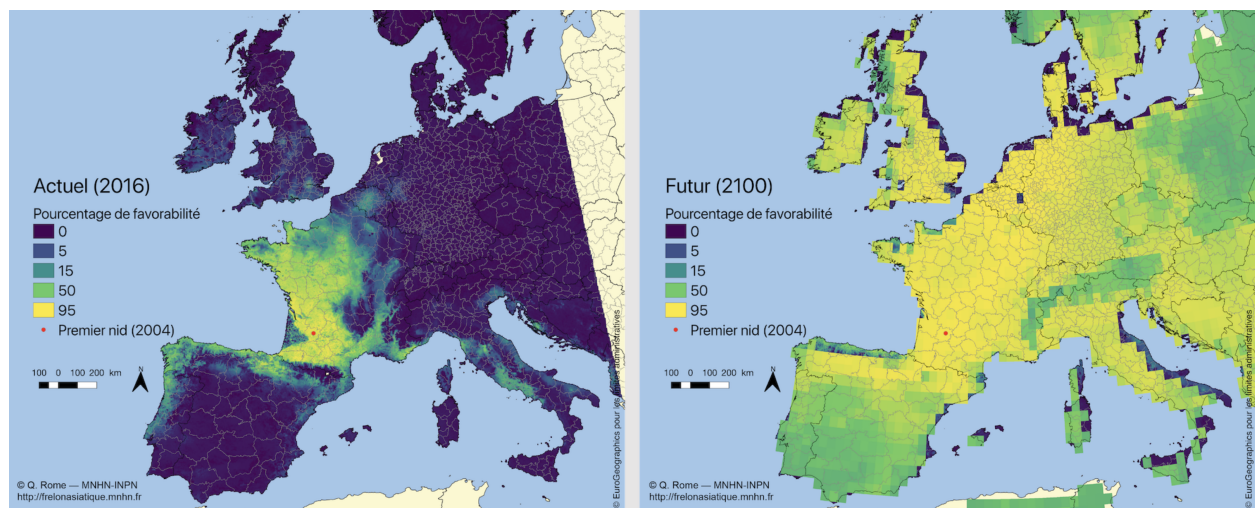
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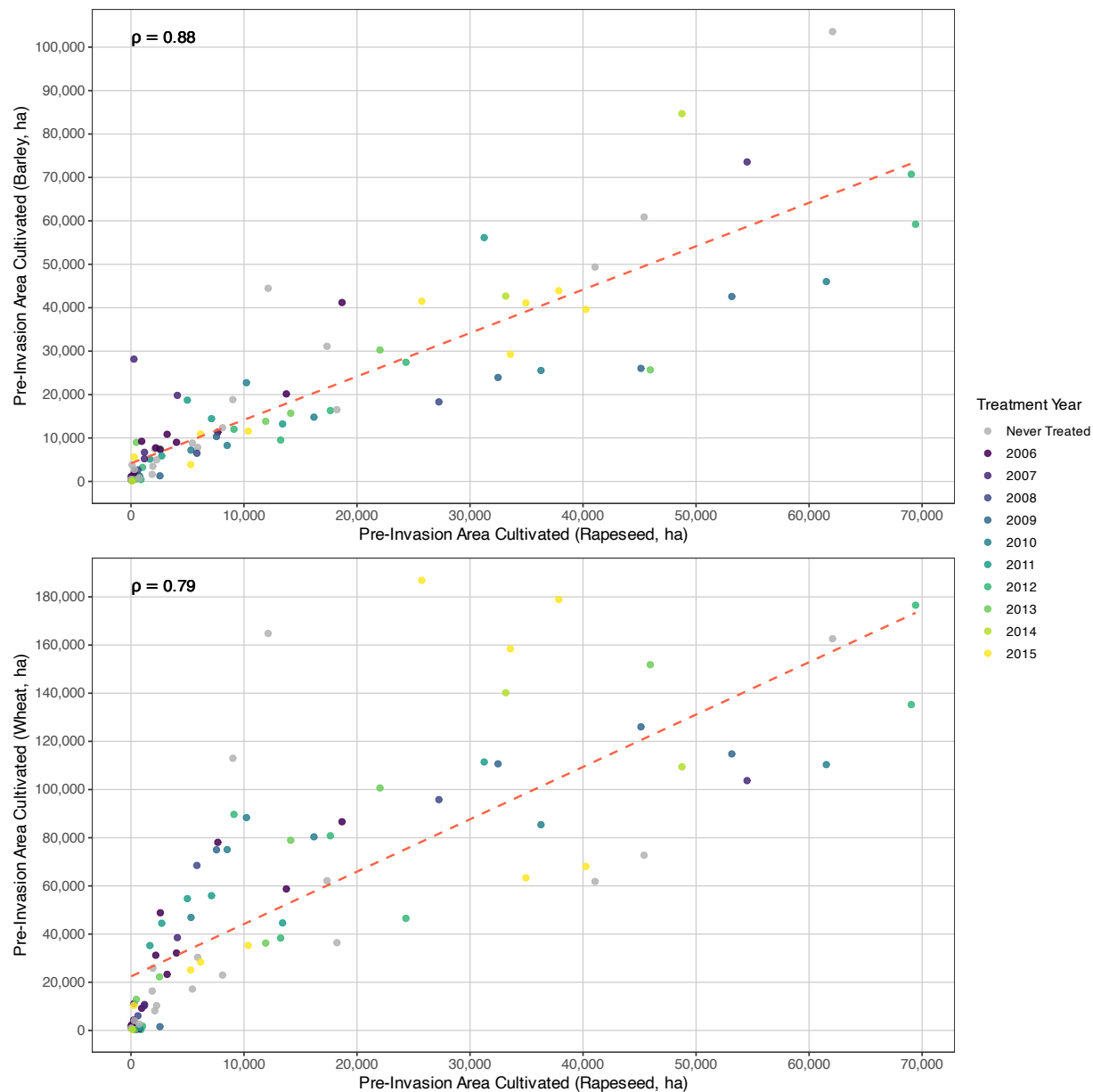
Figures and Tables

Figure 8: Climate Suitability for the Asian Hornet under Current and Future Conditions



Note: Estimated habitat suitability for *Vespa velutina* across Europe based on species distribution (niche) models. The left panel shows favorability as of 2016, using observed climate and land cover data (Fournier et al., 2017). The right panel presents projected suitability for 2100 under climate change scenarios (Barbet-Massin et al., 2013). Color scale indicates the percentage of favorable conditions, from dark blue (least suitable) to yellow (highly suitable). The red dot marks the location of the first recorded nest in France (2004). Image by Quentin Rome, MNHN-INPN, available at <https://frelonasiatique.mnhn.fr/biologie/CartePrediction>.

Figure 9: Correlation Between Baseline Cultivated Area in Rapeseed and Other Non-Pollinated Crops



Note: Each panel shows a scatter plot at the département level. The left panel compares mean baseline area (2000–2003) cultivated in rapeseed (x-axis) and barley (y-axis); the right panel compares rapeseed (x-axis) and wheat (y-axis). Each point corresponds to a département. The color indicates the year of Asian hornet invasion, with grey for départements never invaded. The fitted line is based on a linear regression, and the reported ρ is the Pearson correlation coefficient.

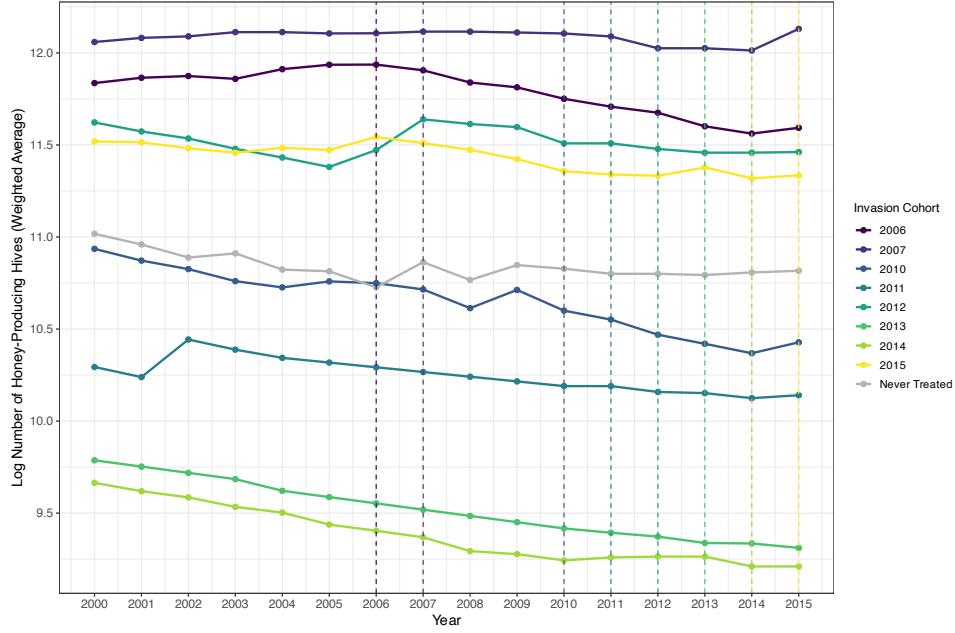
Table 4: Summary Statistics of Crop Production, Area Cultivated, and Yields

Variable	Production (q)		Area (ha)		Yield (q/ha)		N		Pollination
	Mean (SD)	Range	Mean (SD)	Range	Mean (SD)	Range	Obs.	Dépts.	Depdence Index
Cereals, Protein & Oil Seeds									
Total rapeseed	514,142 (658,119)	200–3,587,556	15,615 (19,213)	10–88,100	30.27 (6.15)	8–48	1,392	87	Modest
Total oilseeds	715,294 (772,077)	490–3,621,496	24,191 (25,901)	19–113,078	28.53 (6.37)	10.21–47.94	1,392	87	Modest
Sunflower	201,960 (333,950)	125–2,219,350	8,666 (14,668)	5–88,774	23.94 (4.61)	8–37	1,232	77	Modest
Total soft wheat	3,772,777 (4,156,293)	200–20,065,482	52,180 (51,179)	5–202,718	62.20 (15.50)	18–99.91	1,488	93	No Increase
Total cereals	7,103,948 (5,948,073)	1,740–26,618,089	99,709 (75,320)	46–305,451	65.47 (16.05)	19.93–112.91	1,488	93	No Increase
Total barley	1,166,052 (1,456,113)	539–9,192,496	18,294 (21,093)	14–119,310	56.21 (13.42)	15–94.34	1,472	92	No Increase
Total oats	51,041 (53,110)	300–400,575	1,139 (1,058)	14–7,446	41.75 (11.89)	10–73.46	1,440	90	No Increase
Triticale	194,210 (227,404)	450–1,364,880	3,725 (4,137)	10–20,680	51.92 (11.26)	15–87	1,440	90	No Increase
Spring barley	362,860 (765,408)	200–6,424,635	6,093 (11,496)	5–87,410	47.92 (13.92)	9–80	1,408	88	No Increase
Field peas	120,296 (229,535)	25–1,788,500	2,817 (4,909)	1–36,500	35.34 (10.54)	9–60	1,312	82	No Increase
Rye + mixed grains	16,875 (25,416)	75–208,425	357 (502)	3–3,200	46.51 (11.60)	17–83	1,248	78	No Increase
Non-irrigated grain maize	1,060,590 (1,190,260)	4,950–7,312,000	12,923 (13,543)	81–77,800	78.67 (16.27)	22.06–119.38	1,216	76	No Increase
Winter oats	35,368 (48,446)	164–376,695	743 (911)	4–6,849	44.03 (12.14)	10–75	1,104	69	No Increase
Fruits									
Other apples	110,143 (236,620)	334–1,804,572	288 (537)	2–3,508	316.15 (115.72)	20–994	1,424	89	Great
Golden apples	75,620 (156,245)	66–1,053,750	181 (354)	1–1,877	353.66 (133.49)	19–1,610	1,280	80	Great
Total table pears	23,032 (60,264)	44–605,075	95 (220)	1–2,482	203.80 (83.33)	10.38–515	1,232	77	Great
Autumn pears	8,552 (13,617)	50–94,900	39 (57)	1–370	209.31 (83.20)	35–582	1,088	68	Great
Legumes									
Total strawberries	6,173 (16,133)	29–148,800	40 (88)	1–800	130.59 (56.82)	8–426.71	1,376	86	Modest
Green beans	39,575 (98,622)	56–621,500	341 (825)	1–4,818	100.20 (73.19)	25–976	1,344	84	Little
Zucchini	17,102 (50,061)	110–645,450	37 (75)	1–662	340.63 (147.25)	70–975	1,170	78	Essential
Pumpkins, squash, etc.	22,100 (57,870)	159–758,550	60 (116)	1–952	323.35 (115.85)	81–906	1,125	75	Essential
Turnip	8,000 (15,762)	68–113,820	33 (71)	1–616	216.62 (86.68)	51–682.26	1,125	75	Great

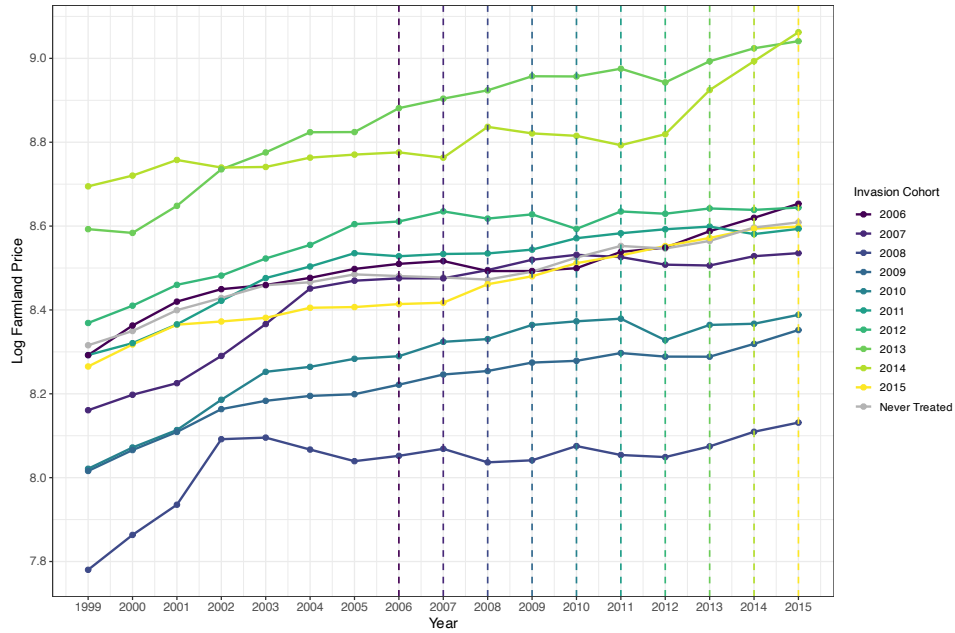
Note: Summary statistics are based on the truncated estimation sample (up to 2015) and are aggregated at the département-year level. For each crop, the table reports means with standard deviations in parentheses and the observed minimum–maximum range. Production is measured in quintals (q), where 1 quintal = 100 kg; cultivated area is measured in hectares (ha), where 1 ha = 10,000 m². N (Obs.) denotes the number of département-year observations; N (Dépts.) indicates the number of unique départements cultivating the crop. The final column provides the pollination dependence index from Klein et al. (2007). The primary crops used in the analysis are *Total soft wheat*, *Total barley*, and *Total rapeseed*.

Figure 10: Trends by Invasion Cohort: Honey-Production Beehives and Farmland Prices

(a) Log Honey-Producing Beehive Count

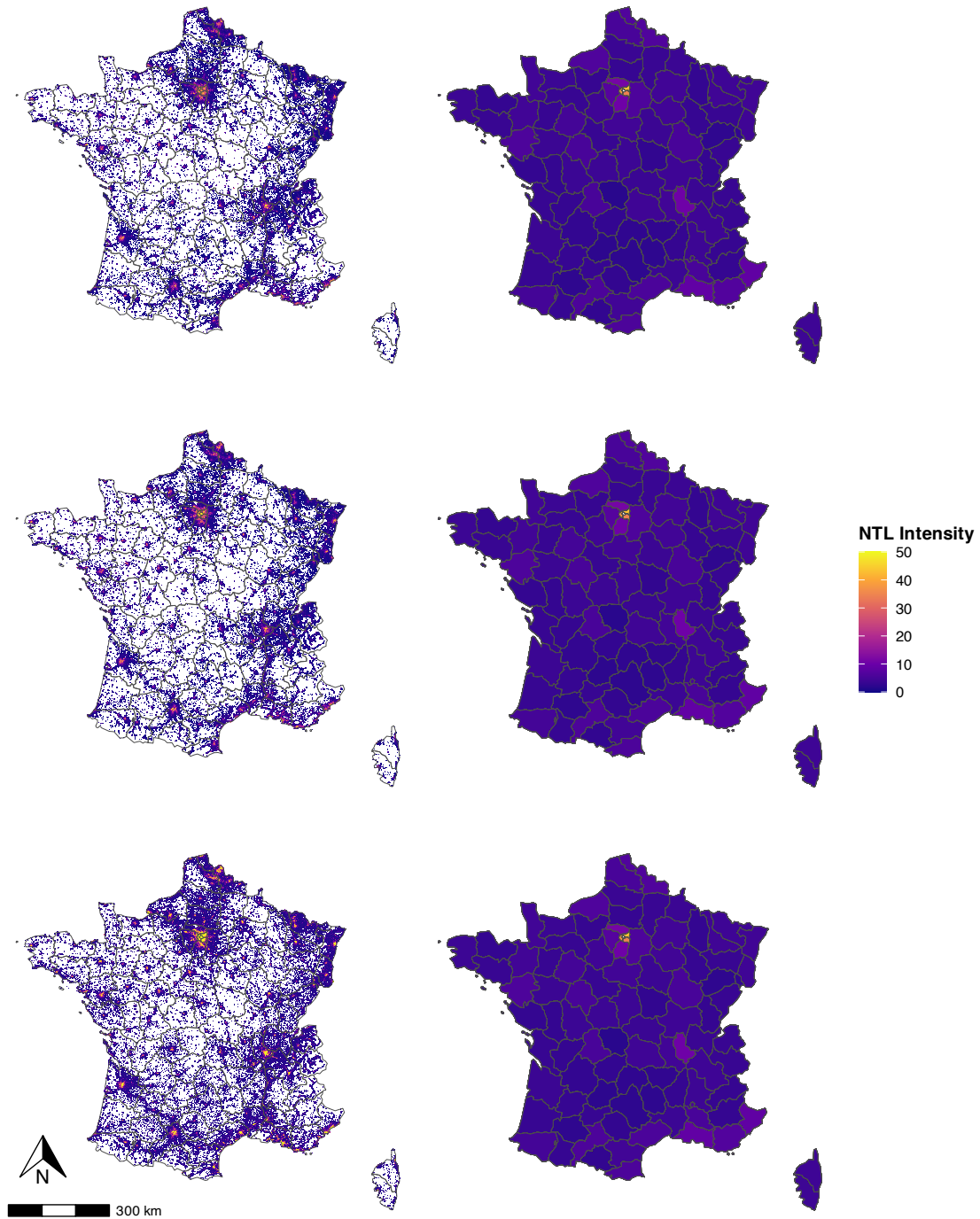


(b) Log Farmland Prices



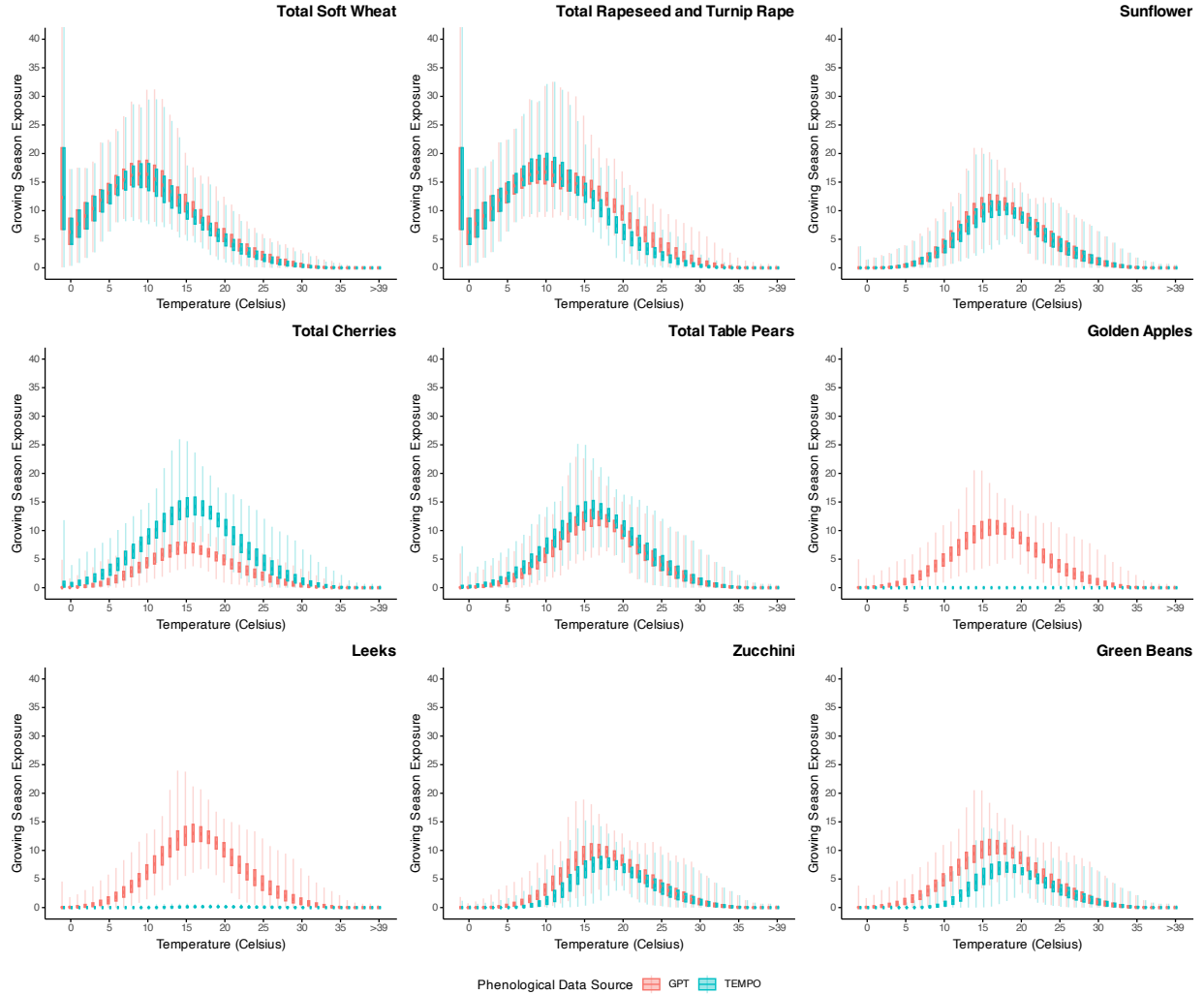
Note: Each line depicts the log average outcome by invasion cohort. Panel (a) shows trends in honey-production beehive counts, where treatment timing is defined by the median year of first invasion among constituent départements. Observations are weighted by baseline beehive counts (2000–2003 average). Panel (b) displays trends in farmland prices, with treatment timing defined at the département level. Observations are unweighted.

Figure 11: Evolution of Night-Time Light Intensity in France: 2000, 2007, and 2015



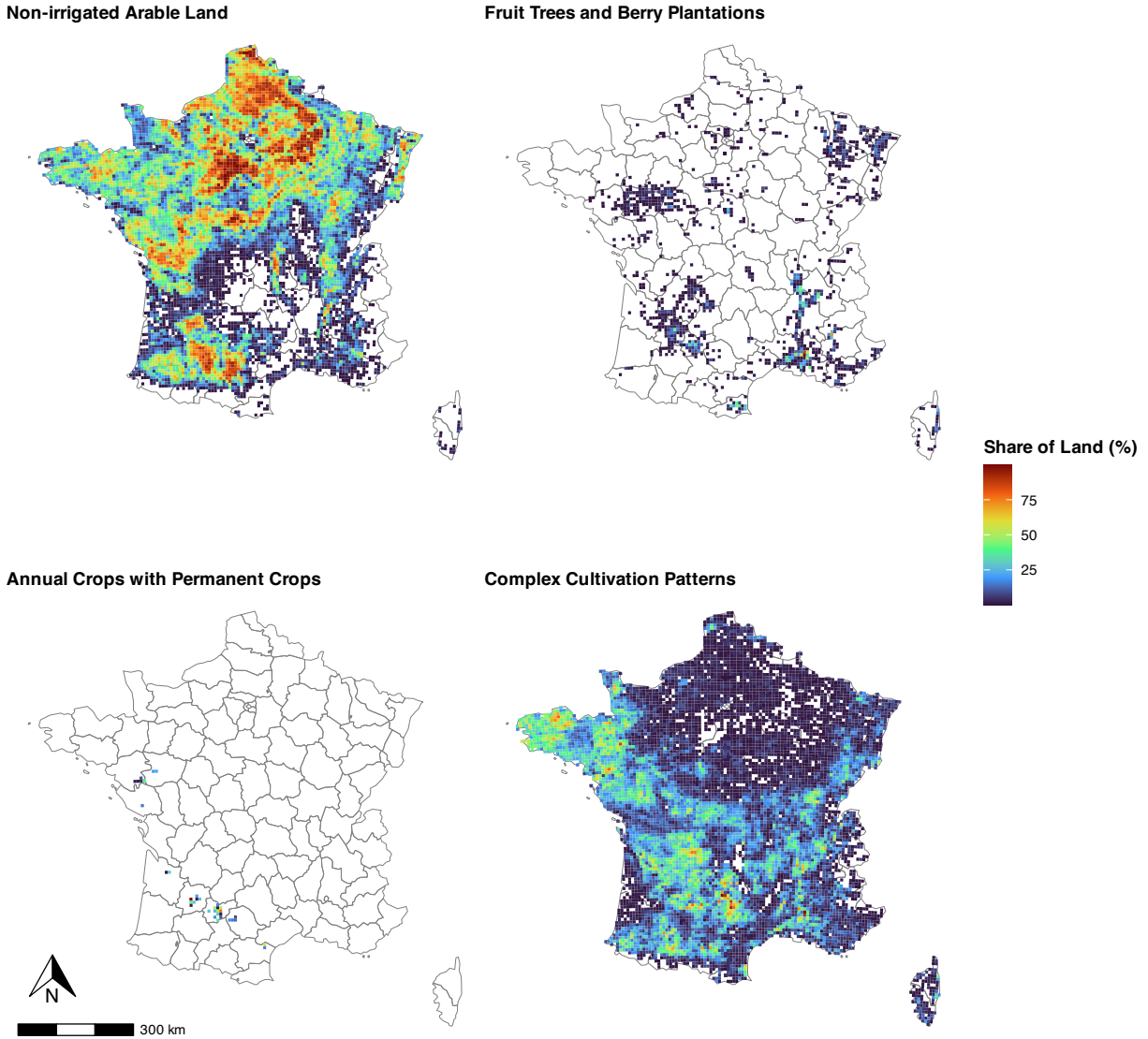
Note: Each row corresponds to a selected year—2000 (top), 2007 (middle), and 2015 (bottom). The left column displays raw night-time light intensity from the Simulated VIIRS Nighttime Light (SVNL) dataset. The right column shows the corresponding département-level averages used as a proxy for urbanization in the empirical analysis. For visualization purposes, night-time light intensity values are top-coded at 50 to account for exceptionally high values in Île-de-France.

Figure 12: Distribution of Crop-Weighted Temperature Exposure During Growing Seasons



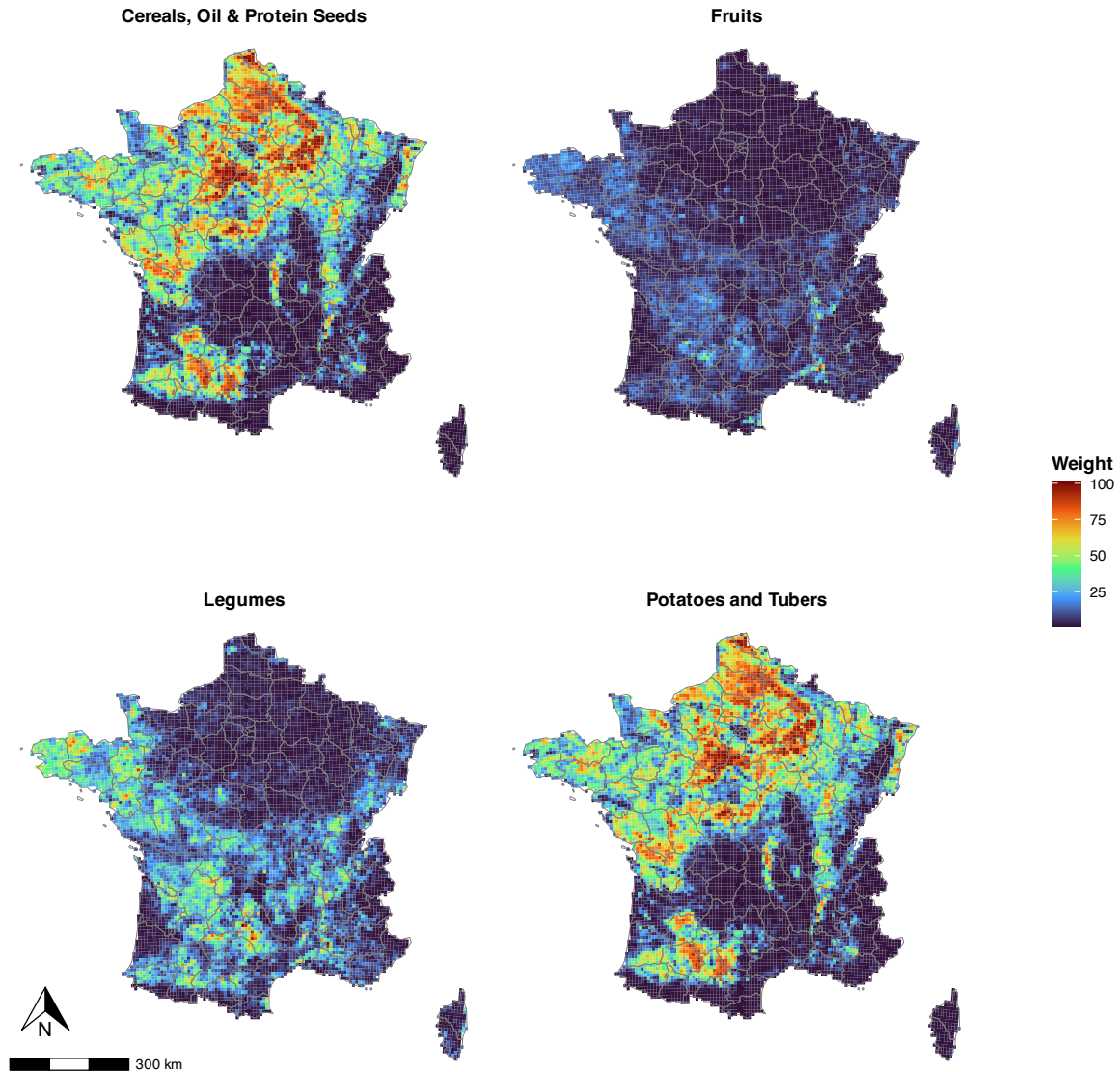
Note: This figure displays the distribution of crop-specific exposure to each 1 degree intervals (in °C) during growing seasons, constructed using either phenological observations from TEMPO (blue) or modeled growing periods from GPT (red). Exposure is calculated as the time crops experienced each temperature bin, weighted spatially by land cover data from the Corine Land Cover dataset. The lowest bin includes all observations below 0°C; the highest includes those above 29°C. For each temperature bin, whiskers show the full range (min–max) across département-year observations, while boxes represent the interquartile range (25th–75th percentiles). No TEMPO match was found for Leeks and Golden Apples. All main crops analyzed in the study are matched to TEMPO. Additional methodological details are provided in Appendix 6.

Figure 13: Spatial Distribution of Land Cover Types in France (Corine 2012)



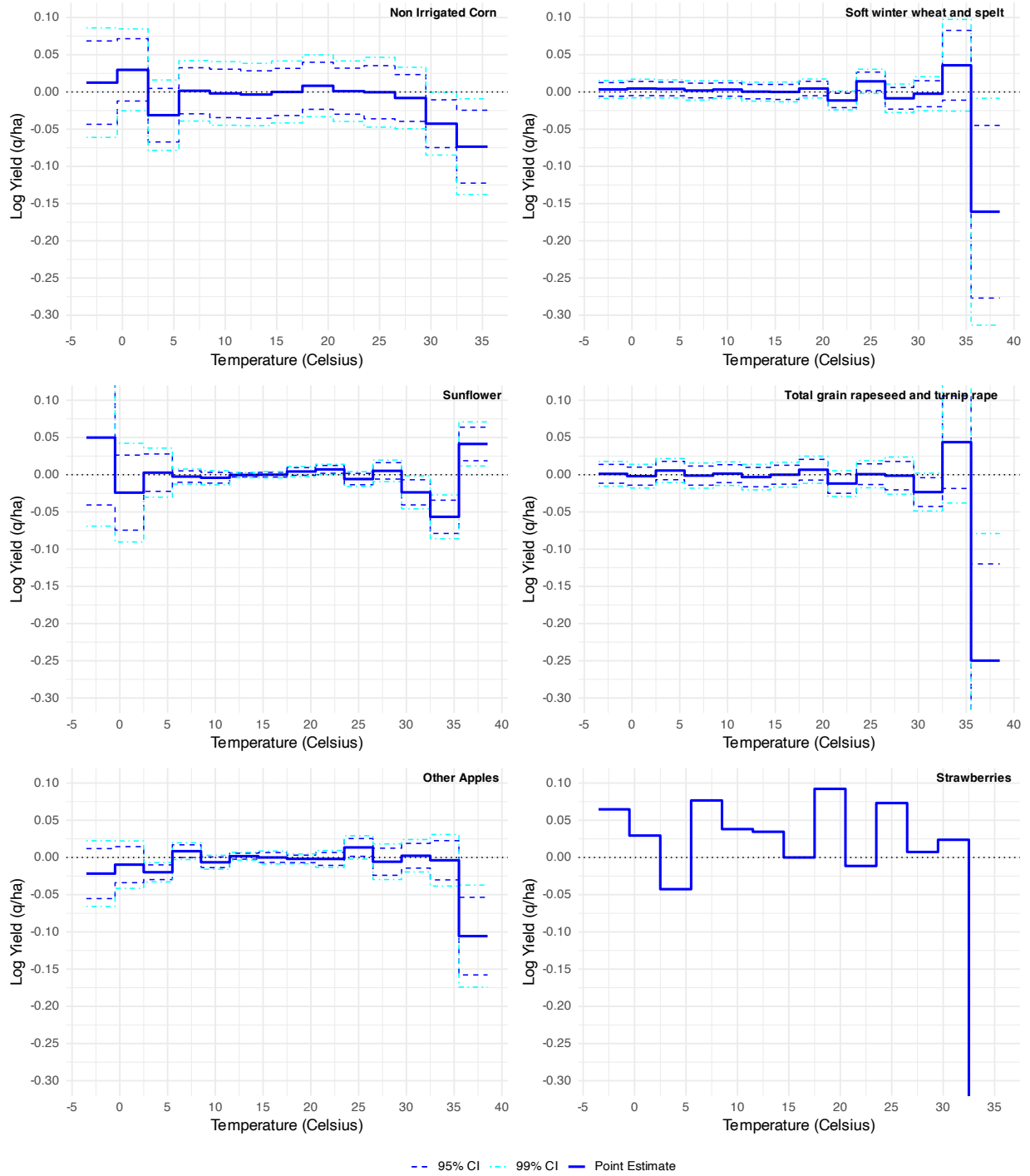
Note: Each panel maps the share of land area in SAFRAN grid cells corresponding to four key land cover categories, based on the 2012 Corine Land Cover dataset produced by Copernicus. These land cover shares are used to construct spatial weights when aggregating agro-climatic exposures at the crop type level. The categories displayed are those most relevant for agricultural land use and pollinator activity. Further methodological details are provided in Appendix 6.

Figure 14: Spatial Weights by Crop Type for Agro-Climatic Exposure



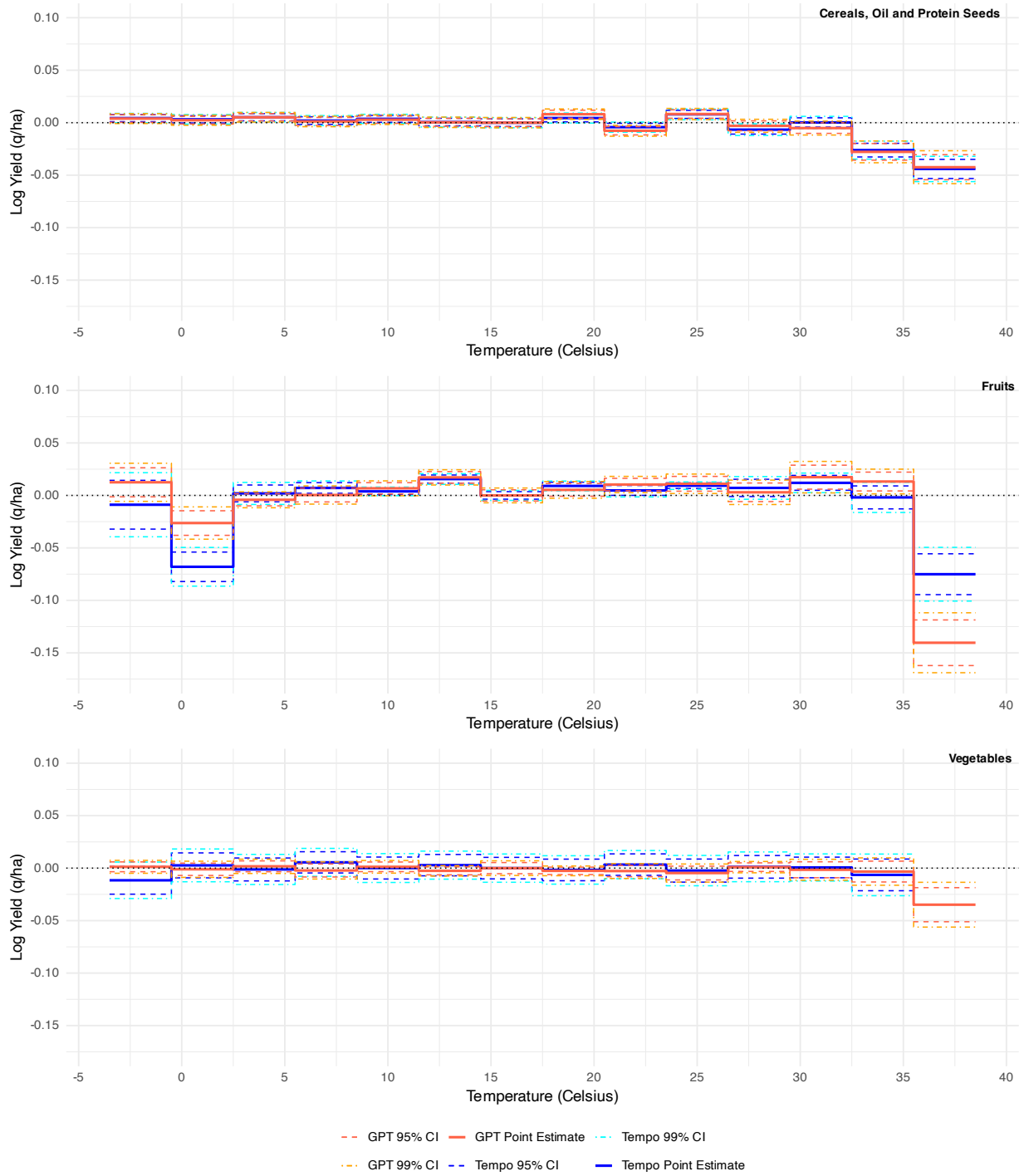
Note: Each panel displays the spatial weights derived from the 2012 Corine Land Cover dataset, reflecting the estimated share of land within each SAFRAN grid cell cultivated across crop types. These weights are used to aggregate agro-climatic exposures at the crop level in the construction of weather controls. Further methodological details are provided in Appendix 6.

Figure 15: Yield–Temperature Response Functions for Selected Crops



Note: Each panel displays estimates from stepwise regressions of crop yields on crop-weighted temperature exposure, aggregated in 3°C intervals using crop-specific spatial weights. All specifications include unit fixed effects, a quadratic in precipitation and region-specific quadratic time trends. Standard errors are clustered at the département level. Confidence intervals for strawberries exceed the plotting bounds and are omitted. Additional methodological details are provided in Appendix 6.

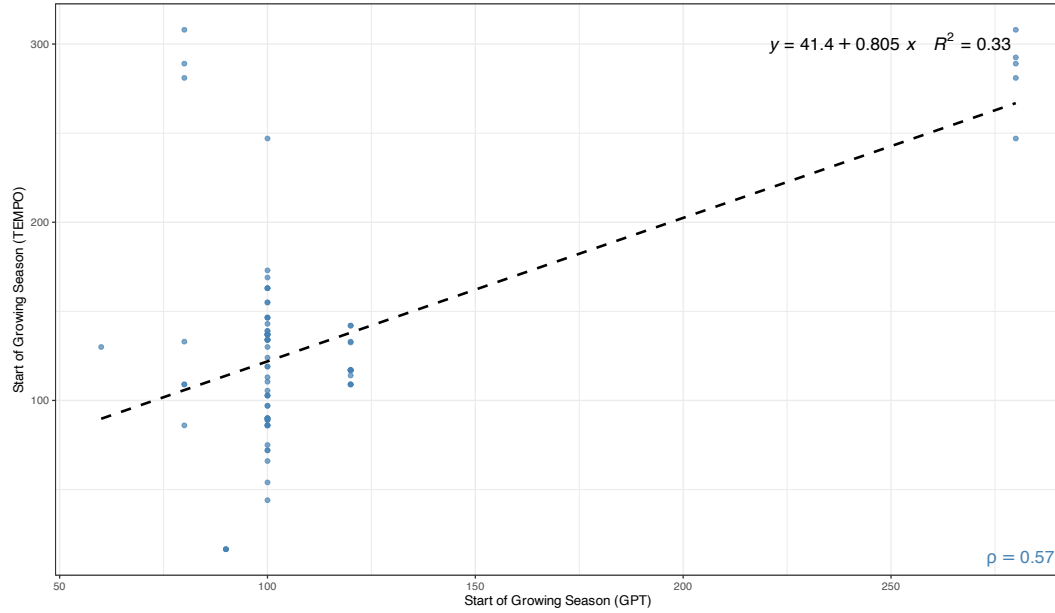
Figure 16: Yield–Temperature Response Functions By Crop Type



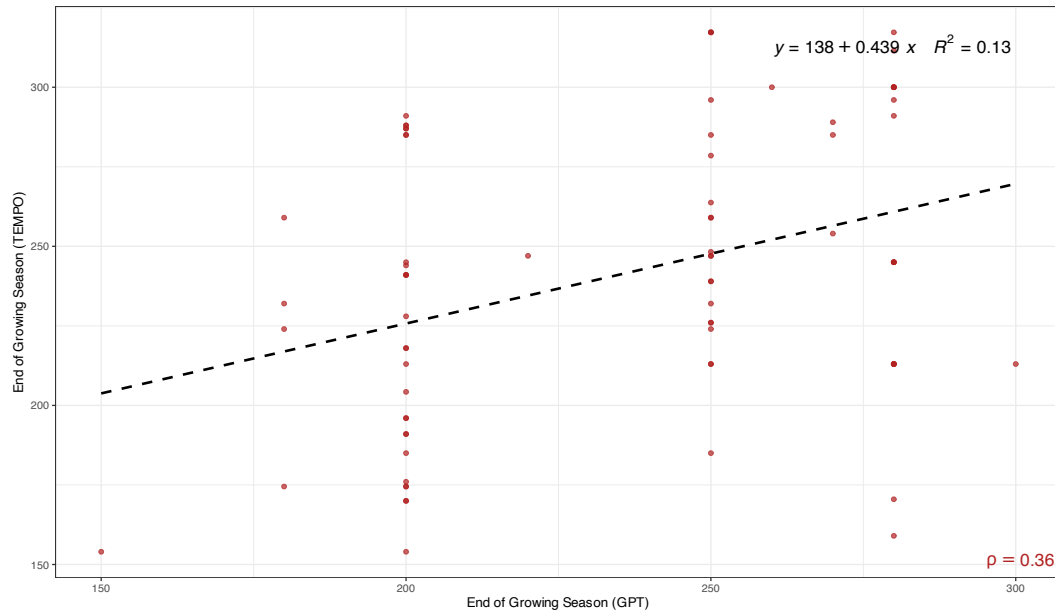
Note: Each panel reports estimated yield–temperature response functions from stepwise regressions using 3°C temperature bins. The red line displays point estimates based on GPT-derived growing season weights, while the blue line reflects estimates using phenological weights from Tempo observations. Exposure is crop-weighted and spatially aggregated using weights derived from the Corine Land Cover data. The regressions pool crop-level data by crop type and include crop-département fixed effects, a quadratic in precipitation, and region-specific quadratic time trends. Standard errors are clustered at the département level. Confidence intervals for strawberries are not shown due to scale constraints. Additional methodological details are provided in Appendix 6.

Figure 17: Comparison of Growing Season Estimates from TEMPO and GPT

(a) Start of the growing season (Day of Year).

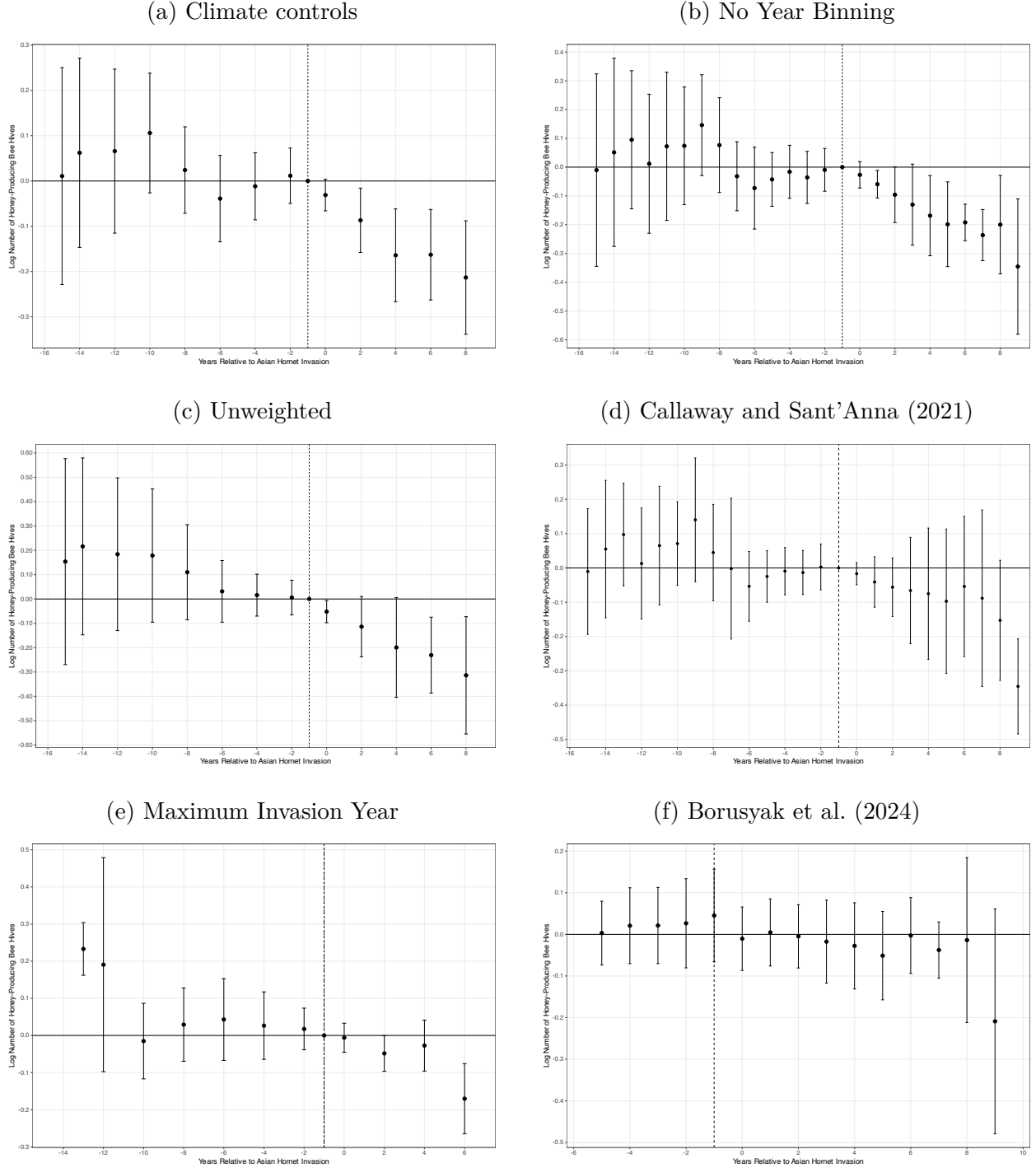


(b) End of the growing season (Day of Year).



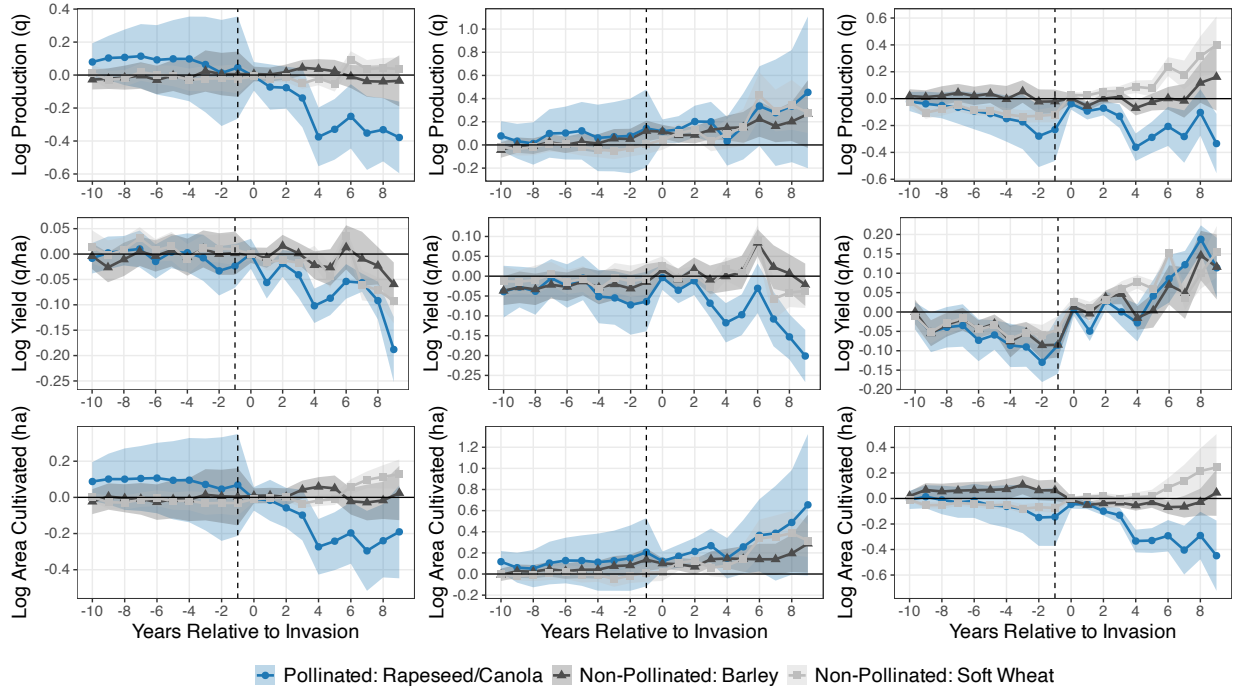
Note: These panels compare growing season estimates derived from TEMPO phenological observations and GPT-based agronomic synthesis. Panel (a) plots the estimated start dates, and panel (b) the estimated end dates, both in terms of day of year. Each point represents a crop matched across the two sources. Dashed lines indicate linear fits. The correlation coefficient (ρ) and regression statistics are annotated in each plot.

Figure 18: Robustness of Event Study: Honey-Production Beehives



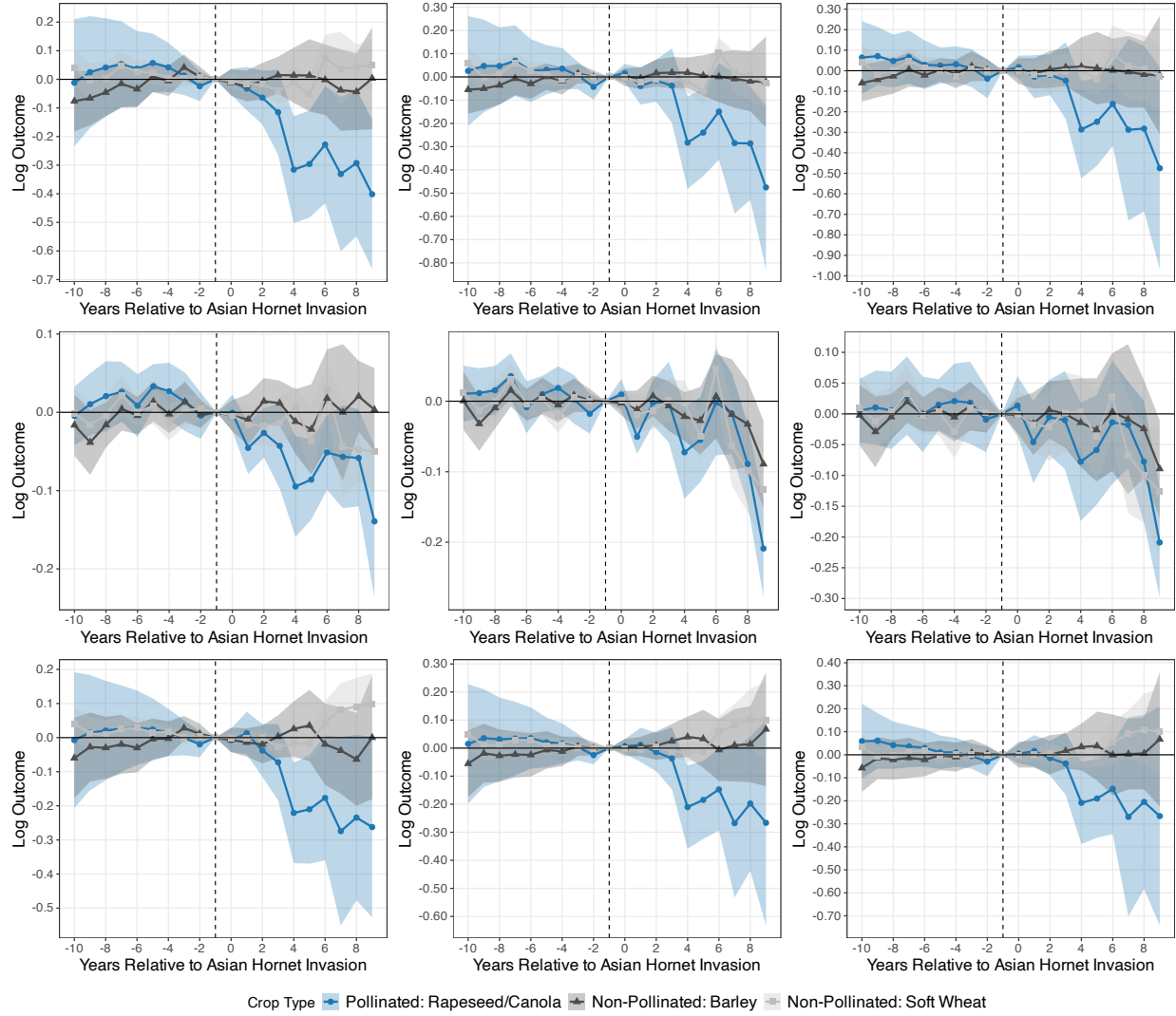
Note: This figure presents robustness checks for event-study estimates of the impact of the Asian hornet invasion on the log number of honey-producing beehives. Panels (a)–(c) and (e) use the estimator of Sun and Abraham (2021) with never-treated regions as the comparison group. Panel (d) applies the estimator of Callaway and Sant’Anna (2021) with not-yet-treated regions as controls. Panel (f) shows estimates from the imputation-based estimator of Borusyak et al. (2024), also using not-yet-treated units as the comparison group. Panel (e) uses the maximum year of first recorded invasion across départements within a region to define treatment timing. Panel (a) includes agro-climatic covariates; however, as noted in email correspondence with `fixest`’s author Laurent Bergé, covariates may not be handled in a way consistent with the original Sun and Abraham (2021) framework. All specifications weight observations by the baseline mean number of honey-producing hives and cluster standard errors at the regional level. Relative time 0 denotes the first year of treatment.

Figure 19: Robustness Checks for Event-Study Estimates on Crop Production



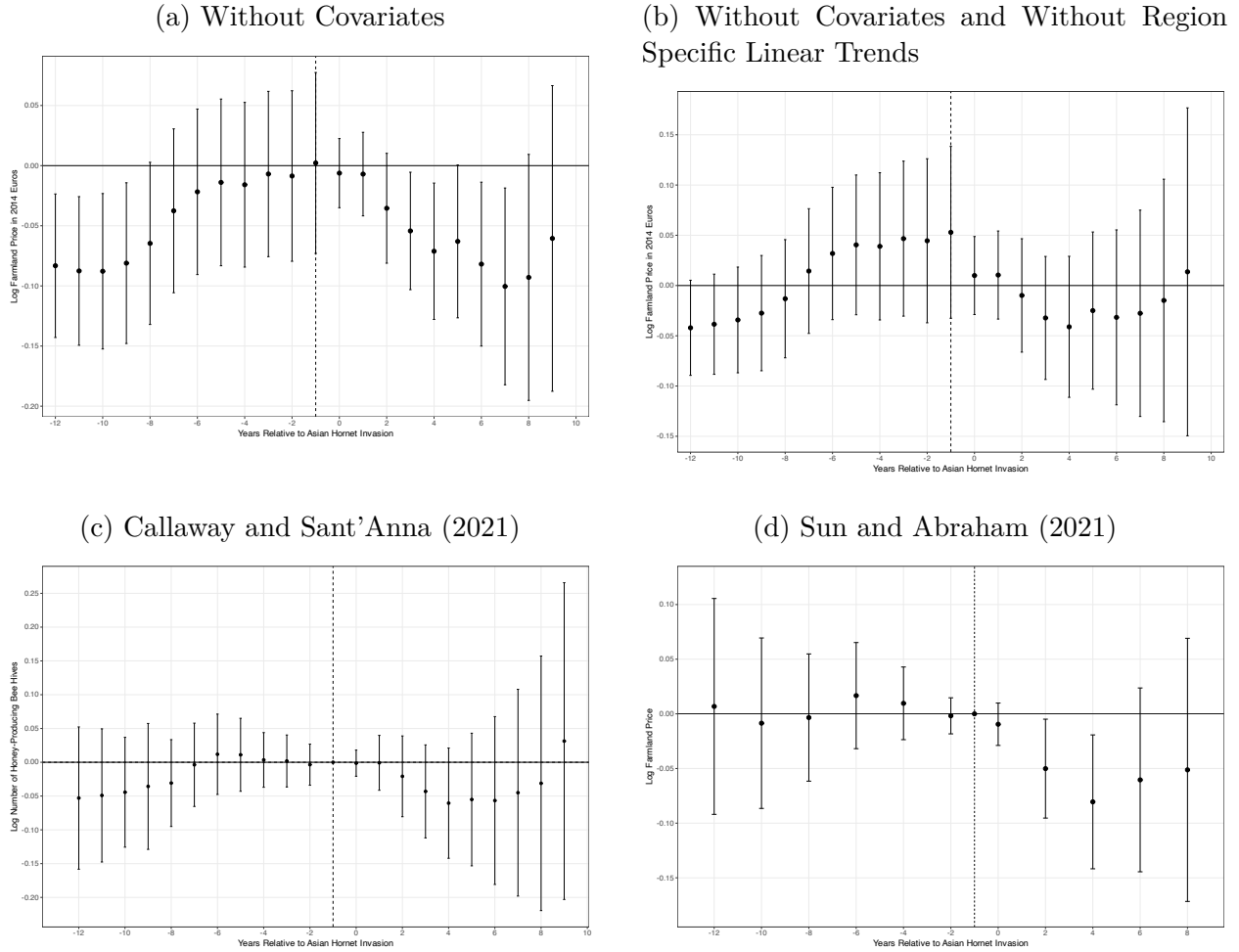
Note: This figure presents robustness checks for the event-study estimates shown in Figure 6. Each row reports logged outcomes for rapeseed (blue), barley (dark gray), and soft wheat (light gray) at the département-year level: total production (top), yield per hectare (middle), and cultivated area (bottom). Each column corresponds to a variant of the baseline specification. Column (1) excludes all time-varying controls. Column (2) removes baseline cultivated area weights. Column (3) augments the baseline with linear region-specific time trends. Estimates are obtained using the imputation estimator of Borusyak et al. (2024), with untreated units as the control group. Standard errors are clustered at the département level. Shaded areas represent 95% confidence intervals.

Figure 20: Estimator Comparison for Event-Study Effects on Crop Production



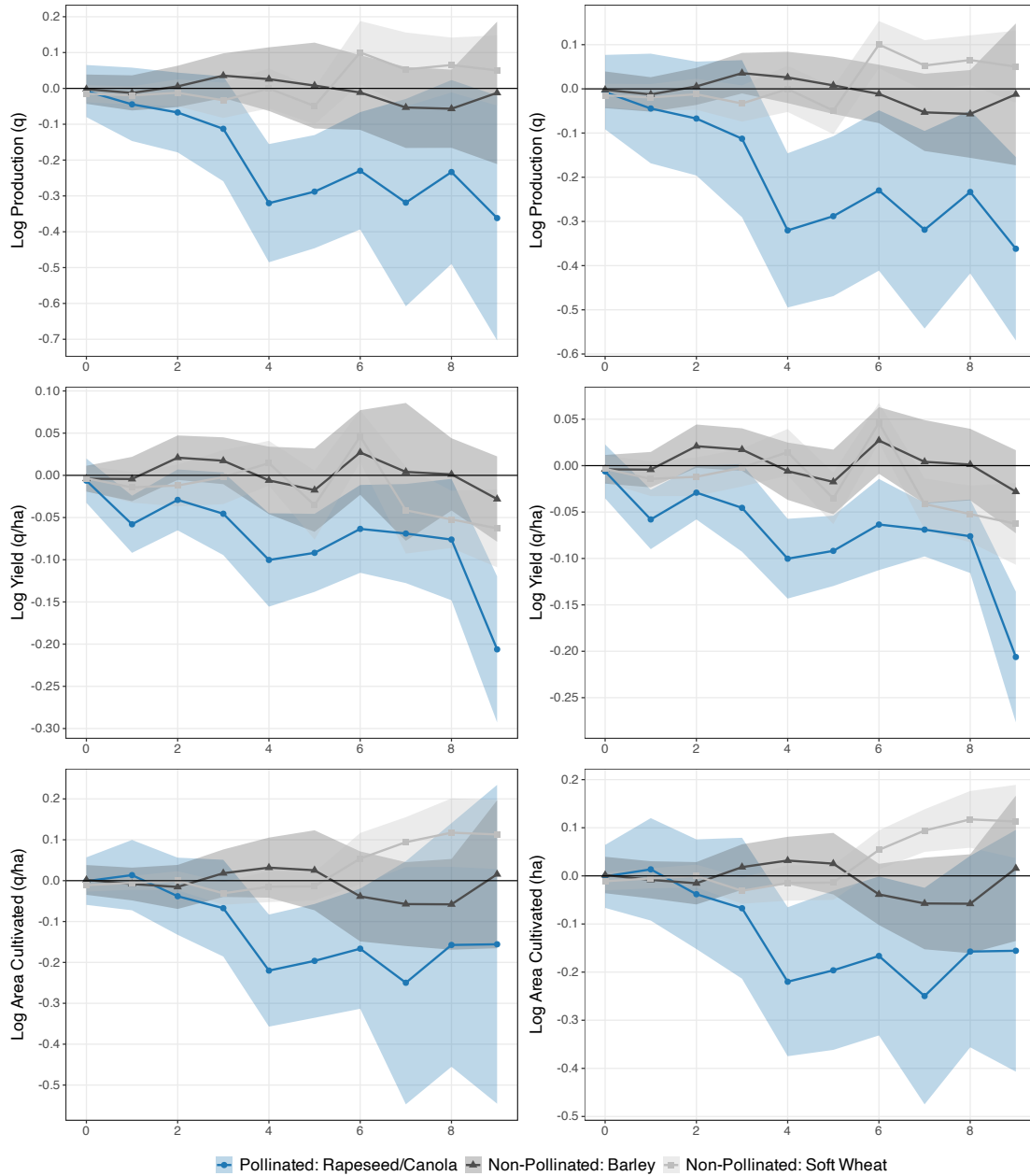
Note: This figure compares event-study estimates of the Asian hornet invasion on crop production across three alternative estimators. Each row displays logged outcomes for rapeseed (blue), barley (dark gray), and soft wheat (light gray) at the département-year level: total production (top), yield per hectare (middle), and cultivated area (bottom). Each column reports results from a different estimation strategy. Column (1) presents a two-way fixed effects (TWFE) model with agro-climatic and night-time light intensity covariates. Column (2) applies the interaction-weighted estimator of Sun and Abraham (2021) using never-treated units as the control group and excludes covariates. Column (3) implements the estimator of Callaway and Sant'Anna (2021) using not-yet-treated units as controls, also without covariates. All specifications include baseline cultivated area weights, and standard errors are clustered at the département level. Shaded regions represent 95% confidence intervals.

Figure 21: Robustness of Event Study: Farmland Prices



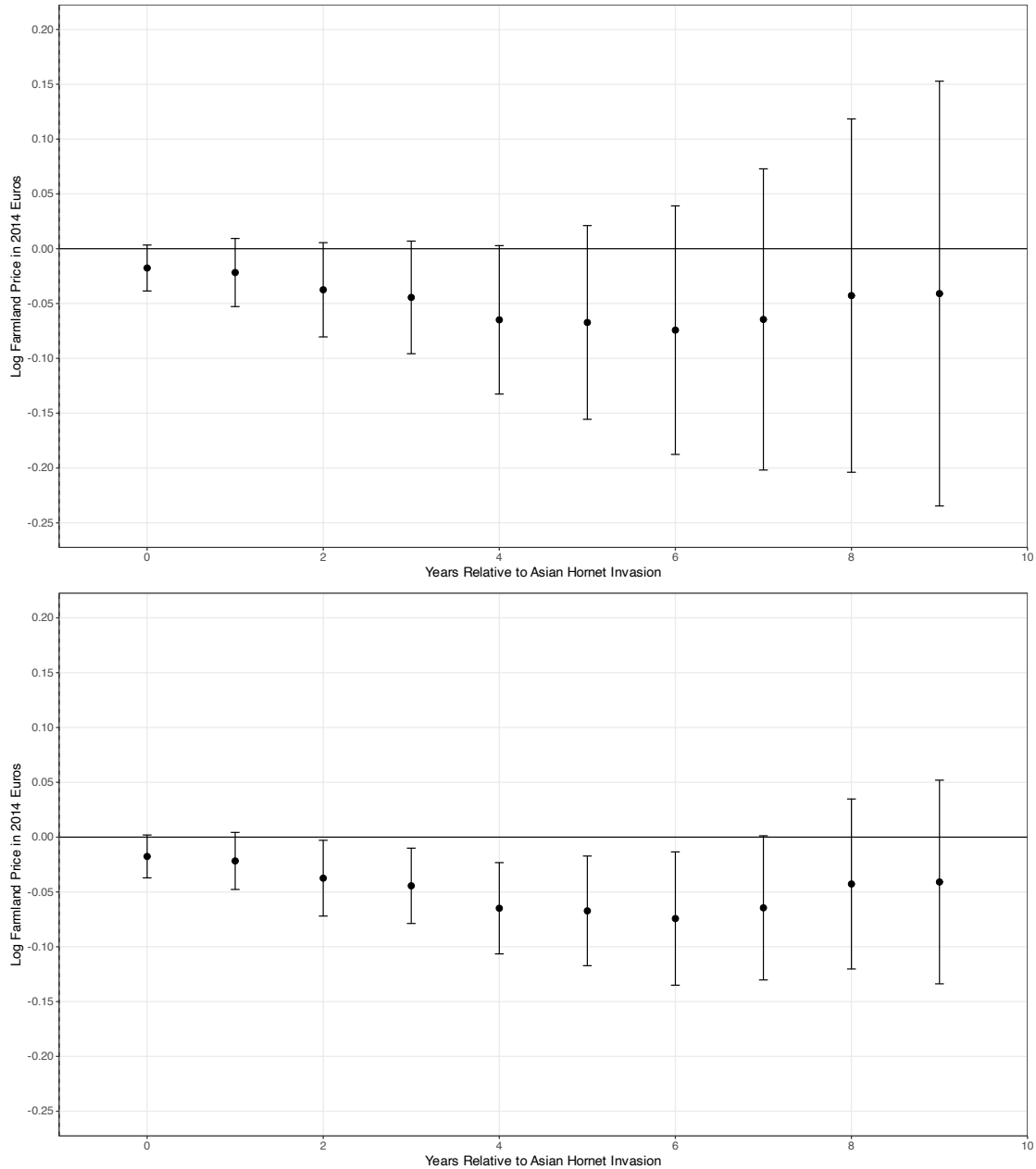
Note: This panel reports event study estimates on farmland prices of alternative specifications. Panel (a) and (b) uses the Borusyak et al. (2024) estimator. Panel (c) uses Callaway and Sant'Anna (2021) with the not-yet-treated as comparison group. Panel (d) uses Sun and Abraham (2021). Observations are unweighted across all specifications and standard errors are clustered at the département level.

Figure 22: Bootstrap vs. Borusyak et al. (2024) Standard Errors for Crop-Level Event-Study Estimates



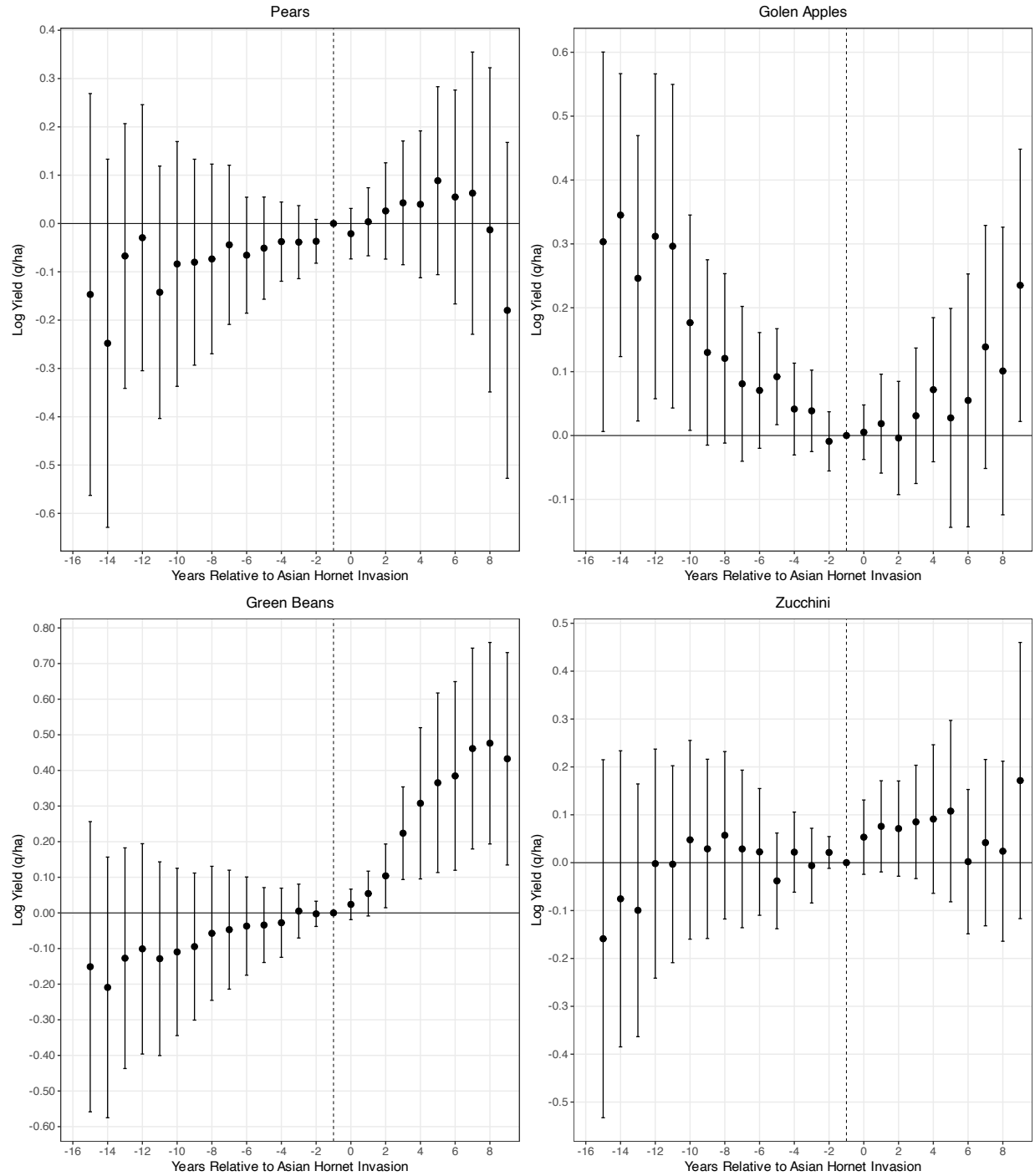
Note: This figure presents a comparison of post-treatment event-study estimates under two standard error computation methods, using the baseline specification. Each row corresponds to a different outcome: log production (top), log yield (middle), and log area cultivated (bottom). Each color indicates a different crop: pollinated (Rapeseed/Canola, blue), and non-pollinated (Barley, dark gray; Soft Wheat, light gray). The left column reports 95% confidence intervals derived from a nonparametric cluster bootstrap procedure in which départements are resampled with replacement and the model re-estimated 1,000 times. The right column displays analytical standard errors computed using the approach of Borusyak et al. (2024). This comparison serves as a consistency check, since bootstrap errors are used in the estimation of cohort-level average treatment effects, for which analytical errors are not available. All specifications weight observations by baseline cultivated area. Further details on the bootstrap procedure are provided in the Appendix 23.

Figure 23: Bootstrap vs. Borusyak et al. (2024) Standard Errors for Farmland Prices Event-Study Estimates



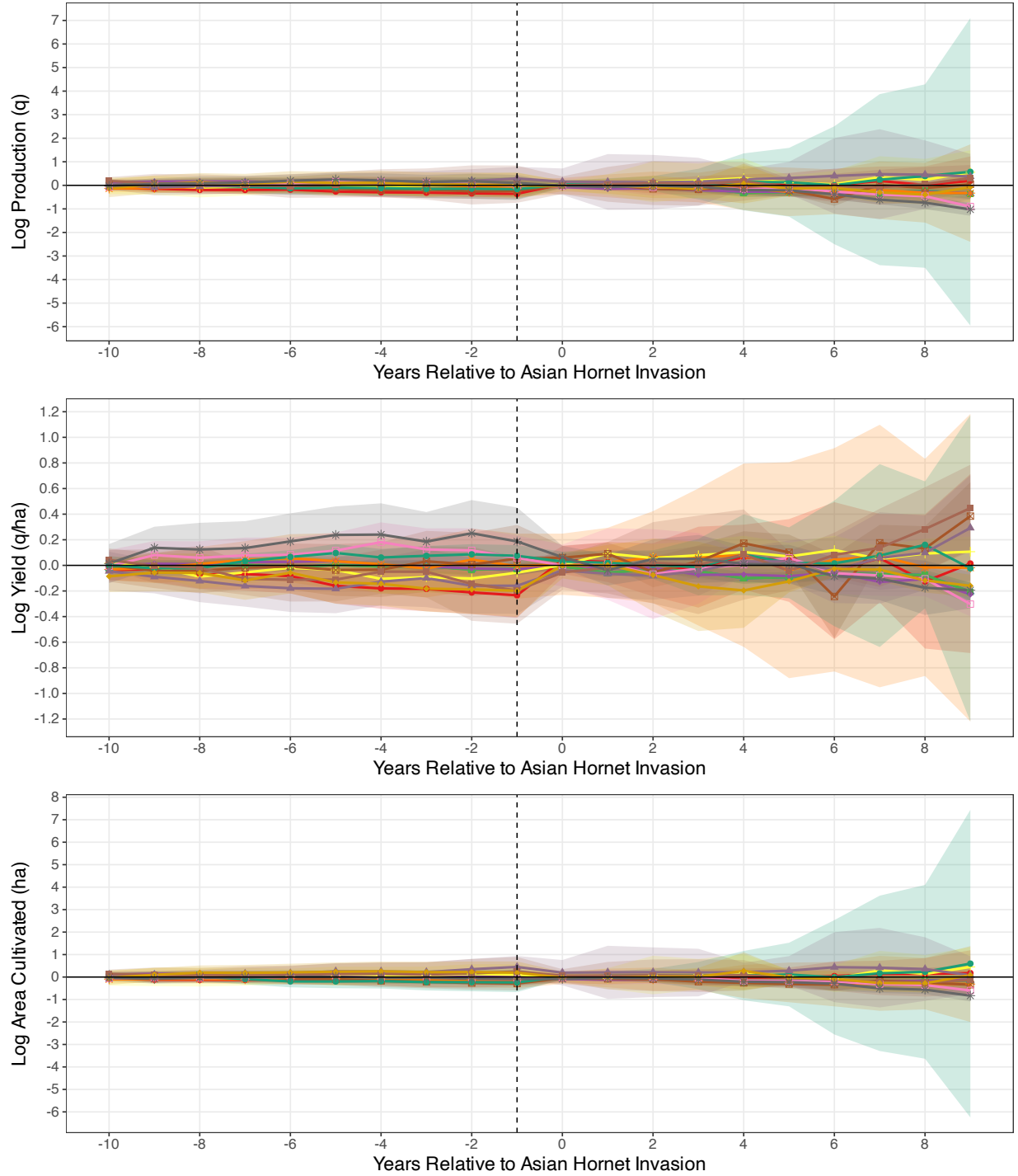
Note: This figure compares post-treatment event-study estimates of farmland prices from the baseline specification using two standard error estimation methods. The top row shows 95% confidence intervals based on a nonparametric cluster bootstrap (1,000 resamples at the département level). The bottom row displays analytical standard errors following Borusyak et al. (2024). This serves as a consistency check, as bootstrap errors are used for cohort-level ATT estimates, for which analytical standard errors are not available. Further details on the bootstrap procedure are provided in Appendix 23.

Figure 24: Event Study Estimates on Fruit and Legumes Crops



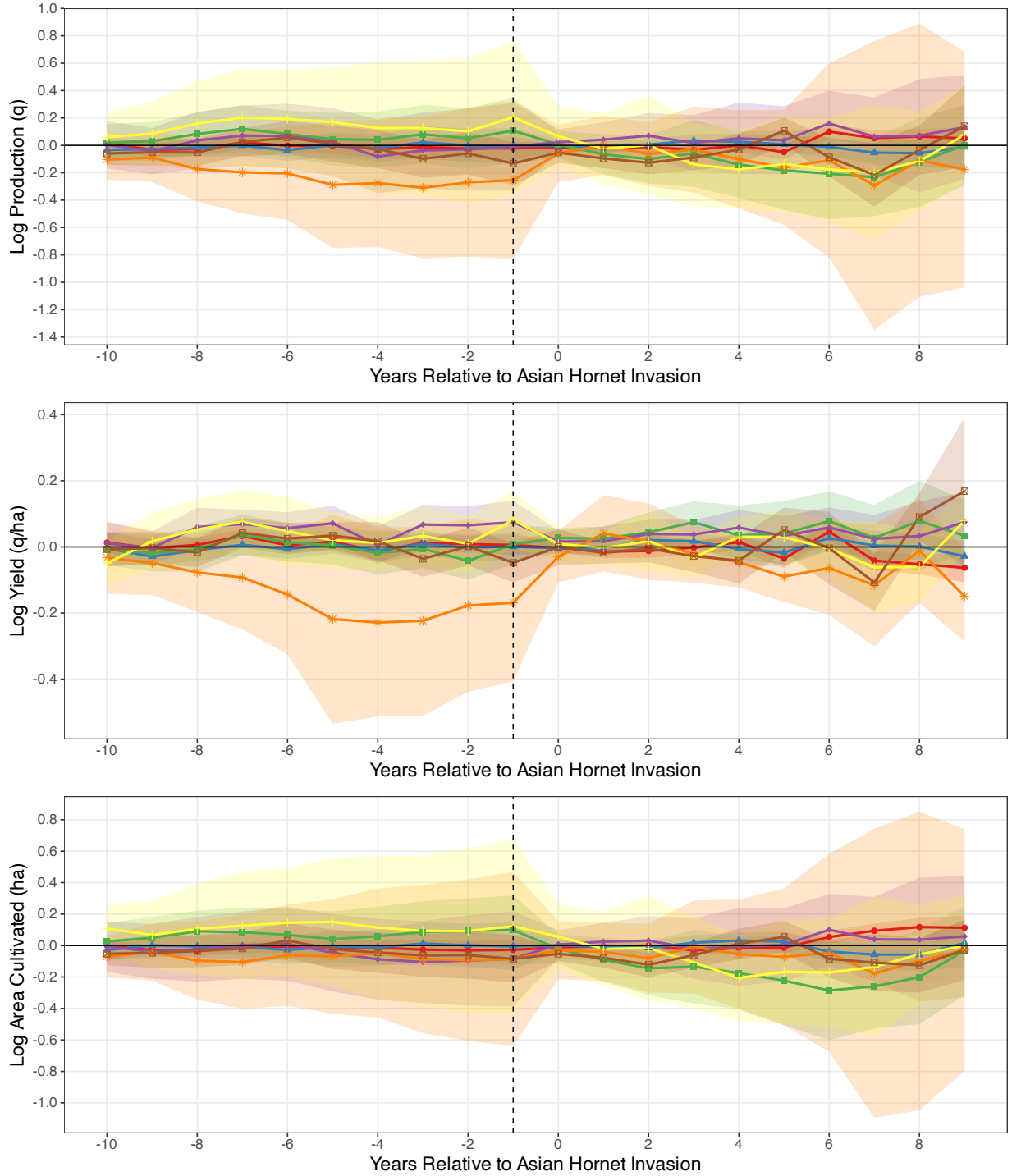
Note: This figure presents event study estimates of the impact of the Asian hornet invasion on the yield of a selected set of fruit and legume crops, using the estimator proposed by Sun and Abraham (2021). The dependent variable is the logarithm of crop yield. Observations are weighted by baseline cultivated area, and standard errors are clustered at the département level.

Figure 25: Event-Study Estimates by Crop: Pollination-Dependent Species



Note: This figure presents event-study estimates of the effect of the Asian hornet invasion on agricultural outcomes for all crops classified as pollination-dependent. Each line represents a separate estimation for a different crop, identified by color. The rows display logged outcomes at the département-year level: production in quintals (top), yield per hectare (middle), and area cultivated in hectares (bottom). All estimates are obtained using the estimator of Borusyak et al. (2024), with untreated units as the control group and a 2015-truncated balanced panel. Specifications include controls for weighted growing and hot degree days (GDD, HDD), a quadratic in precipitation, and average night-time light intensity. Observations are weighted by baseline average cultivated area. Standard errors are clustered at the département level, and shaded bands represent 95% confidence intervals.

Figure 26: Event-Study Estimates by Crop: Non-Pollination-Dependent Species



Note: This figure displays event-study estimates of the effect of the Asian hornet invasion on agricultural outcomes for crops classified as non-pollination-dependent. Each line represents a separate crop-level estimate, distinguished by color. The three panels show logged outcomes at the département-year level: production in quintals (top), yield per hectare (middle), and cultivated area in hectares (bottom). All specifications use the estimator of Borusyak et al. (2024), with untreated units as the control group, based on a 2015-truncated balanced panel. Covariates include weighted growing and hot degree days (GDD, HDD), a quadratic in precipitation, and average night-time light intensity. Observations are weighted by baseline average cultivated area. Standard errors are clustered at the département level; shaded bands represent 95% confidence intervals.

Table 5: Summary Statistics for Farmland Prices and Beehives in Honey Production

Variable	Mean (Std.Dev.)	Range	<i>N</i> (Obs.)	<i>N</i> (Units)
<i>Farmland</i>				
Average Market Price per ha (€)	4,962 (3,463)	1,030–45,570	6,323	91
<i>Beekeeping Activity</i>				
Number of hives	65,445 (53,439)	10,000–185,500	208	13
<i>Urbanisation Rate Proxy</i>				
Night Time Light Intensity	5.28 (3.99)	1.58–51.81	1,504	94

Note: Summary statistics are reported for farmland prices (in 2014 euros) and beehive counts in honey production. Farmland prices are measured at the *Nouvelle Région Agricole*–year level; beehive data are at the region–year level. Means are shown with standard deviations in parentheses; ranges indicate observed minimum and maximum values. *N* (Obs.) denotes the number of observations, and *N* (Units) the number of unique spatial units. Night-time light intensity is included as a proxy for urbanization at the département–year level.

Appendix

A. Construction of Agro-Climatic Variables

To identify the causal impact of the Asian Hornet, it is crucial to control for agro-climatic variation that directly affects crop yields. In particular, changes in the effect of temperature and precipitation during crop growing seasons may confound estimates if they are spatially correlated with the Asian hornet’s spread. For instance, if climate change has led to greater warming or altered precipitation patterns in the South-West – where the Asian hornet first appeared – relative to the North-East. I therefore construct crop-specific, département-level agro-climatic exposure variables using gridded daily weather data and agronomic calendars following the methodology of Schlenker and Roberts (2009).

Data Source and Structure. I obtain minimum and maximum daily temperature in Celsius as well as total daily liquid precipitation in millimeters from the SAFRAN model developed by Météo-France. The data covers the period 1999–2023 at the grid-cell level (8,602 regular cells across metropolitan France) and are accessed via the INRAE SICLIMA platform.

Table 6: Summary Statistics of Agro-Climate Conditions by Crop

	GDD		HDD		Precipitation (mm)	
	Mean (SD)	Range	Mean (SD)	Range	Mean (SD)	Range
Weighted Agro-Climate						
Total rapeseed	2069.98 (317.04)	1189.56–3152.91	1.11 (1.62)	0–12.57	712.77 (180.65)	396.20–1520.35
Total oilseeds	1975.69 (183.38)	1459.61–2509.88	5.14 (7.75)	0–58.77	351.23 (98.19)	63.01–741.61
Sunflower	1880.83 (159.92)	1376.02–2315.09	5.59 (8.01)	0–58.77	317.94 (94.23)	59.75–741.48
Total soft wheat	1673.03 (279.44)	916.68–2627.40	1.15 (1.79)	0–13.33	628.69 (163.62)	296.19–1431.53
Total cereals	—	—	—	—	—	—
Total barley	1439.22 (273.04)	748.30–2369.78	0.23 (0.52)	0–5.15	611.88 (165.26)	288.27–1454.42
Total oats	—	—	—	—	—	—
Triticale	1417.90 (264.81)	718.45–2327.95	0.46 (1.13)	0–11.43	587.13 (155.40)	252.79–1419.17
Spring barley	1429.27 (264.93)	748.30–2369.78	0.22 (0.50)	0–5.15	612.53 (164.13)	290.93–1454.42
Field peas	1069.70 (170.84)	614.81–1594.09	0.41 (1.00)	0–9.19	336.61 (102.36)	116.29–943.81
Rye and mixed grains	—	—	—	—	—	—
Non-irrigated grain maize	2283.54 (198.63)	1683.88–2854.68	4.60 (7.24)	0–58.77	427.68 (110.07)	192.49–840.49
Winter oats	—	—	—	—	—	—
Other apples	2599.62 (305.22)	1810.99–3501.69	4.96 (7.64)	0–69.30	554.69 (145.67)	228.79–1157.66
Golden apples	—	—	—	—	—	—
Total table pears	2361.73 (254.88)	1748.03–3151.59	4.97 (7.62)	0–69.30	438.70 (123.16)	146.54–918.90
Autumn pears	2351.86 (253.94)	1748.03–3151.59	4.89 (7.64)	0–69.30	436.37 (117.52)	146.54–916.65
Zucchini	1397.80 (135.05)	1017.80–1791.36	5.14 (7.87)	0–68.05	211.25 (83.02)	22.28–578.44
Pumpkins, squash, etc.	1405.40 (132.87)	1061.97–1791.36	5.28 (7.96)	0–68.05	207.80 (80.98)	22.28–608.58
Turnip	—	—	—	—	—	—
Green beans	1247.55 (116.56)	905.36–1604.11	4.86 (7.54)	0–67.56	175.59 (71.67)	13.12–524.42
Total strawberries	272.44 (48.12)	74.91–401.56	0.03 (0.13)	0–1.79	58.26 (34.79)	0.64–239.59
Unweighted Agro-Climate						
—	2,875 (437)	1,361–4,181	4.39 (6.86)	0–59.22	860.88 (208.44)	351.55–1,874.90

Note: This table reports summary statistics for crop-weighted Growing Degree Days (GDD), Hot Degree Days (HDD), and total precipitation. Observations are at the département–year level. Further details on variable construction are provided in Appendix 6. Standard deviations are in parentheses. Ranges indicate the minimum and maximum values in the estimation sample. Missing values are denoted by em dashes.

Table 7: Balance of Agro-Climatic, Agricultural, and Farmland Variables by Ever-Treatment Status

Variable	Never-Treated Mean	Ever-Treated Mean	SMD	N (N-T)	N (E-T)
Agro-Climate and Nighttime Lights					
GDD	2659.28	2963.86	0.668	11 607	40 796
HDD	5.08	8.31	0.313	11 607	40 796
Precipitation	998.58	907.33	0.387	11 607	40 796
NTL	5.04	5.64	0.119	11 607	40 796
Crop Groups					
<i>Cereals, Oil & Protein Seeds</i>					
Production (q)	424 955.19	559 916.02	0.073	3098	10 915
Area (ha)	6279.53	9308.56	0.120	3098	10 915
Yield (q/ha)	46.85	43.41	0.164	3098	10 915
<i>Fruits</i>					
Production (q)	4129.14	14 512.75	0.174	3188	11 241
Area (ha)	28.52	82.42	0.174	3188	11 241
Yield (q/ha)	112.41	133.96	0.190	3188	11 241
<i>Industrial Crops</i>					
Production (q)	584 637.26	349 123.18	0.068	815	2827
Area (ha)	785.36	532.00	0.057	815	2827
Yield (q/ha)	222.15	139.79	0.312	815	2827
<i>Legumes</i>					
Production (q)	11 633.08	14 631.49	0.037	3868	13 602
Area (ha)	50.18	74.75	0.055	3868	13 602
Yield (q/ha)	344.82	314.96	0.054	3868	13 602
<i>Potatoes and Tubers</i>					
Production (q)	639 996.72	248 520.08	0.228	638	2211
Area (ha)	1647.62	712.69	0.229	638	2211
Yield (q/ha)	295.48	275.30	0.215	638	2211
Honey Production					
N. Hives in Production	45 038.38	74 343.68	0.643	8	44
Farmland Prices					
Price (ha)	3920.48	3694.51	0.112	355	1540

Note: This table reports means and standardized mean differences (SMDs) between ever-treated and never-treated units for key variables. SMDs are calculated as the mean difference divided by the pooled standard deviation. **Bolded SMD values** denote differences exceeding 0.4, indicating potential imbalance. **Bolded area values** flag cases where the average cultivated area is below 50 hectares in either group.

Table 8: Covariate Balance by Ever-Treatment Status for Pollination Dependent Crops

Variable	Control Mean	Treated Mean	SMD	N (C)	N (T)
Other apples					
Production (q)	31 220.76	124 363.34	0.537	78	292
Area (ha)	118.49	346.89	0.526	78	292
Yield (q/ha)	212.04	290.19	0.756	78	292
Zucchini					
Production (q)	12 914.91	18 485.72	0.106	57	279
Area (ha)	28.44	40.44	0.169	57	279
Yield (q/ha)	351.58	310.07	0.293	57	279
Golden apples					
Production (q)	48 337.15	96 005.47	0.287	66	268
Area (ha)	129.67	234.63	0.279	66	268
Yield (q/ha)	247.56	331.04	0.795	66	268
Green beans					
Production (q)	31 695.27	39 015.50	0.079	84	293
Area (ha)	255.65	367.47	0.144	84	293
Yield (q/ha)	120.02	84.44	0.289	84	293
Garden turnip					
Production (q)	9105.99	9201.64	0.005	74	258
Area (ha)	29.77	38.51	0.118	74	258
Yield (q/ha)	218.59	221.26	0.033	74	258
Autumn pears					
Production (q)	11 920.09	8519.17	0.180	55	262
Area (ha)	56.00	42.48	0.166	55	262
Yield (q/ha)	180.07	183.37	0.041	55	262
Pumpkins, squashes, etc.					
Production (q)	16 890.26	27 045.54	0.151	57	280
Area (ha)	42.11	63.06	0.189	57	280
Yield (q/ha)	358.25	314.33	0.344	57	280
Total rapeseed					
Production (q)	316 720.24	388 229.63	0.150	72	277
Area (ha)	10 668.39	13 054.34	0.151	72	277
Yield (q/ha)	29.07	27.77	0.249	72	277
Total strawberries					
Production (q)	2904.22	6823.56	0.285	73	284
Area (ha)	26.03	47.15	0.246	73	284
Yield (q/ha)	95.40	122.80	0.687	73	284
Total oilseeds					
Production (q)	339 187.72	636 944.24	0.509	76	280
Area (ha)	11 551.18	23 757.43	0.576	76	280
Yield (q/ha)	27.82	26.02	0.326	76	280
Total table pears					
Production (q)	21 111.19	27 303.35	0.099	68	266
Area (ha)	94.31	123.26	0.116	68	266
Yield (q/ha)	160.95	186.58	0.313	68	266
Sunflower					
Production (q)	29 676.33	237 834.40	0.811	66	262
Area (ha)	1069.02	10 219.55	0.827	66	262
Yield (q/ha)	25.44	23.08	0.465	66	262

Note: This table reports means and standardized mean differences (SMDs) between ever-treated and never-treated units for pollination dependent crops. SMDs are calculated as the mean difference divided by the pooled standard deviation. **Bolded SMD values** denote differences exceeding 0.4, indicating potential imbalance. **Bolded area values** flag cases where the average cultivated area is below 50 hectares in either group.

Table 9: Covariate Balance by Ever-Treatment Status for Non-Pollination Dependent Crops

Variable	Control Mean	Treated Mean	SMD	N (C)	N (T)
Winter oats					
Production (q)	8611.79	43 032.99	0.830	48	273
Area (ha)	188.94	924.63	0.943	48	273
Yield (q/ha)	44.46	42.40	0.160	48	273
Non-irrigated grain maize					
Production (q)	1 057 070.37	909 639.10	0.116	71	258
Area (ha)	12 529.23	12 098.45	0.029	71	258
Yield (q/ha)	80.84	72.13	0.539	71	258
Protein peas					
Production (q)	139 501.75	208 804.38	0.218	72	280
Area (ha)	3003.90	4731.07	0.251	72	280
Yield (q/ha)	37.60	37.25	0.036	72	280
Rye and mixed grains					
Production (q)	8702.62	15 831.58	0.432	73	283
Area (ha)	179.38	364.39	0.481	73	283
Yield (q/ha)	47.64	43.95	0.323	73	283
Total oats					
Production (q)	39 182.63	64 845.69	0.481	76	292
Area (ha)	873.53	1471.12	0.571	76	292
Yield (q/ha)	41.50	40.98	0.046	76	292
Total soft wheat					
Production (q)	2 950 474.19	3 678 249.63	0.182	80	295
Area (ha)	39 025.95	53 045.69	0.283	80	295
Yield (q/ha)	64.09	59.07	0.326	80	295
Total barley					
Production (q)	1 147 084.31	1 050 070.03	0.066	84	292
Area (ha)	17 863.75	17 583.11	0.013	84	292
Yield (q/ha)	54.65	52.22	0.178	84	292
Total cereals					
Production (q)	5 325 181.45	6 959 385.41	0.284	84	295
Area (ha)	71 293.25	102 525.92	0.426	84	295
Yield (q/ha)	66.91	61.60	0.321	84	295
Triticale					
Production (q)	63 606.11	161 174.36	0.676	76	288
Area (ha)	1193.17	3311.66	0.758	76	288
Yield (q/ha)	54.34	48.78	0.458	76	288

Note: This table reports means and standardized mean differences (SMDs) between ever-treated and never-treated units for non-pollination dependent crops. SMDs are calculated as the mean difference divided by the pooled standard deviation. **Bolded SMD values** denote differences exceeding 0.4, indicating potential imbalance. **Bolded area values** flag cases where the average cultivated area is below 50 hectares in either group.

Table 10: Metadata Summary of Datasets Used

Dataset	Description	Source	Temporal Unit	Spatial Unit	Key Variables	Transformed	URL
SAA	Agricultural production statistics	AGRESTE	Annual	Département	Production (q), Area (ha), Yield (q/ha), Hive count (Region level)	No	Link
VVT	Farmland market prices	AGRESTE	Annual	Nouvelle région agricole	Farmland prices (€/ha)	No	Link
POLLINATION	Pollination dependence index	Klein et al. (2006)	N/A	N/A	Pollination dependence index by crop	No	Link
SAFRAN	Historical daily weather	Météo-France	Daily	SAFRAN grid cell (8×8 km)	Min/Max temperature (°C), Precipitation (mm)	Yes	Link
CORINE	Agricultural land cover classification	Copernicus	N/A	Grid cell (100 m), aggregated to 8×8 km	Share of agricultural land cover	Yes	Link
TEMPO	Phenological stage observations	TEMPO portal	Daily	GPS coordinates	BBCH development stages	Yes	Link
SVNL	Night-time light intensity (urbanisation proxy)	Chen et al. (2024)	Annual	SVNL grid cell	Annual mean light intensity (0–63 DN)	Yes	Link

Note: “Transformed” indicates whether the dataset was preprocessed (e.g., aggregated, cleaned, or spatially merged). “N/A” denotes that the dataset does not follow a regular spatial or temporal resolution. SVNL = Simulated VIIRS Night-time Lights; BBCH = Biologische Bundesanstalt, Bundessortenamt, and CHemical industry scale for crop phenology.

Temperature Interpolation and Binning. For each grid-day, I interpolate the hourly temperature profile using a sinusoidal curve between the observed daily minimum and maximum temperatures. This method allows me to estimate the fraction of the day during which temperature falls within each degree bins.

Unweighted Aggregation. The first approach assumes uniform spatial distribution of crops within each département and uniform phenological timing across the calendar year. For each degree bin, I compute the mean daily exposure across all grid cells in the département and sum these exposures over the calendar year. This yields an unweighted estimate of the number of département-days spent within each temperature bin. Precipitation is aggregated analogously by averaging daily values across grid cells within each département and summing these averages over the calendar year.

Crop-Specific Spatial Weighting. To account for within-département variation in crop location, I construct crop-specific spatial weights using land cover data from the 2012 Copernicus CORINE inventory. Each grid cell is assigned a weight based on the proportion of relevant land cover types, mapped to each crop category. Due to known limitations in satellite-based land cover data – particularly underreporting of perennial crop areas such as orchards – and the absence of standardized weighting protocols, I adopt a conservative

approach. Specifically, I assign a baseline weight of 1 to all grid cells and rescale crop-specific weights to lie between 1 and 100. This bounding ensures that no single grid cell receives disproportionately high influence in the weighted aggregation. The resulting grid-level weights are then used to compute département-level agro-climatic exposure measures. Figure 14 shows the spatial distribution of weights for selected crops.

Crop-Specific Temporal Weighting. I estimate crop growing seasons using phenological data from the TEMPO database, which reports in-situ observations on the BBCH scale for a wide range of cultivated species. For each crop, I define the growing season as the interval between the first quartile of early-stage (BBCH 0–1) observations and the third quartile of late-stage (BBCH 8–9) observations, based on data from 2000 to 2023. This conservative approach minimizes the influence of outliers and captures the typical phenological window across départements.

For crops not directly represented in the TEMPO database (e.g., Golden Apples), I impute growing seasons using ChatGPT-generated estimates. These are cross-validated against TEMPO-based values for crops with dual coverage, yielding moderate correlations of 0.57 for start dates and 0.36 for end dates – imperfect but without evidence of systematic bias (Figure 17). Robustness checks using unweighted agro-climatic metrics yield qualitatively similar results.

For crops with growing seasons that span two calendar years, I assign the start date to the previous year and the end date to the current one. I then aggregate the crop-specific daily département-level temperature exposure over the corresponding growing season. This yields a more agronomically meaningful estimate of the number of days within each temperature bin during the period most critical to crop development. Summary statistics are reported for a selected set of crops in Figure 12.

Final Aggregation. For each crop-département-year combination, I compute the spatially weighted daily mean temperature exposure over the crop-specific growing season and aggregate these to recover growing-season exposure per degree bin. Precipitation is aggregated similarly.

Summary Metrics. To reduce dimensionality and capture agronomically relevant aspects of thermal exposure, I construct two standard summary measures:

- **Growing Degree Days (GDD):** Calculated as the sum of temperature degrees above a base threshold of 4°C and below an upper threshold of 30°C. Specifically, each daily temperature observation contributes $(T - 4)$ if $4^\circ \leq T \leq 30^\circ$, and 0 otherwise.

- **Hot Degree Days (HDD):** Captures exposure to extreme heat, computed as the sum of temperature degrees exceeding 30°C. Each observation contributes $(T - 30)$ if $T > 30^\circ$, and 0 otherwise.

These metrics are computed using both unweighted and crop-weighted daily temperature profiles, aggregated over the crop-specific growing season and spatial grid. They are used as key climatic controls in the empirical analysis. Summary statistics for each crop-specific dataset are reported in Table 6.

B. Yield–Temperature Response Function

To validate the construction of the agro-climatic exposure variables, I replicate the empirical strategy of Schlenker and Roberts (2009), who document a robust nonlinear relationship between crop yields and high-temperature exposure. Their seminal study finds sharp yield declines for maize, soybean, and cotton in the U.S. when daily temperatures exceed 30°C during the growing season.

I estimate an analogous step-function model:

$$\ln(y_{cdrt}) = \sum_k \beta_k \text{temp}_{cdt}^{(k)} + \gamma_1 p_{cdt} + \gamma_2 p_{cdt}^2 + \alpha_i + \delta_r t + \delta_r^2 t^2 + \varepsilon_{cdt}$$

The dependent variable is the logarithm of yield for crop c in département d , region r , and year t . The main regressors $\text{temp}_{cdt}^{(k)}$ denote the total number of crop-weighted growing-season days in each 3°C bin⁴².

Total crop-weighted precipitation p_{cdt} is included with a quadratic term. The specification controls for département fixed effects in crop-level regressions and crop–département fixed effects in pooled regressions. Region-specific quadratic time trends allow for heterogeneous technological and productivity evolution. Standard errors are clustered at the département level.

Among the crops analyzed by Schlenker and Roberts (2009), only maize is widely cultivated in France. The estimated temperature–yield profile for maize in my data replicates their main finding: yields remain stable at moderate temperatures but decline sharply above 30°C, with steeper losses at higher thresholds (see Figure 15).

Results for other major field crops—including wheat and rapeseed (colza)—exhibit similarly nonlinear patterns (see Figure 16). These findings are consistent with recent farm-level estimates for France reported by Puy and Shrader (2024). The response function for sunflower,

⁴²All temperatures below 0°C and above 35°C are grouped into the lowest and highest bins, respectively.

by contrast, shows a muted response and in some specifications a slight increase in yields at high temperatures, possibly reflecting differing heat tolerance or cropping practices.

Finally, I estimate the temperature–yield relationship for one fruit crop (apples) and one legume (strawberries). Both exhibit declining yields at elevated temperatures, particularly above 30°C. However, the strawberry estimates are noisy due to low coverage in high-temperature départements and greater production heterogeneity.

C. Conceptual Framework

In this study, we are interested in the effect of the successful establishment of the Asian Hornet on agricultural outcomes. This setting is a case of binary-staggered treatment adoption with treatment timing heterogeneity. Formally, we may define the treatment variable as:

$$D_{i,t} = 1\{t \geq g_i\},$$

where $D_{i,t}$ indicates whether unit i is treated in year $t \in \{\underline{T}, \dots, \bar{T}\}$, and g_i is the first year of treatment where $g \in \{2004, \dots, 2018, 2020, 2023\}$. If unit i is never treated, set $g_i = 0$.

Relying on the potential outcome framework, let us define the unit-time specific treatment effect:

$$\text{TE}_{i,t} = E[Y_{i,t} - Y_{i,t}(\mathbf{0}_t)] = E[Y_{i,t}(\mathbf{0}_{g_i-1}, \mathbf{1}_{t-(g_i-1)}) - Y_{i,t}(\mathbf{0}_t)].$$

that is, the expected effect of having the Asian Hornet present from g_i through time t , compared to not having been invaded.

Re-expressing in terms of relative time $\ell = t - g_i$, where $\ell = 0$ corresponds to the first year of treatment, the relative treatment effect is:

$$\text{TE}_{i,\ell}^r = E[Y_{i,g_i+\ell}(\mathbf{0}_{g_i-1}, \mathbf{1}_{\ell+1}) - Y_{g,g_i+\ell}(\mathbf{0}_{g_i+\ell})].$$

The first parameter of interest is the average treatment effect of having been exposed to the Asian Hornet for ℓ periods:

$$\text{ATT}_\ell = \frac{1}{G_\ell} \sum_{i: g_i > 0, g_i + \ell \leq T} \text{TE}_{g,\ell}^r,$$

where N_ℓ is the number of units for which relative period to first treatment ℓ is observed before the last year in the sample T .

The second parameter is the average treatment effect on the treated across all exposed units

and time periods in the estimation window:

$$\text{ATT} = \frac{1}{\sum_i \sum_t D_{it}} \sum_{\ell} N_{\ell} \text{ATT}_{\ell},$$

These parameters serve two purposes. First, the dynamic effects ATT_{ℓ} allow us to examine treatment dynamics and assess the plausibility of identifying assumptions. Second, ATT summarizes the overall impact of the invasion over the estimation window. However, both should be interpreted with caution. Under treatment effect heterogeneity, variation in ATT_{ℓ} may reflect both dynamic effects and compositional changes in the treated population as we do not expose all units for relatively longer ℓ . For the same reasons, units with longer exposure durations receive greater weight in the estimation of ATT.

To address these concerns, I additionally report cohort-specific average treatment effects in the Appendix, defined by the first year of treatment:

$$\text{ATT}_g = \frac{1}{N_g} \sum_{i: g_i=g, g \leq t \leq T} \text{TE}_{i,t},$$

where N_g is the number of units first treated in year g .

D. Heterogeneity-Robust Estimators

A growing literature proposes alternative estimators for binary, staggered treatment settings with treatment effect heterogeneity (e.g., Callaway and Sant’Anna (2021) and Gardner (2022)). These estimators differ in the choice of comparison groups, the estimation method and the incorporation of covariates leading to distinct identifying assumptions and statistical properties. In this study, I rely primarily on the imputation-based estimator proposed by Borusyak et al. (2024) which I concisely present below⁴³.

Let Ω denote the set of observations in the estimation sample, and define $\Omega_1 = \{(i, t) \in \Omega : D_{it} = 1\}$ as the set of treated observations and $\Omega_0 = \{(i, t) \in \Omega : D_{it} = 0\}$ as the untreated observations.

Identification of the target parameters rests on the following assumptions:

Assumption 1 (Conditional Parallel Trends). *There exist non-stochastic unit and time effects α_i and β_t , and a parameter vector γ' , such that the untreated potential outcome satisfies*

$$E[Y_{it}(0)] = \alpha_i + \delta_t + \gamma' \mathbf{X}_{it} \quad \forall (i, t) \in \Omega.$$

⁴³Note that the two-stage estimator proposed by Gardner (2022), which shares similar features, yields results that are nearly identical in practice and serves as a robustness check.

This is a standard conditional parallel trends assumption, allowing potential outcomes to evolve according to a two-way fixed effects structure, conditional on observed covariates $\mathbf{X}_{it}$.

Assumption 2 (No Anticipation). *For all $(i, t) \in \Omega_0$, $Y_{it} = Y_{it}(0)$.*

This implies that units do not alter their behavior in anticipation of future treatment.

Under these two assumptions, the imputation-based estimator is an unbiased estimator of the target parameters ATT and ATT_ℓ . Specifically, the estimation procedure is as follows:

1. *Estimate by OLS the untreated potential outcomes using only untreated observations $(i, t) \in \Omega_0$:*

$$Y_{it} = \alpha_i + \delta_t + \gamma' \mathbf{X}_{it} \delta + \varepsilon_{it},$$

2. *Impute untreated potential outcomes for treated observations:* For each $(i, t) \in \Omega_1$, predict $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\delta}_{tt} + \hat{\gamma}' \mathbf{X}_{it}$, and compute the estimated treatment effect $\hat{TE}_{i,t} = Y_{it} - \hat{Y}_{it}(0)$.
3. *Aggregate treatment effects:* Compute a weighted average of $\hat{\tau}_{it}$ to estimate the target parameters⁴⁴:

$$\begin{aligned} \hat{ATT}_\ell &= \frac{1}{G_\ell} \sum_{i: g_i > 0, g_i + \ell \leq T} \hat{\tau}_{it}, \\ \hat{ATT} &= \frac{1}{\sum_i \sum_t D_{it}} \sum_\ell N_\ell \hat{ATT}_\ell \end{aligned}$$

The estimator proposed by Borusyak et al. (2024) is particularly well-suited to this setting for several reasons.

First, it flexibly accommodates unit-specific time-varying covariates, enhancing the credibility of the parallel trends assumption. Unlike doubly robust estimators based on weighted differences across treatment cohorts (e.g., Callaway and Sant’Anna (2021) and Sun and Abraham (2021)), which require sufficient within-cohort variation and may suffer from multicollinearity in finite samples, the imputation-based approach estimates covariate effects directly via OLS on untreated units. This makes it more stable and tractable, particularly in settings with limited variation or small sample sizes.

Second, the estimator is the most efficient linear heterogeneity robust estimator under classical error assumptions. While the assumption of spherical disturbances is strong, Monte Carlo simulations and empirical exercises in Borusyak et al. (2024) suggest that the efficiency advantages relative to alternative robust estimators—such as those of Callaway and Sant’Anna (2021) or de Chaisemartin and D’Haultfoeuille (2022)—are maintained even in

⁴⁴Note that more generally, this is an unbiased estimator of any transformation of the unit-time specific treatment effects including ATT_g .

the presence of heteroskedasticity and serial correlation. In my application, standard errors are clustered at the treatment level to account for correlated shocks within départements.

Third, the imputation framework generalizes easily to triple-difference designs by modifying the specification of the untreated potential outcome. This is particularly attractive in my context, as it enables comparisons across crop types that are subject to similar agro-climatic and economic conditions but differ in their dependence on insect pollination. This design offers a way to partially relax the parallel trends assumption by introducing an internal placebo.

A potential limitation of the imputation-based estimator, however, is its reliance on the correct specification and consistent estimation of untreated potential outcomes. In settings with small samples or model misspecification—such as structural differences between treated and untreated units—imputation error may introduce bias or additional variance. Given the relatively small number of départements, the spatial correlation in invasion dynamics, and heterogeneity in agricultural production structures, I complement the main results with estimators that do not require correct specification of the untreated potential outcome.

Specifically, I also implement the dynamic interaction-weighted estimator proposed by Sun and Abraham (2021). This approach addresses the issue of forbidden comparisons in staggered adoption settings by estimating cohort-specific average treatment effects at each event time. They first estimate the following saturated model via OLS:

$$y_{it} = \alpha_i + \delta_t + \sum_{g_i > 0} \sum_{\ell \neq -1} \delta_{g,\ell} \cdot \left(1\{g_i = g\} \cdot D_{it}^\ell\right) + \varepsilon_{it},$$

where $D_{it}^\ell = 1\{t - g_i = \ell\}$ is a relative time indicator, and cohort g is excluded from the omitted period $\ell = -1$ for normalization.

The resulting cohort- and event-time-specific treatment effects are then aggregated using weights reflecting the number of units contributing to each estimate. Under the assumptions of parallel trends across cohorts and no anticipation, this procedure yields unbiased estimates of ATT_ℓ by relative time, or a single average treatment effect, ATT , pooled across cohorts and periods.

E. Standard Errors Estimation for Cohort ATT – Imputation Estimator

The `did_imputation` R package, which implements the imputation-based estimator of Borusyak et al. (2024), does not support inference for cohort-specific average treatment effects, although such estimation is theoretically possible.

To address this limitation, I estimate cohort-specific ATT manually, following the same first-stage residualization strategy used by `did_imputation`, and validate the approach by recovering the overall ATT and event-study estimates produced by the package.

For inference, I implement a nonparametric bootstrap clustered at the département level with 1,000 replications. In each iteration, I re-estimate the first-stage model, compute residualized treatment effects, and average these within cohorts. I then calculate standard errors as the standard deviation of the resulting bootstrap distribution and construct confidence intervals using the normal approximation.

This bootstrap-based approach accommodates arbitrary heteroskedasticity and intra-cluster correlation. Figure 23 compares standard errors from `did_imputation` with the bootstrapped estimates. The two are highly correlated, but the bootstrapped estimates are consistently slightly larger. This suggests that the bootstrap approach may yield conservative standard errors, though further investigation would be needed to formally assess the degree and conditions under which this occurs.